

Where irrigation exists is globally contested

3. Main analysis

Arnald Puy

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1 Preliminary

```
# Load libraries -----

sensobol::load_packages(c("data.table", "scales", "cowplot", "tidyverse", "sensobol",
                          "here", "terra", "rnatrualearth", "sf", "countrycode",
                          "wesanderson", "benchmarkme", "parallel"))

# Create custom theme -----

theme_AP <- function() {
  theme_bw() +
    theme(panel.grid.major = element_blank(),
          panel.grid.minor = element_blank(),
          legend.background = element_rect(fill = "transparent", color = NA),
          legend.key = element_rect(fill = "transparent", color = NA),
          legend.key.width = unit(0.4, "cm"),
          legend.key.height = unit(0.5, "lines"),
          legend.key.spacing.y = unit(0, "lines"),
          legend.box.spacing = unit(0, "pt"),
          legend.spacing.y = unit(0.1, "cm"),
          legend.text = element_text(size = 6),
          legend.title = element_text(size = 7),
          axis.text.x = element_text(size = 7),
          axis.text.y = element_text(size = 7),
          axis.title.x = element_text(size = 7.3),
          axis.title.y = element_text(size = 7.3),
          axis.title = element_text(size = 10),
          plot.title = element_text(size = 8),
          strip.text.x = element_text(size = 7.4),
          strip.text.y = element_text(size = 7.4),
          strip.background = element_rect(fill = "white"),
          strip.text = element_text(margin = margin(t = 1.5, b = 1.5)))
}

# Source all .R files in the "functions" folder -----

r_functions <- list.files(path = here("functions"),
                          pattern = "\\R$", full.names = TRUE)

lapply(r_functions, source)

# Set seed -----

seed <- 123
```

```

# Load countries as sf (medium scale is a good balance) -----
countries_sf <- ne_countries(scale = "medium", returnclass = "sf")

# Keep only what is needed -----
countries_sf <- countries_sf[, c("name", "geometry")]

# Define colors and palettes and plotting datasets -----

tau_palette <- c("Agreement"= "#2ECC71",
  "Low disagreement" = "#F7DC6F",
  "Moderate disagreement" = "#F39C12",
  "High disagreement" = "#E74C3C",
  "Extreme disagreement" = "red")

res_palette <- c("0.2deg" = "lightblue",
  "0.4deg" = "#117A65",
  "1deg" = "darkblue")

```


2 Uncertainty and sensitivity analysis

```
# LOAD ALL DATASETS #####

# List all resolution files -----

files <- list.files("./datasets/irrigated_areas_regridded",
                    pattern = "irrigated_areas_regridded_.*\\.csv",
                    full.names = TRUE)

# Load them, bind them and label them -----

dt <- rbindlist(lapply(files, function(f) {
  x <- fread(f)
  res <- sub(".*_(\\d+)\\.csv", "\\1", f)
  x[, resolution:= paste0(as.numeric(res) / 10, "deg")]
  x
}))

# Load cropland mask -----

crop_path <- "./irrigated_area_datasets/GirrEO/asap_mask_crop_v02.tif"
crop_native <- rast(crop_path)

# Retrieve only cells within cropland -----

dt <- filter_long_dt_by_crop_mask_aggregated(dt = dt, crop_native = crop_native,
                                             rule = "any")

# DEFINE TAU GRID #####

# Vector with the datasets used -----

dataset_names <- unique(dt$dataset)
new_dataset_names <- c("GAEZ+2015", "GIAM", "GMIA", "GRIPC", "LUH2", "Meier",
                      "MIRCA 2000", "MIRCA OS", "Nagaraj", "SPAM2010")

# Define a dense grid of tau values from 1 ha (10^0) to 1e5 ha (10^5) -----

log_tau_min <- 0
log_tau_max <- 5
log_step <- 0.1

taus_ha_nonzero <- 10^(seq(log_tau_min, log_tau_max, by = log_step))
taus_ha <- c(0, taus_ha_nonzero)
taus_mha <- taus_ha * 1e-6
```

```

# Labels: "any_positive", then "1ha", "1.26ha", ..., "1e+05ha"---
tau_labels <- ifelse(taus_ha == 0, "any_positive", sprintf("%gha", signif(taus_ha, 3)))

# Define tau grid -----

tau_grid <- data.table(tau_label = tau_labels, tau_ha = taus_ha, tau_mha = taus_mha)

# TRANSFORM THE SAMPLE MATRIX #####

# Define the settings -----

N <- 2^13
params <- c("resolution", "tau", "exclusion")
matrices <- c("A", "B", "AB")
R <- 10^3
boot <- TRUE

# Resolution vector -----

resolution_vec <- c("0.2deg", "0.4deg", "1deg")
n_resolution_vec <- length(resolution_vec)

# Tau vector -----

tau_vec <- tau_grid$tau_ha
n_tau_vec <- length(tau_vec)

# Dataset exclusion vector -----

dataset_vec <- c("all", dataset_names)
n_dataset_vec <- length(dataset_vec)

# Create the sample matrix -----

mat <- data.table(sobol_matrices(matrices = matrices, N = N, params = params))

# Transform resolution column -----

mat[, resolution:= pmin(n_resolution_vec, 1L + floor(resolution * n_resolution_vec))]
mat[, resolution:= resolution_vec[resolution]]

# Transform tau column -----

mat[, tau:= pmin(n_tau_vec, 1L + floor(tau * n_tau_vec))]
mat[, tau:= tau_vec[tau]]

```

```

# Transform exclusion column -----

mat[, exclusion:= pmin(n_dataset_vec, 1L + floor(exclusion * n_dataset_vec))]
mat[, exclusion:= dataset_vec[exclusion]]

# PRE-COMPUTE ALL COMBINATIONS #####

# Compute all possibilities by looping over resolutions and tau values -----

# Set parallel backend -----

resolutions <- sort(unique(dt$resolution))
ncores <- max(1, parallel::detectCores() - 1)

# loop over resolutions and scenarios in parallel

tau_results <- rbindlist(
  parallel::mclapply(resolutions, function(res) {

    dt_res_long <- dt[resolution == res]

    # datasets actually present at this resolution
    ds_res <- sort(unique(dt_res_long$dataset))

    rbindlist(
      lapply(dataset_vec, function(scen) {

        # which datasets to use -----

        if (scen == "all") {

          use_ds <- ds_res

        } else {

          use_ds <- setdiff(ds_res, scen)
        }

        # if only one dataset left, skip -----
        if (length(use_ds) < 2L) {

          return(NULL)
        }

        # reshape to wide -----

        dt_res_wide <- dcast(dt_res_long[dataset %in% use_ds],

```

```

    lon + lat + country + code + continent ~ dataset,
    value.var = "mha", fill = 0
  )

dt_res_wide[, `:=`(A_min = do.call(pmin, c(.SD, list(na.rm = TRUE))),
                  A_max = do.call(pmax, c(.SD, list(na.rm = TRUE)))),
            .SDcols = use_ds]

# compute tau -----

out <- tau_grid[, compute_tau_fun(dt_res_wide, tau_mha, use_ds),
               .(tau_label, tau_ha, tau_mha)]

# tag with resolution + scenario -----

out[, `:=`(resolution = res, scenario = scn)]

  out
},
use.names = TRUE, fill = TRUE
)
}, mc.cores = ncores),
use.names = TRUE, fill = TRUE)

# ARRANGE OUTPUT #####

# Rename tau_ha to tau to match mat -----

setnames(tau_results, "tau_ha", "tau")

# Join all three disagreement metrics onto mat -----

mat[tau_results, on = .(resolution, tau, exclusion = scenario),
    `:=`(frac_disagree = i.frac_disagree,
          frac_major_disagree = i.frac_major_disagree,
          frac_minor_disagree = i.frac_minor_disagree,
          share_existence = i.share_existence,
          share_marginal = i.share_marginal)]

# Reshape just these three for plotting -----

mat_long <- melt(mat, id.vars = c("resolution", "tau", "exclusion"),
                measure.vars = c("frac_disagree", "frac_major_disagree",
                                "frac_minor_disagree", "share_existence",
                                "share_marginal"),
                variable.name = "agreement", value.name = "frac_value")

```

2.1 Composite figure

```
# PREPARE DATA TO PLOT UNCERTAINTY #####

# Calculate mean and quantiles by tau x agreement -----

mat_sum <- mat_long[, .(frac_mean = mean(frac_value, na.rm = TRUE),
    frac_median = median(frac_value, na.rm = TRUE),
    frac_lwr = quantile(frac_value, 0.025, na.rm = TRUE),
    frac_upr = quantile(frac_value, 0.975, na.rm = TRUE)),
    .(tau, agreement)]

# Get values at tau = 0, 1, 10, 500, 1000, 1600, 2000, 5000, 10000, 12000 -----

target_tau <- c(0, 1, 10, 500, 1000, 1600, 2000, 5000, 10000, 12000)

mat_sum[, {dt_sub <- .SD
idx <- sapply(target_tau, function(x) {
  which.min(abs(log10(tau + 1e-12) - log10(x + 1e-12)))
})
dt_sub[idx]
},
agreement
] %>%
[, .(agreement, tau, frac_mean, frac_lwr, frac_upr)]
```

##	agreement	tau	frac_mean	frac_lwr	frac_upr
##	<fctr>	<num>	<num>	<num>	<num>
## 1:	frac_disagree	0.0000	0.7725147940	0.5965542885	0.9048508038
## 2:	frac_disagree	1.0000	0.7641095924	0.5935795736	0.8913352054
## 3:	frac_disagree	10.0000	0.7458169499	0.5872583044	0.8574304934
## 4:	frac_disagree	501.1872	0.5696037035	0.4990357022	0.6046632124
## 5:	frac_disagree	1000.0000	0.5101819636	0.4207578610	0.5563956371
## 6:	frac_disagree	1584.8932	0.4773297398	0.3893680378	0.5323500248
## 7:	frac_disagree	1995.2623	0.4499272243	0.3558259018	0.5173525037
## 8:	frac_disagree	5011.8723	0.3573094252	0.2308683309	0.4640555280
## 9:	frac_disagree	10000.0000	0.2758302271	0.1276267473	0.4086514626
## 10:	frac_disagree	12589.2541	0.2498516315	0.1079319283	0.3922905305
## 11:	frac_major_disagree	0.0000	0.5459749341	0.3824987605	0.6639383467
## 12:	frac_major_disagree	1.0000	0.5345828870	0.3787803669	0.6428472244
## 13:	frac_major_disagree	10.0000	0.5066465761	0.3694843827	0.5942080415
## 14:	frac_major_disagree	501.1872	0.3422024147	0.2610007097	0.3931581557
## 15:	frac_major_disagree	1000.0000	0.2967305803	0.2158021415	0.3670054536
## 16:	frac_major_disagree	1584.8932	0.2702110404	0.1959761780	0.3446950917
## 17:	frac_major_disagree	1995.2623	0.2537648037	0.1781096677	0.3362667328
## 18:	frac_major_disagree	5011.8723	0.1922679985	0.1100610979	0.2802429351
## 19:	frac_major_disagree	10000.0000	0.1453529051	0.0641605209	0.2356222112

```

## 20: frac_major_disagree 12589.2541 0.1288693376 0.0554201253 0.2146752603
## 21: frac_minor_disagree      0.0000 0.2265398599 0.1985622211 0.2863424569
## 22: frac_minor_disagree      1.0000 0.2295267055 0.1991819534 0.2922131021
## 23: frac_minor_disagree     10.0000 0.2391703737 0.2035200793 0.3034221002
## 24: frac_minor_disagree    501.1872 0.2274012888 0.1963311849 0.2613710001
## 25: frac_minor_disagree   1000.0000 0.2134513833 0.1893901834 0.2318943438
## 26: frac_minor_disagree   1584.8932 0.2071186994 0.1860436292 0.2213102108
## 27: frac_minor_disagree   1995.2623 0.1961624206 0.1772688000 0.2109764073
## 28: frac_minor_disagree   5011.8723 0.1650414267 0.1208072330 0.1941001487
## 29: frac_minor_disagree  10000.0000 0.1304773220 0.0634662264 0.1818294497
## 30: frac_minor_disagree  12589.2541 0.1209822939 0.0525118030 0.1823252355
## 31:      share_existence      0.0000 1.0000000000 1.0000000000 1.0000000000
## 32:      share_existence      1.0000 0.9998842092 0.9993735644 1.0000000000
## 33:      share_existence     10.0000 0.9987395427 0.9958634953 0.9994211288
## 34:      share_existence    501.1872 0.9265090021 0.8442199776 0.9606578698
## 35:      share_existence   1000.0000 0.8878074738 0.7835751053 0.9334183673
## 36:      share_existence   1584.8932 0.8549958612 0.7312409812 0.9107164472
## 37:      share_existence   1995.2623 0.8301866190 0.6886404631 0.8912397072
## 38:      share_existence   5011.8723 0.7100854672 0.4888273553 0.7785060526
## 39:      share_existence  10000.0000 0.5726446689 0.2750825646 0.6224474129
## 40:      share_existence  12589.2541 0.5234532434 0.2116215677 0.5748619682
## 41:      share_marginal      0.0000 0.0000000000 0.0000000000 0.0000000000
## 42:      share_marginal      1.0000 0.0001157908 0.0000000000 0.0006264356
## 43:      share_marginal     10.0000 0.0012604573 0.0005788712 0.0041365047
## 44:      share_marginal    501.1872 0.0734909979 0.0393421302 0.1557800224
## 45:      share_marginal   1000.0000 0.1121925262 0.0665816327 0.2164248947
## 46:      share_marginal   1584.8932 0.1450041388 0.0892835528 0.2687590188
## 47:      share_marginal   1995.2623 0.1698133810 0.1087602928 0.3113595369
## 48:      share_marginal   5011.8723 0.2899145328 0.2214939474 0.5111726447
## 49:      share_marginal  10000.0000 0.4273553311 0.3775525871 0.7249174354
## 50:      share_marginal  12589.2541 0.4765467566 0.4251380318 0.7883784323
##
##          agreement          tau      frac_mean      frac_lwr      frac_upr
##          <fctr>          <num>          <num>          <num>          <num>

```

```

# Plot uncertainty -----

agreement_cols <- c(frac_disagree = "black",
                    frac_major_disagree = "#B2182B",
                    frac_minor_disagree = "#2166AC" )

plot_uncertainty1 <- mat_sum[!agreement %in% c("share_existence", "share_marginal")] %>%
  ggplot(.,
    aes(x = tau, y = frac_median, colour = agreement, fill = agreement)) +
  geom_ribbon(aes(ymin = frac_lwr, ymax = frac_upr), alpha = 0.18, colour = NA) +
  geom_line(linewidth = 1.1) +
  scale_x_log10(labels = label_log(digits = 2)) +
  scale_colour_manual(values = agreement_cols,
                      breaks = c("frac_disagree",

```

```

        "frac_major_disagree",
        "frac_minor_disagree"),
    labels = c("Total",
              "Major",
              "Minor"),
    name = "Disagreement") +
scale_fill_manual(values = agreement_cols,
                  breaks = c("frac_disagree",
                            "frac_major_disagree",
                            "frac_minor_disagree"),
                  labels = c("Total",
                            "Major",
                            "Minor"),
                  name = "Disagreement") +
labs(x = expression(tau~"(\text{ha})"), y = "Fraction of cells") +
theme_AP() +
theme(legend.position = c(0.65, 0.77))

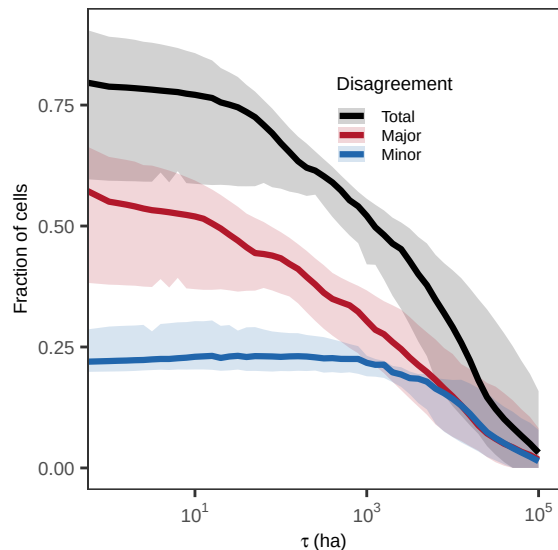
```

plot_uncertainty1

```

## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation introduced :
## log-10 transformation introduced infinite values.

```



```

# Set colors for existential and marginal disagreement -----
exist_cols <- c("share_existence" = "#1f78b4", "share_marginal" = "#e66101")

plot_agreement1 <- mat_sum[agreement %in% c("share_existence", "share_marginal")] %>%
  ggplot(., aes(x = tau, y = frac_median, colour = agreement, fill = agreement)) +
  geom_ribbon(aes(ymin = frac_lwr, ymax = frac_upr), alpha = 0.18, colour = NA) +
  geom_line(linewidth = 1.1) +
  scale_colour_manual(values = exist_cols,

```

```

      labels = c(share_existence = "Existential",
                 share_marginal = "Marginal"),
      name = "Disagreement") +
scale_fill_manual(values = exist_cols,
                  labels = c(share_existence = "Existential",
                             share_marginal = "Marginal"),
                  name = "Disagreement") +
scale_x_log10(labels = label_log(digits = 2)) +
labs(x = expression(tau~"(ha)"), y = "Fraction of \ndisagreeing cells") +
theme_AP() +
theme(legend.position = c(0.37, 0.7))

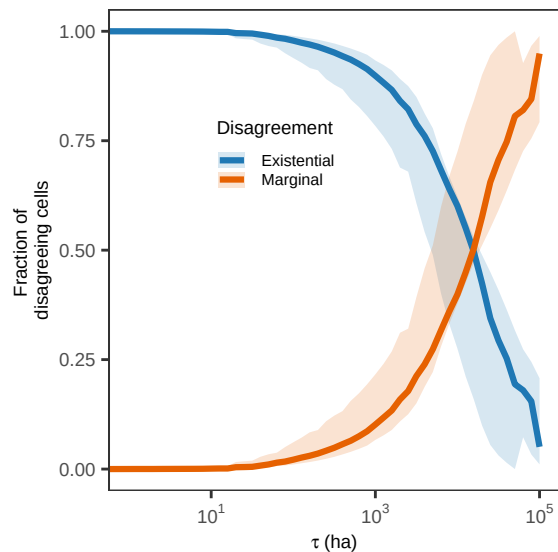
```

plot_agreement1

```

## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation introduced :
## log-10 transformation introduced infinite values.

```



```

# Around tau > 10^4 ha, the existential and marginal components intersect. That
# intersection has a clear conceptual meaning:
# -It is the tolerance level at which disagreement about whether irrigation exists
# at all becomes less dominant than disagreement about how much irrigation exists.
# -Still small relative to grid-cell area. 1000 ha is only 2-3% of a 0.2° grid cell.
# That makes it a conservative tolerance as we are not allowing massive within-cell
# divergence.

```

SENSITIVITY ANALYSIS

Sobol' indices -----

```

ind <- sobol_indices(matrices = matrices, N = N, params = params, Y
                     = mat[, frac_disagree], R = R, boot = TRUE)

```



```

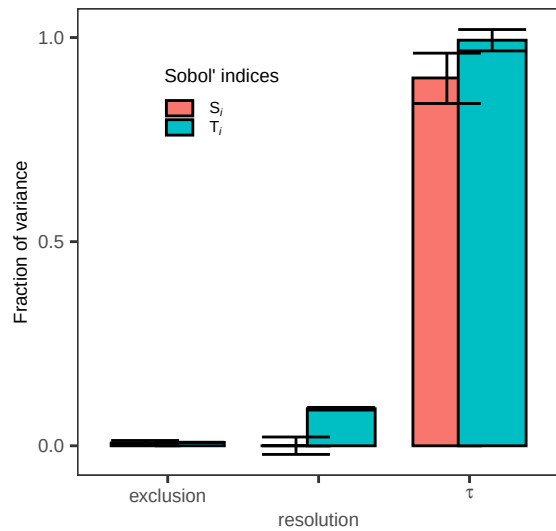
print(ind)

##
## First-order estimator: saltelli | Total-order estimator: jansen
##
## Total number of model runs: 40960
##
## Sum of first order indices: 0.9079082
##      original      bias    std.error    low.ci    high.ci
##      <num>      <num>    <num>      <num>      <num>
## 1: 0.0001506532 -3.275755e-05 0.0108794512 -0.0211399217 0.021506743
## 2: 0.9010276594 9.404230e-04 0.0314407547 0.8384644897 0.961709983
## 3: 0.0067298400 -2.710929e-06 0.0031939823 0.0004724605 0.012992641
## 4: 0.0907715303 2.194818e-06 0.0016776382 0.0874812249 0.094057446
## 5: 0.9935216113 6.024186e-05 0.0133459914 0.9673037070 1.019619032
## 6: 0.0083001208 8.542363e-06 0.0001909142 0.0079173935 0.008665763
##      sensitivity parameters
##      <char>      <char>
## 1:      Si resolution
## 2:      Si      tau
## 3:      Si exclusion
## 4:      Ti resolution
## 5:      Ti      tau
## 6:      Ti exclusion

plot_indices1 <- plot(ind) +
  labs(x = "", y = "Fraction of variance") +
  theme_AP() +
  scale_x_discrete(labels = c(exclusion = "exclusion",
                              resolution = "resolution",
                              tau = expression(tau)),
    guide = guide_axis(n.dodge = 2)) +
  theme(legend.position = c(0.3, 0.8))

plot_indices1

```

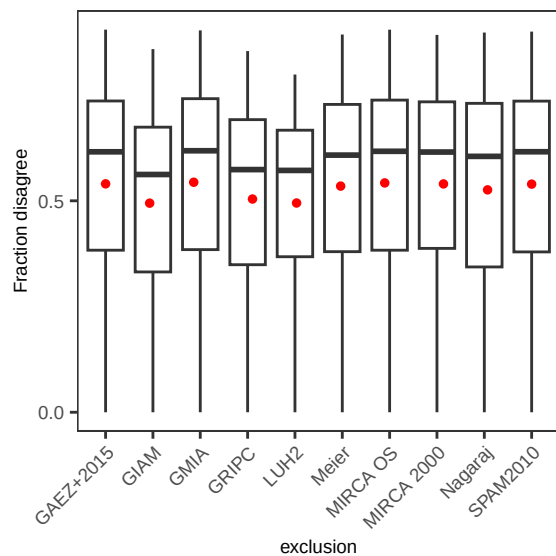


SCATTERPLOTS OF OUTPUT AGAINST INPUT

```
label_map <- setNames(new_dataset_names, dataset_names)

a1 <- ggplot(mat[!exclusion == "all", c("exclusion", "frac_disagree")],
  aes(exclusion, frac_disagree)) +
  geom_boxplot() +
  stat_summary_bin(fun = "mean", geom = "point",
    colour = "red", size = 1) +
  scale_x_discrete(labels = label_map) +
  scale_y_continuous(breaks = breaks_pretty(n = 3)) +
  theme_AP() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(x = "exclusion", y = "Fraction disagree")
```

a1

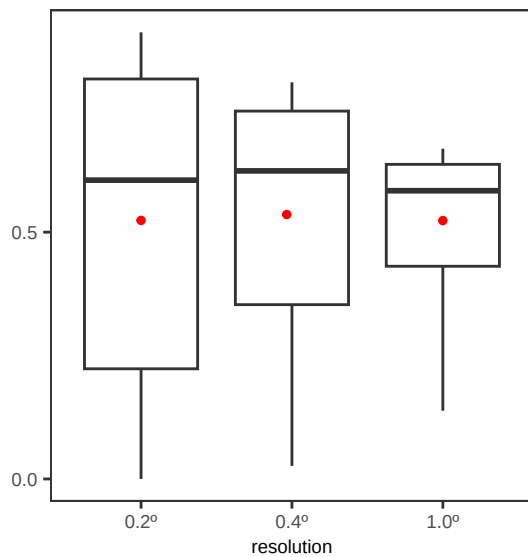


```

b1 <- ggplot(mat[1:N, c("resolution", "frac_disagree")],
  aes(resolution, frac_disagree)) +
  geom_boxplot() +
  scale_x_discrete(labels = c(`0.2deg` = "0.2°",
    `0.4deg` = "0.4°",
    `1deg` = "1.0°")) +
  scale_y_continuous(breaks = breaks_pretty(n = 3)) +
  stat_summary_bin(fun = "mean", geom = "point",
    colour = "red", size = 1) +
  theme_AP() +
  labs(x = "resolution", y = "")

```

b1



```

c1 <- ggplot(mat[1:N, c("tau", "frac_disagree")],
  aes(tau, frac_disagree)) +
  geom_point(size = 0.1, alpha = 0.1) +
  stat_summary_bin(fun = "mean", geom = "point",
    colour = "red", size = 1) +
  scale_y_continuous(breaks = breaks_pretty(n = 3)) +
  theme_AP() +
  scale_x_log10(labels = label_log(digits = 2)) +
  labs(x = expression(tau~"(\u0027ha\u0027)"), y = "")

```

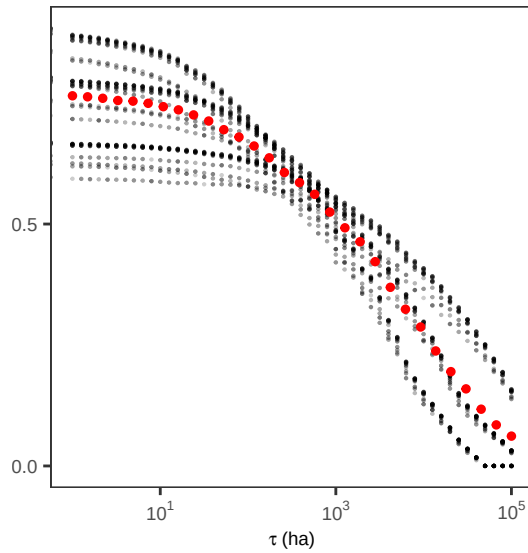
c1

```

## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation introduced :
## log-10 transformation introduced infinite values.

## Warning: Removed 157 rows containing non-finite outside the scale range
## (`stat_summary_bin()`).

```



```
# PLOT DATASETS #####
```

```
tmp <- dt[resolution == "0.2deg"] %>%
  .[, .(mha = sum(mha)), dataset] %>%
  .[, continent := "GLOBAL"]

dt[resolution == "0.2deg"] %>%
  .[, .(mha = sum(mha)), .(continent, dataset)] %>%
  rbind(., tmp)
```

##	continent	dataset	mha
##	<char>	<char>	<num>
## 1:	Americas	gaez_v4	46.2015416
## 2:	Europe	gaez_v4	23.8489672
## 3:	Africa	gaez_v4	13.1101914
## 4:	Asia	gaez_v4	194.2821806
## 5:	Oceania	gaez_v4	2.8450580
## 6:	Americas	giam	45.5089223
## 7:	Europe	giam	38.6557015
## 8:	Africa	giam	6.9337475
## 9:	Asia	giam	233.9728781
## 10:	Oceania	giam	10.1185604
## 11:	Americas	gmia	40.2113502
## 12:	Europe	gmia	11.9504917
## 13:	Africa	gmia	10.7517031
## 14:	Asia	gmia	180.0792101
## 15:	Oceania	gmia	2.8368249
## 16:	Americas	gripc	51.2475536
## 17:	Europe	gripc	23.0350045
## 18:	Africa	gripc	12.2038719
## 19:	Asia	gripc	212.3461235
## 20:	Oceania	gripc	7.1657891

```

## 21: Americas      luh2  40.0523677
## 22:   Europe      luh2  13.7826118
## 23:   Africa      luh2  10.8576084
## 24:    Asia      luh2 195.5457561
## 25:  Oceania      luh2   2.1157916
## 26: Americas     meier  50.4449965
## 27:   Europe     meier  26.0018526
## 28:   Africa     meier  14.8522635
## 29:    Asia     meier 259.4777712
## 30:  Oceania     meier   3.8551223
## 31: Americas  mirca2000  34.4909117
## 32:   Europe  mirca2000  15.3232002
## 33:   Africa  mirca2000   9.4163731
## 34:    Asia  mirca2000 148.8331835
## 35:  Oceania  mirca2000   2.4615961
## 36: Americas  mirca_os  40.1775798
## 37:   Europe  mirca_os  10.9262841
## 38:   Africa  mirca_os  12.9762031
## 39:    Asia  mirca_os 199.6637856
## 40:  Oceania  mirca_os   1.7670877
## 41: Americas  nagaraj  53.2348690
## 42:   Europe  nagaraj  32.0035173
## 43:   Africa  nagaraj   9.0617581
## 44:    Asia  nagaraj 188.5240888
## 45:  Oceania  nagaraj   2.9588799
## 46: Americas    spam  28.3824799
## 47:   Europe    spam  10.9216630
## 48:   Africa    spam  11.8448877
## 49:    Asia    spam 167.5769329
## 50:  Oceania    spam   0.8283964
## 51:  GLOBAL  gaez_v4 280.2879388
## 52:  GLOBAL    giam 335.1898098
## 53:  GLOBAL    gmia 245.8295799
## 54:  GLOBAL  gripc 305.9983426
## 55:  GLOBAL    luh2 262.3541356
## 56:  GLOBAL    meier 354.6320061
## 57:  GLOBAL  mirca2000 210.5252647
## 58:  GLOBAL  mirca_os 265.5109403
## 59:  GLOBAL  nagaraj 285.7831131
## 60:  GLOBAL    spam 219.5543599
##      continent  dataset      mha
##      <char>    <char>      <num>

```

```

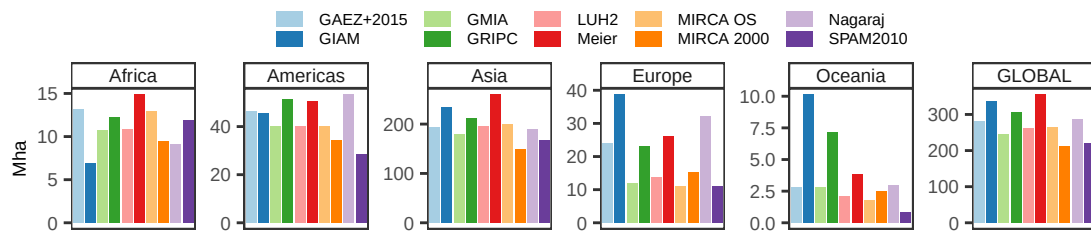
dt[resolution == "0.2deg"] %>%
  .[, .(mha = sum(mha)), .(continent, dataset)] %>%
  rbind(., tmp) %>%
  .[, continent := factor(continent, levels = c("Africa", "Americas",
                                                "Asia", "Europe", "Oceania",

```

```

"GLOBAL"))] %>%
ggplot(., aes(dataset, mha, fill = dataset)) +
geom_bar(stat = "identity") +
scale_fill_brewer(palette = "Paired", label = label_map, name = "") +
facet_wrap(~continent, ncol = 6, scales = "free_y") +
theme_AP() +
labs(x = "", y = "Mha") +
theme(legend.position = "top") +
theme(axis.text.x = element_blank(),
axis.ticks.x = element_blank())

```



```

# MERGE UA/SA PLOTS #####

```

```

top <- plot_grid(plot_uncertainty1, plot_agreement1, plot_indices1,
ncol = 3, labels = "auto", rel_widths = c(0.4, 0.3, 0.3))

```

```

## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation introduced :
## log-10 transformation introduced infinite values.
## log-10 transformation introduced infinite values.
## log-10 transformation introduced infinite values.

```

```

bottom <- plot_grid(a1, b1, c1, ncol = 3, rel_widths = c(0.5, 0.25, 0.25), labels = "d")

```

```

## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation introduced :
## log-10 transformation introduced infinite values.

```

```

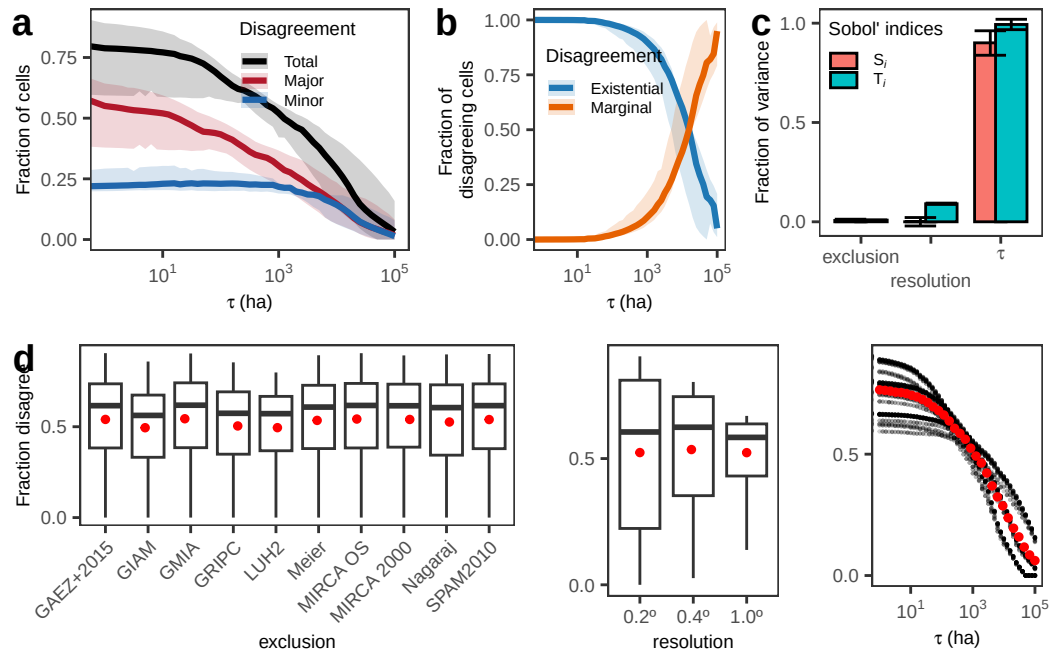
## Warning: Removed 157 rows containing non-finite outside the scale range
## (`stat_summary_bin()`).

```

```

plot_grid(top, bottom, ncol = 1)

```



2.2 Does coarsening reduce disagreement?

```
# DOES ZOOMING OUT REDUCE DISAGREEMENT? #####

effect_resolution_dt <- tau_results[scenario == "all", .(tau, frac_disagree, resolution)]

# Order resolution -----

effect_resolution_dt[, resolution:= factor(resolution,
                                           levels = c("0.2deg", "0.4deg", "1deg"))]

# To wide table -----

wide_dt <- dcast(effect_resolution_dt, tau ~ resolution, value.var = "frac_disagree")

# Rename -----

setnames(wide_dt, old = c("0.2deg", "0.4deg", "1deg"), new = c("res_0.2", "res_0.4", "res_1"))

# Compute absolute change (fine - coarse) -----

wide_dt[, abs_change:= res_0.2 - res_1]

# Compute elasticity (log-log) -----

# Area ratio from 0.2 to 1
area_ratio <- 25
wide_dt[, elasticity:= ifelse(res_0.2 > 0 & res_1 > 0,
```

```

log(res_1 / res_0.2) / log(area_ratio), NA_real_)]

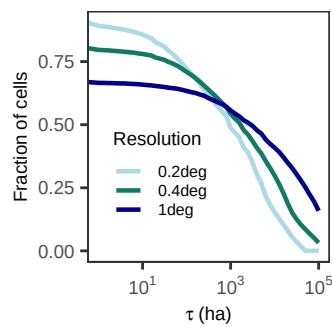
# Plot disagreement curves -----

p1 <- ggplot(effect_resolution_dt, aes(tau, frac_disagree, colour = resolution)) +
  geom_line(linewidth = 0.8) +
  scale_x_log10(labels = label_log(digits = 2)) +
  scale_color_manual(values = res_palette) +
  labs(x = expression(tau~"(ha)"),
       y = "Fraction of cells",
       colour = "Resolution") +
  theme_AP() +
  theme(legend.position = c(0.3, 0.35))

p1

```

Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation
introduced infinite values.



```

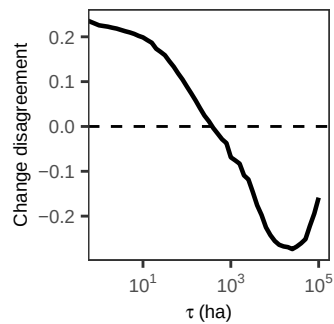
# Plot absolute change -----

p2 <- ggplot(wide_dt, aes(tau, abs_change)) +
  geom_hline(yintercept = 0, linetype = "dashed") +
  geom_line(linewidth = 0.8) +
  scale_x_log10(labels = label_log(digits = 2)) +
  labs(x = expression(tau~"(ha)"),
       y = "Change disagreement") +
  theme_AP()

p2

```

Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation
introduced infinite values.

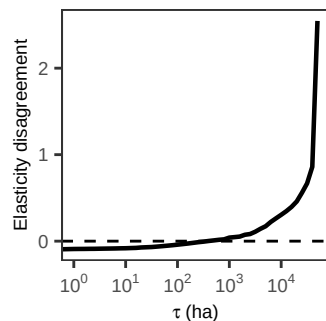


```
# Plot elasticity -----
```

```
p3 <- ggplot(wide_dt[!is.na(elasticity)],
  aes(x = tau, y = elasticity)) +
  geom_hline(yintercept = 0, linetype = "dashed") +
  geom_line(linewidth = 0.8) +
  scale_x_log10(labels = label_log(digits = 2)) +
  labs(x = expression(tau~"(ha)"),
    y = "Elasticity disagreement") +
  theme_AP()
```

p3

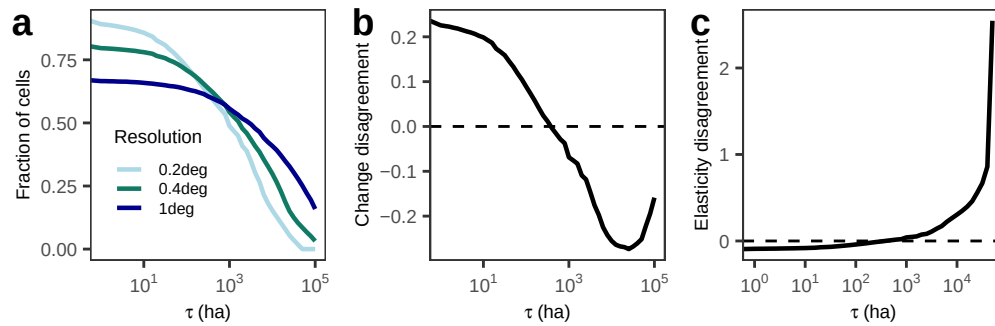
```
## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation
## introduced infinite values.
```



```
# Merge -----
```

```
plot_grid(p1, p2, p3, ncol = 3, labels = "auto")
```

```
## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation introduced :
## log-10 transformation introduced infinite values.
## log-10 transformation introduced infinite values.
```



Increasing spatial aggregation from 0.2 to 1 (25x coarsening) reduces
existential disagreement by c20 percentage points at near-zero
thresholds. this effect vanishes around $\tau = 300-500$ ha, beyond which
aggregation increases disagreement. At higher thresholds disagreement is
greater at 1 than at 0.2. Coarse-scale aggregation can
amplify rather than resolve structural spatial inconsistencies between datasets.

These results refute the criticism of "if you zoom out, the maps converge".

ALSO: Increasing cell area by a factor of 25 (0.2 to 1) yields an elasticity
of approximately -0.07 at $\tau = 0$, implying that a 1% increase in spatial
aggregation reduces disagreement by only 0.07%. Elasticity approaches zero
at moderate thresholds and becomes positive at higher thresholds.

2.3 UA/SA by excluding mapping paradigms

##	agreement	tau	frac_mean	frac_median	frac_lwr
##	<fctr>	<num>	<num>	<num>	<num>
## 1:	frac_disagree	0.0000	0.7430530248	0.7755168771	0.5669310858
## 2:	frac_disagree	1.0000	0.7333800619	0.7621964390	0.5613534953
## 3:	frac_disagree	10.0000	0.7137910743	0.7146835560	0.5469757065
## 4:	frac_disagree	501.1872	0.5389292065	0.5637516586	0.4301616935
## 5:	frac_disagree	1000.0000	0.4806388477	0.5117712159	0.3618199772
## 6:	frac_disagree	1584.8932	0.4489805921	0.4738165904	0.3457740612
## 7:	frac_disagree	5011.8723	0.3349236070	0.3436038030	0.2032739840
## 8:	frac_disagree	10000.0000	0.2572035781	0.2772682201	0.1073456352
## 9:	frac_major_disagree	0.0000	0.5031937420	0.4994215001	0.3299454636
## 10:	frac_major_disagree	1.0000	0.4896659751	0.4891594145	0.3238720873
## 11:	frac_major_disagree	10.0000	0.4602356204	0.4695407214	0.3087506197
## 12:	frac_major_disagree	501.1872	0.3058621258	0.3168921978	0.2007899528
## 13:	frac_major_disagree	1000.0000	0.2646720627	0.2671663565	0.1717144444
## 14:	frac_major_disagree	1584.8932	0.2411082172	0.2412093164	0.1586694233
## 15:	frac_major_disagree	5011.8723	0.1711008105	0.1681171085	0.0920017280
## 16:	frac_major_disagree	10000.0000	0.1291880616	0.1322249610	0.0532755269
## 17:	frac_minor_disagree	0.0000	0.2398592828	0.2369856222	0.2027764006
## 18:	frac_minor_disagree	1.0000	0.2437140868	0.2374814080	0.2047595439
## 19:	frac_minor_disagree	10.0000	0.2535554539	0.2493837718	0.2100892414
## 20:	frac_minor_disagree	501.1872	0.2330670807	0.2306643530	0.2063708478
## 21:	frac_minor_disagree	1000.0000	0.2159667850	0.2171889934	0.1901055328
## 22:	frac_minor_disagree	1584.8932	0.2078723749	0.2062469013	0.1787816289
## 23:	frac_minor_disagree	5011.8723	0.1638227966	0.1626339353	0.1112722560
## 24:	frac_minor_disagree	10000.0000	0.1280155164	0.1342340109	0.0540701083
## 25:	share_existence	0.0000	1.0000000000	1.0000000000	1.0000000000
## 26:	share_existence	1.0000	0.9997672523	0.9998374407	0.9993735644
## 27:	share_existence	10.0000	0.9982965766	0.9988512349	0.9958634953
## 28:	share_existence	501.1872	0.9168285618	0.9332604417	0.8442199776
## 29:	share_existence	1000.0000	0.8703566888	0.8809219223	0.7835751053
## 30:	share_existence	1584.8932	0.8326718403	0.8422081094	0.7312409812
## 31:	share_existence	5011.8723	0.6665122263	0.6842837274	0.4888273553
## 32:	share_existence	10000.0000	0.5232850394	0.5448511469	0.2750825646
## 33:	share_marginal	0.0000	0.0000000000	0.0000000000	0.0000000000
## 34:	share_marginal	1.0000	0.0002327477	0.0001625593	0.0000000000
## 35:	share_marginal	10.0000	0.0017034234	0.0011487651	0.0004279575
## 36:	share_marginal	501.1872	0.0831714382	0.0667395583	0.0382878568
## 37:	share_marginal	1000.0000	0.1296433112	0.1190780777	0.0644334054
## 38:	share_marginal	1584.8932	0.1673281597	0.1577918906	0.0856892626
## 39:	share_marginal	5011.8723	0.3334877737	0.3157162726	0.2163691902
## 40:	share_marginal	10000.0000	0.4767149606	0.4551488531	0.3759911779
##	agreement	tau	frac_mean	frac_median	frac_lwr
##	<fctr>	<num>	<num>	<num>	<num>
##	frac_upr				
##	<num>				

```

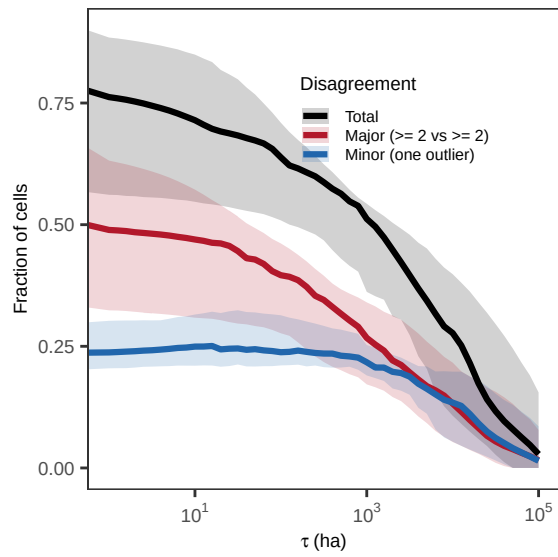
## 1: 0.8997747400
## 2: 0.8846468356
## 3: 0.8495463943
## 4: 0.5901705317
## 5: 0.5430094199
## 6: 0.5178482895
## 7: 0.4510411502
## 8: 0.3983639068
## 9: 0.6584379918
## 10: 0.6317153701
## 11: 0.5720677631
## 12: 0.3672533466
## 13: 0.3403569658
## 14: 0.3210213188
## 15: 0.2595438770
## 16: 0.2165344571
## 17: 0.2993566205
## 18: 0.3024038016
## 19: 0.3153176783
## 20: 0.2842595735
## 21: 0.2525534607
## 22: 0.2416298948
## 23: 0.2086018840
## 24: 0.1979424888
## 25: 1.0000000000
## 26: 1.0000000000
## 27: 0.9995720425
## 28: 0.9617121432
## 29: 0.9355665946
## 30: 0.9143107374
## 31: 0.7836308098
## 32: 0.6240088221
## 33: 0.0000000000
## 34: 0.0006264356
## 35: 0.0041365047
## 36: 0.1557800224
## 37: 0.2164248947
## 38: 0.2687590188
## 39: 0.5111726447
## 40: 0.7249174354
##          frac_upr
##          <num>

```

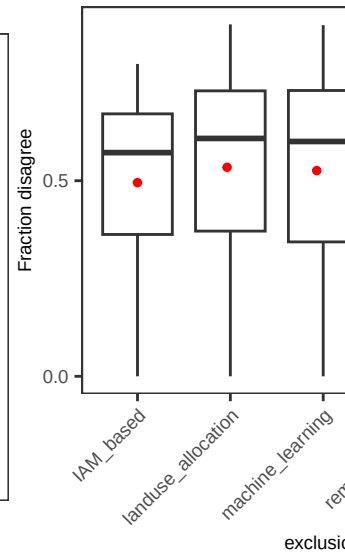
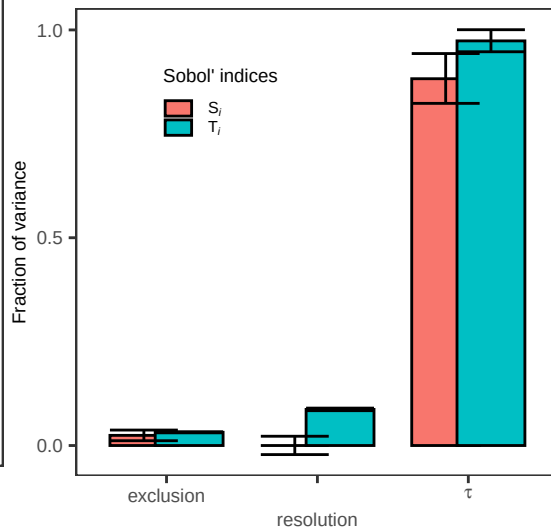
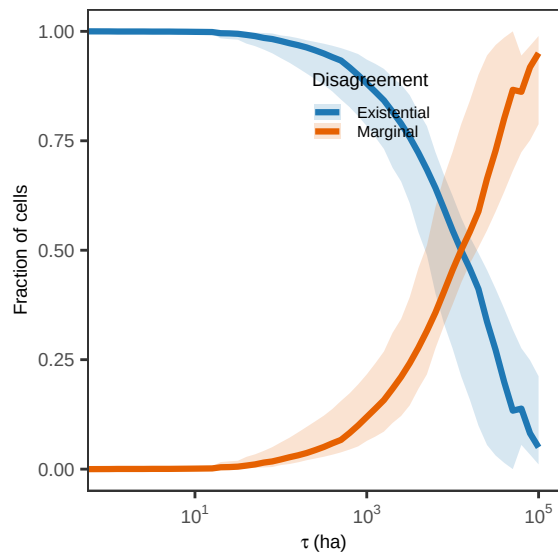
```

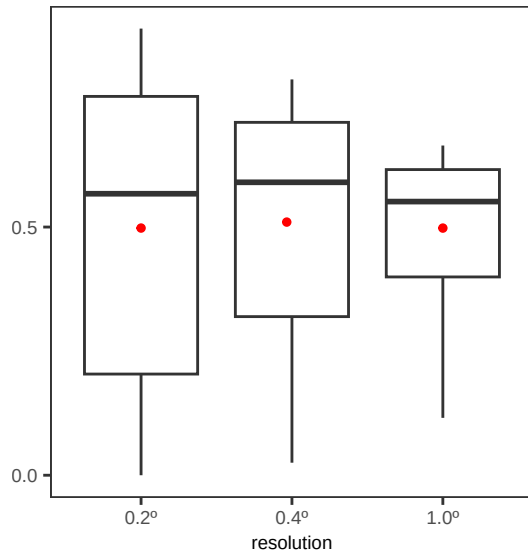
## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation introduced :
## log-10 transformation introduced infinite values.

```



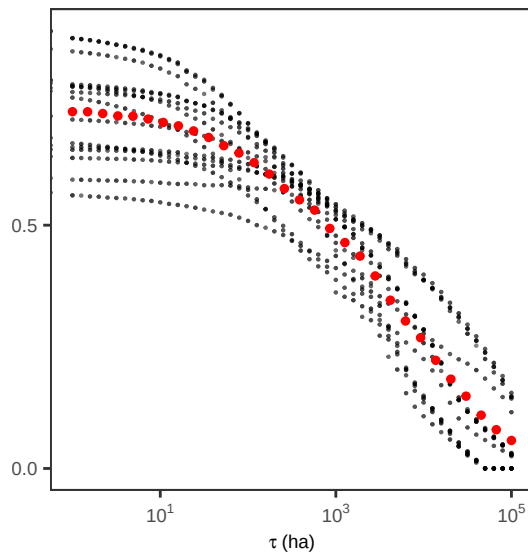
```
## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation introduced :
## log-10 transformation introduced infinite values.
```





```
## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation introduced :
## log-10 transformation introduced infinite values.

## Warning: Removed 157 rows containing non-finite outside the scale range
## (`stat_summary_bin()`).
```



```
# MERGE UA/SA PLOTS #####
```

```
top <- plot_grid(plot_uncertainty, plot_agreement, plot_indices2,
  ncol = 3, labels = "auto", rel_widths = c(0.4, 0.3, 0.3))
```

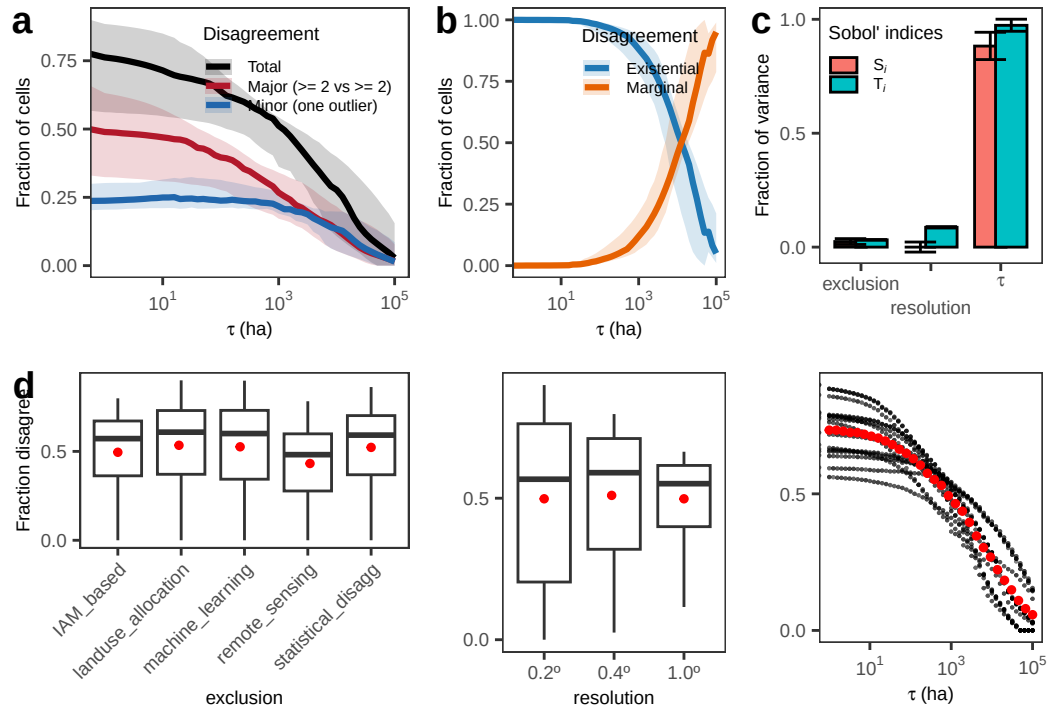
```
## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation introduced :
## log-10 transformation introduced infinite values.
## log-10 transformation introduced infinite values.
## log-10 transformation introduced infinite values.
```

```
bottom <- plot_grid(a2, b2, c2, ncol = 3, rel_widths = c(0.4, 0.3, 0.3), labels = "d")
```

```
## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation introduced :
## log-10 transformation introduced infinite values.
```

```
## Warning: Removed 157 rows containing non-finite outside the scale range
## (`stat_summary_bin()`).
```

```
plot_grid(top, bottom, ncol = 1)
```

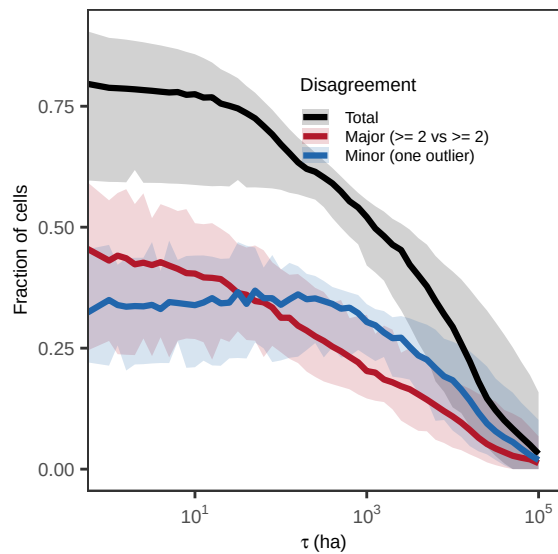


2.4 UA/SA by weighting newer datasets more

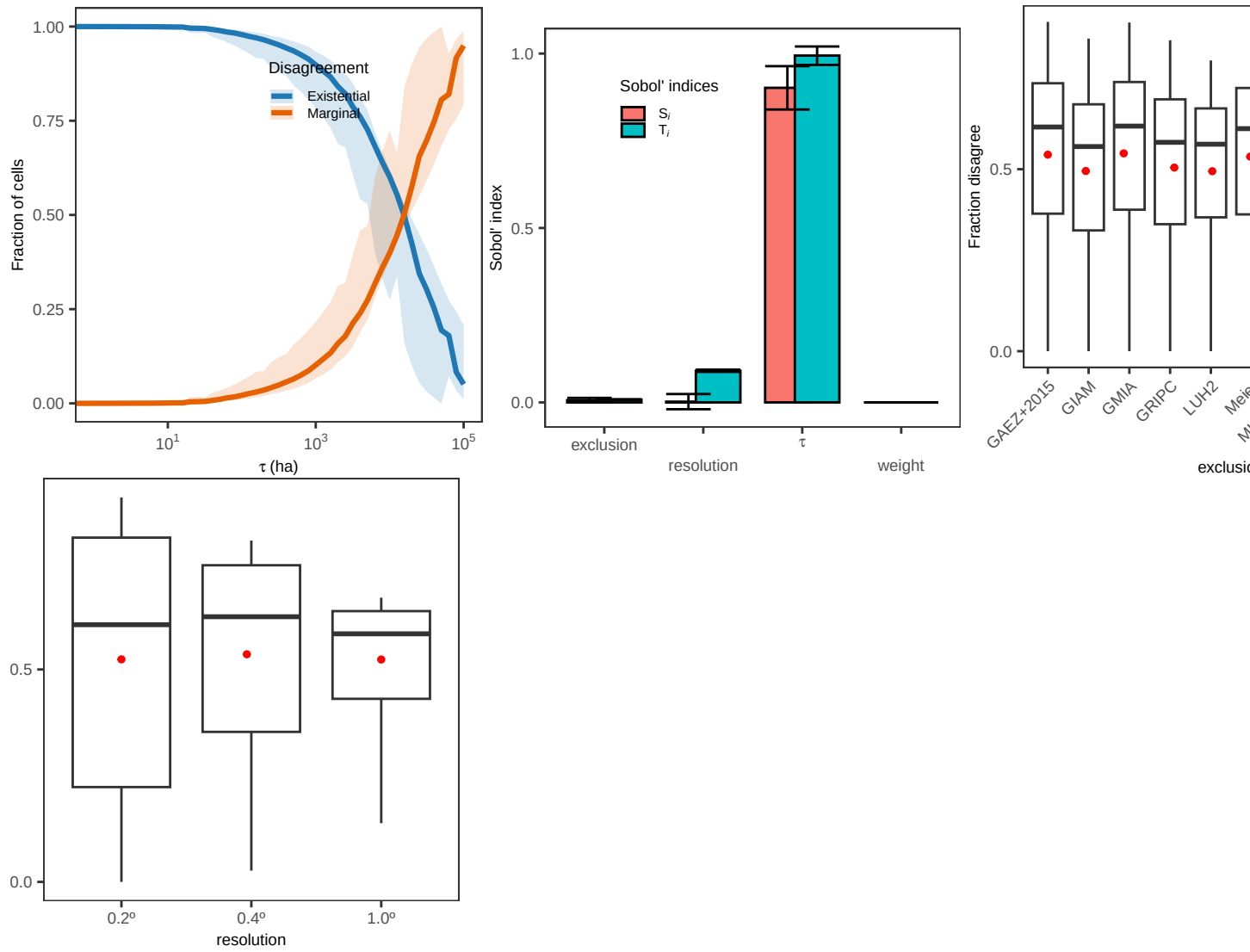
##	tau	agreement	frac_mean	frac_median	frac_lwr
##	<num>	<fctr>	<num>	<num>	<num>
## 1:	10	frac_disagree	0.7447800994	7.748378e-01	0.5872583044
## 2:	100	frac_disagree	0.6666677100	6.712284e-01	0.5772186415
## 3:	1	frac_disagree	0.7650881937	7.880427e-01	0.5935795736
## 4:	0	frac_disagree	0.7716544945	7.962423e-01	0.5965542885
## 5:	1000	frac_disagree	0.5103718881	5.209518e-01	0.4207578610
## 6:	10	frac_major_disagree	0.3947745529	4.040948e-01	0.2688398612
## 7:	100	frac_major_disagree	0.3177585471	3.128409e-01	0.2415218934
## 8:	1	frac_major_disagree	0.4230902508	4.310257e-01	0.2649138572
## 9:	0	frac_major_disagree	0.4405374142	4.551409e-01	0.2451660882
## 10:	1000	frac_major_disagree	0.2096229915	2.026456e-01	0.1460315287
## 11:	10	frac_minor_disagree	0.3500055465	3.388696e-01	0.2105850273
## 12:	100	frac_minor_disagree	0.3489091629	3.395040e-01	0.2245909767
## 13:	1	frac_minor_disagree	0.3419979429	3.495649e-01	0.2138076351
## 14:	0	frac_minor_disagree	0.3311170803	3.235827e-01	0.2205658156
## 15:	1000	frac_minor_disagree	0.3007488966	3.040855e-01	0.2345066931
## 16:	10	share_existence	0.9987395006	9.990102e-01	0.9958634953
## 17:	100	share_existence	0.9749187906	9.785986e-01	0.9388543231
## 18:	1	share_existence	0.9998888226	9.999368e-01	0.9993735644
## 19:	0	share_existence	1.0000000000	1.000000e+00	1.0000000000
## 20:	1000	share_existence	0.8881529264	8.982820e-01	0.7835751053
## 21:	10	share_marginal	0.0012604994	9.897843e-04	0.0006042296
## 22:	100	share_marginal	0.0250812094	2.140140e-02	0.0104125963
## 23:	1	share_marginal	0.0001111774	6.324311e-05	0.0000000000
## 24:	0	share_marginal	0.0000000000	0.000000e+00	0.0000000000
## 25:	1000	share_marginal	0.1118470736	1.017180e-01	0.0665816327
##	tau	agreement	frac_mean	frac_median	frac_lwr
##	<num>	<fctr>	<num>	<num>	<num>
##	frac_upr				
##	<num>				
## 1:	0.8574304934				
## 2:	0.7225445120				
## 3:	0.8913352054				
## 4:	0.9047505169				
## 5:	0.5563956371				
## 6:	0.5082584548				
## 7:	0.4138286634				
## 8:	0.5555861863				
## 9:	0.5927423088				
## 10:	0.3187902826				
## 11:	0.4413451381				
## 12:	0.4123195332				
## 13:	0.4634183973				
## 14:	0.4544157127				
## 15:	0.3409766981				


```
## 16: 0.9993957704
## 17: 0.9895874037
## 18: 1.0000000000
## 19: 1.0000000000
## 20: 0.9334183673
## 21: 0.0041365047
## 22: 0.0611456769
## 23: 0.0006264356
## 24: 0.0000000000
## 25: 0.2164248947
##          frac_upr
##          <num>
```

```
## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation introduced :
## log-10 transformation introduced infinite values.
```

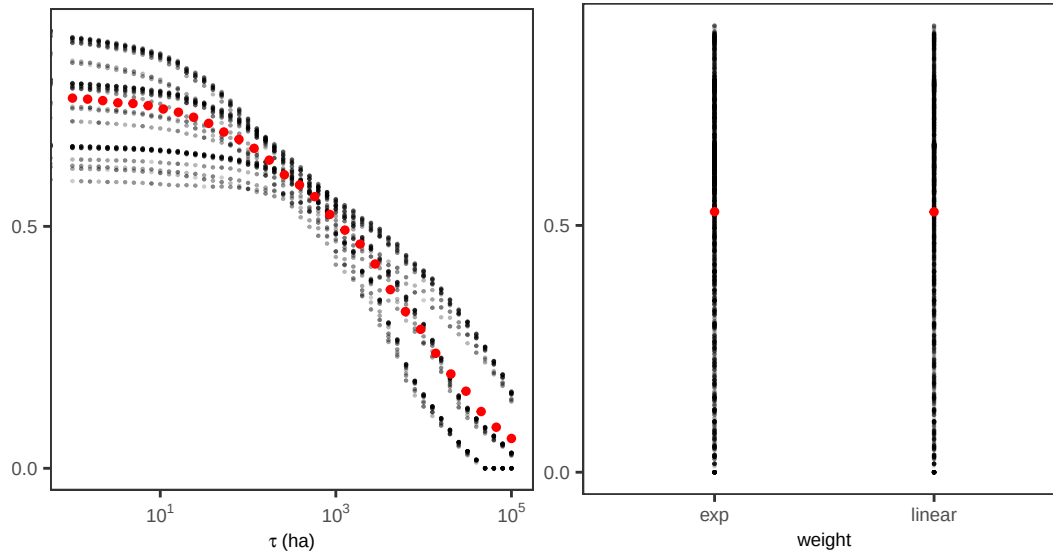


```
## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation introduced :
## log-10 transformation introduced infinite values.
```



```
## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation introduced :
## log-10 transformation introduced infinite values.
```

```
## Warning: Removed 157 rows containing non-finite outside the scale range
## (`stat_summary_bin()`).
```



```
# MERGE UA/SA PLOTS #####
```

```
top <- plot_grid(plot_uncertainty, plot_agreement, plot_indices3,
  ncol = 3, labels = "auto", rel_widths = c(0.4, 0.3, 0.3))
```

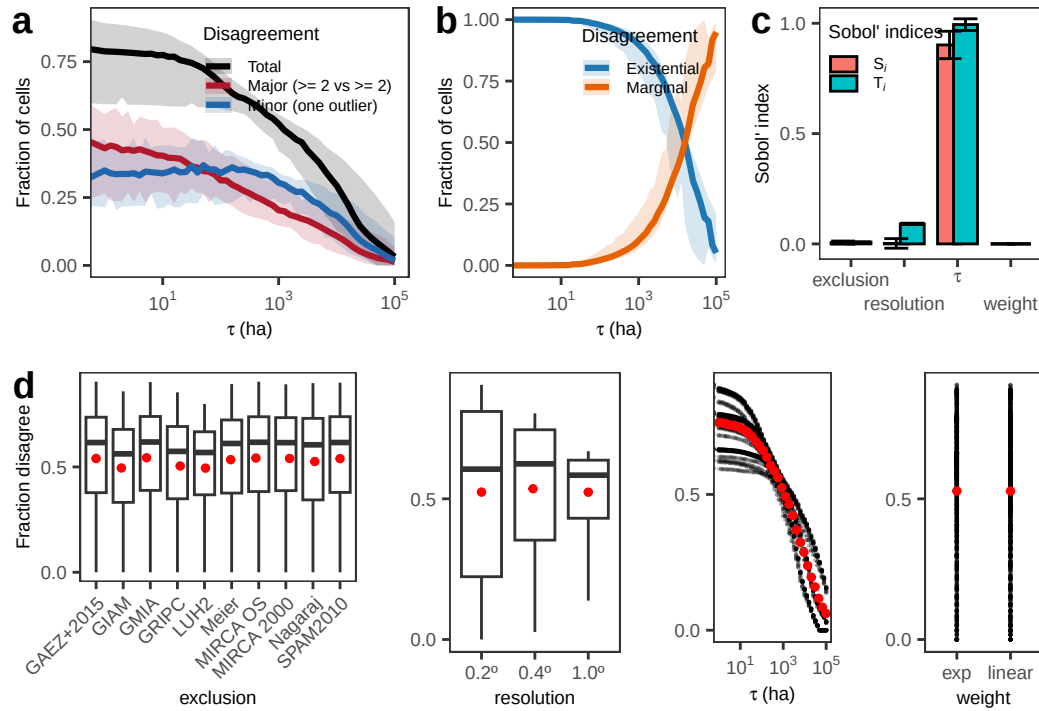
```
## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation introduced :
## log-10 transformation introduced infinite values.
## log-10 transformation introduced infinite values.
## log-10 transformation introduced infinite values.
```

```
bottom <- plot_grid(a3, b3, c3, d3, ncol = 4, rel_widths = c(0.35, 0.25, 0.2, 0.2), labels = "auto")
```

```
## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation introduced :
## log-10 transformation introduced infinite values.
```

```
## Warning: Removed 157 rows containing non-finite outside the scale range
## (`stat_summary_bin()`).
```

```
plot_grid(top, bottom, ncol = 1)
```



2.5 UA/SA on mapping variables

```
# MAPPING PARADIGM EXCLUSION UA/SA #####

ds_groups <- list(AEI = c("meier", "nagaraj", "gaez_v4", "spam", "luh2"),
                 AAI = c("giam", "gmia", "gripc", "mirca2000", "mirca_os"))

mapping_paradigm_vec <- names(ds_groups)
n_mapping_paradigm_vec <- length(mapping_paradigm_vec)

# Create the sample matrix -----

params <- c("resolution", "tau", "exclusion")
mat <- data.table(sobol_matrices(matrices = matrices, N = N, params = params))

# Transform resolution column -----

mat[, resolution:= pmin(n_resolution_vec, 1L + floor(resolution * n_resolution_vec))]
mat[, resolution:= resolution_vec[resolution]]

# Transform tau column -----

mat[, tau:= pmin(n_tau_vec, 1L + floor(tau * n_tau_vec))]
mat[, tau:= tau_vec[tau]]

# Transform exclusion column -----
```

```

mat[, exclusion:= pmin(n_mapping_paradigm_vec, 1L + floor(exclusion * n_mapping_paradigm_vec))
mat[, exclusion:= mapping_paradigm_vec[exclusion]]

# PRE-COMPUTE ALL COMBINATIONS #####

# Compute all possibilities by looping over resolutions and tau values -----

# Set parallel backend -----

resolutions <- sort(unique(dt$resolution))
ncores <- max(1, parallel::detectCores() - 1)

# ds_groups: named list of dataset keys to exclude by mapping category
# mat: data.table with columns resolution, tau (in ha), exclusion (category name)

idcols <- c("lon", "lat", "country", "code", "continent")

tau_results <- rbindlist(
  parallel::mclapply(resolutions, function(res) {

    dt_res_long <- dt[resolution == res]
    if (nrow(dt_res_long) == 0L) return(NULL)

    # scenarios to run for this resolution
    mat_res <- mat[resolution == res]
    if (nrow(mat_res) == 0L) return(NULL)

    # datasets present at this resolution
    ds_res <- sort(unique(dt_res_long$dataset))

    # ---- CAST ONCE per resolution ----

    dt_res_wide_all <- dcast(dt_res_long,
                           lon + lat + country + code + continent ~ dataset,
                           value.var = "mha", fill = 0)

    # loop exclusion groups (few) instead of mat rows (many)
    excl_levels <- unique(mat_res$exclusion)

    rbindlist(lapply(excl_levels, function(excl_grp) {

      # datasets to exclude for this group
      excl_ds <- ds_groups[[excl_grp]]
      if (is.null(excl_ds)) excl_ds <- character(0)

      use_ds <- setdiff(ds_res, excl_ds)
      if (length(use_ds) < 2L) return(NULL)
    })
  })

```

```

# taus used for THIS (resolution, exclusion-group)
taus_ha <- unique(mat_res[exclusion == excl_grp, tau])
taus_mha <- taus_ha * 1e-6

# ---- compute A_min/A_max ONCE per exclusion group ----
dt_w <- dt_res_wide_all[, c(idcols, use_ds), with = FALSE]

dt_w[, `:=`(A_min = do.call(pmin, c(.SD, list(na.rm = TRUE))),
           A_max = do.call(pmax, c(.SD, list(na.rm = TRUE))),
           .SDcols = use_ds)]

# ---- now only loop over taus
out_grp <- rbindlist(lapply(seq_along(taus_mha), function(k) {

  tau_ha <- taus_ha[k]
  tau_mha <- taus_mha[k]

  out <- compute_tau_fun(dt_w, tau_mha, use_ds)

  out[, `:=`(
    tau_label = paste0(format(tau_ha, scientific = FALSE, trim = TRUE), "ha"),
    tau_ha = tau_ha,
    tau_mha = tau_mha,
    resolution = res,
    exclusion = excl_grp
  )]

  out
}), use.names = TRUE, fill = TRUE)

out_grp
}), use.names = TRUE, fill = TRUE)

}, mc.cores = ncores),
use.names = TRUE, fill = TRUE
)

# ARRANGE OUTPUT #####

# Rename tau_ha to tau to match mat -----

setnames(tau_results, "tau_ha", "tau")

# Join all three disagreement metrics onto mat -----

mat[tau_results, on = .(resolution, tau, exclusion = exclusion),
  `:=`(frac_disagree = i.frac_disagree,

```

```

    frac_major_disagree = i.frac_major_disagree,
    frac_minor_disagree = i.frac_minor_disagree,
    share_existence = i.share_existence,
    share_marginal = i.share_marginal)]

# Reshape just these three for plotting -----

mat_long <- melt(mat, id.vars = c("resolution", "tau", "exclusion"),
  measure.vars = c("frac_disagree", "frac_major_disagree",
    "frac_minor_disagree", "share_existence",
    "share_marginal"),
  variable.name = "agreement", value.name = "frac_value")

# PREPARE DATA TO PLOT UNCERTAINTY #####

# Calculate mean and quantiles by tau x agreement -----

mat_sum <- mat_long[, .(frac_mean = mean(frac_value, na.rm = TRUE),
  frac_median = median(frac_value, na.rm = TRUE),
  frac_lwr = quantile(frac_value, 0.025, na.rm = TRUE),
  frac_upr = quantile(frac_value, 0.975, na.rm = TRUE)),
  .(tau, agreement)]

mat_sum[, {dt_sub <- .SD
  idx <- sapply(target_tau, function(x) {
    which.min(abs(log10(tau + 1e-12) - log10(x + 1e-12)))
  })
  dt_sub[idx]
},
agreement
]

```

##	agreement	tau	frac_mean	frac_median	frac_lwr
##	<fctr>	<num>	<num>	<num>	<num>
## 1:	frac_disagree	0.0000	0.632763385	0.6411539816	0.4969013386
## 2:	frac_disagree	1.0000	0.626999490	0.6402736556	0.4967773922
## 3:	frac_disagree	10.0000	0.606632456	0.6313949394	0.4939266237
## 4:	frac_disagree	501.1872	0.442056757	0.4188292653	0.4095025766
## 5:	frac_disagree	1000.0000	0.390105738	0.3894397620	0.3426651649
## 6:	frac_disagree	1584.8932	0.360526361	0.3355802606	0.2924753911
## 7:	frac_disagree	1995.2623	0.339223484	0.3515121468	0.2679050822
## 8:	frac_disagree	5011.8723	0.264034661	0.2756426380	0.1703027124
## 9:	frac_disagree	10000.0000	0.201851184	0.2132652548	0.1015444194
## 10:	frac_disagree	12589.2541	0.181480564	0.1767443030	0.0863702287
## 11:	frac_major_disagree	0.0000	0.315958611	0.3597766487	0.1761279127
## 12:	frac_major_disagree	1.0000	0.308318422	0.2884702450	0.1798463064

## 13:	frac_major_disagree	10.0000	0.284691982	0.2813270285	0.1813336639
## 14:	frac_major_disagree	501.1872	0.182193883	0.1683201160	0.1477149998
## 15:	frac_major_disagree	1000.0000	0.159155914	0.1683192861	0.1287839047
## 16:	frac_major_disagree	1584.8932	0.143161615	0.1294833744	0.1065587682
## 17:	frac_major_disagree	1995.2623	0.134430265	0.1443976202	0.0949177647
## 18:	frac_major_disagree	5011.8723	0.100141504	0.1012877911	0.0557904157
## 19:	frac_major_disagree	10000.0000	0.075214507	0.0770410986	0.0366356034
## 20:	frac_major_disagree	12589.2541	0.066654489	0.0598118618	0.0314283951
## 21:	frac_minor_disagree	0.0000	0.316804774	0.3207734259	0.2500000000
## 22:	frac_minor_disagree	1.0000	0.318681069	0.3295744083	0.2516113039
## 23:	frac_minor_disagree	10.0000	0.321940474	0.3500679109	0.2547099653
## 24:	frac_minor_disagree	501.1872	0.259862874	0.2538357060	0.2374814080
## 25:	frac_minor_disagree	1000.0000	0.230949824	0.2211204760	0.2138812602
## 26:	frac_minor_disagree	1584.8932	0.217364746	0.2147375567	0.1859166230
## 27:	frac_minor_disagree	1995.2623	0.204793220	0.2071145265	0.1729873176
## 28:	frac_minor_disagree	5011.8723	0.163893157	0.1743548468	0.1145122967
## 29:	frac_minor_disagree	10000.0000	0.126636677	0.1362241561	0.0635819422
## 30:	frac_minor_disagree	12589.2541	0.114826075	0.1169324413	0.0526969482
## 31:	share_existence	0.0000	1.000000000	1.000000000	1.000000000
## 32:	share_existence	1.0000	0.997429207	0.9996449844	0.9927644711
## 33:	share_existence	10.0000	0.990364277	0.9854812471	0.9796737767
## 34:	share_existence	501.1872	0.848206846	0.9383355414	0.7107206519
## 35:	share_existence	1000.0000	0.778994714	0.6961660551	0.6343482906
## 36:	share_existence	1584.8932	0.725210004	0.8147204317	0.5797872340
## 37:	share_existence	1995.2623	0.686576017	0.5903933691	0.5496726445
## 38:	share_existence	5011.8723	0.516795763	0.4410986404	0.3880684907
## 39:	share_existence	10000.0000	0.363638334	0.3178324366	0.2300267719
## 40:	share_existence	12589.2541	0.315040530	0.2801972062	0.1763865546
## 41:	share_marginal	0.0000	0.000000000	0.000000000	0.000000000
## 42:	share_marginal	1.0000	0.002570793	0.0003550156	0.0002296211
## 43:	share_marginal	10.0000	0.009635723	0.0145187529	0.0009438414
## 44:	share_marginal	501.1872	0.151793154	0.0616644586	0.0468247000
## 45:	share_marginal	1000.0000	0.221005286	0.3038339449	0.0996180777
## 46:	share_marginal	1584.8932	0.274789996	0.1852795683	0.1437288136
## 47:	share_marginal	1995.2623	0.313423983	0.4096066309	0.1815937941
## 48:	share_marginal	5011.8723	0.483204237	0.5589013596	0.3539786710
## 49:	share_marginal	10000.0000	0.636361666	0.6821675634	0.5255298651
## 50:	share_marginal	12589.2541	0.684959470	0.7198027938	0.5815244826
##	agreement	tau	frac_mean	frac_median	frac_lwr
##	<fctr>	<num>	<num>	<num>	<num>
##	frac_upr				
##	<num>				
##	1:	0.739917302			
##	2:	0.726077699			
##	3:	0.692334989			
##	4:	0.498038131			
##	5:	0.464055528			
##	6:	0.442736738			


```

## 7: 0.435423897
## 8: 0.375061973
## 9: 0.322508676
## 10: 0.301685672
## 11: 0.395709260
## 12: 0.379223625
## 13: 0.337519672
## 14: 0.222772775
## 15: 0.201660882
## 16: 0.193232524
## 17: 0.191497273
## 18: 0.157535944
## 19: 0.124938027
## 20: 0.114154685
## 21: 0.361087111
## 22: 0.359783380
## 23: 0.359166230
## 24: 0.280490828
## 25: 0.262394646
## 26: 0.249504214
## 27: 0.243926624
## 28: 0.217526029
## 29: 0.197570649
## 30: 0.187530987
## 31: 1.000000000
## 32: 0.999770379
## 33: 0.999056159
## 34: 0.953175300
## 35: 0.900381922
## 36: 0.856271186
## 37: 0.818406206
## 38: 0.646021329
## 39: 0.474470135
## 40: 0.418475517
## 41: 0.000000000
## 42: 0.007235529
## 43: 0.020326223
## 44: 0.289279348
## 45: 0.365651709
## 46: 0.420212766
## 47: 0.450327356
## 48: 0.611931509
## 49: 0.769973228
## 50: 0.823613445
##      frac_upr
##      <num>

```

```

# Plot uncertainty -----

agreement_cols <- c(frac_disagree = "black",
                    frac_major_disagree = "#B2182B",
                    frac_minor_disagree = "#2166AC" )

plot_uncertainty <- mat_sum[!agreement %in% c("share_existence", "share_marginal")] %>%
  ggplot(.,
    aes(x = tau, y = frac_median, colour = agreement, fill = agreement)) +
  geom_ribbon(aes(ymin = frac_lwr, ymax = frac_upr), alpha = 0.18, colour = NA) +
  geom_line(linewidth = 1.1) +
  scale_x_log10(labels = label_log(digits = 2)) +
  scale_colour_manual(values = agreement_cols,
    breaks = c("frac_disagree",
               "frac_major_disagree",
               "frac_minor_disagree"),
    labels = c("Total",
               "Major (\u2265 2 vs \u2265 2)",
               "Minor (one outlier)"),
    name = "Disagreement") +
  scale_fill_manual(values = agreement_cols,
    breaks = c("frac_disagree",
               "frac_major_disagree",
               "frac_minor_disagree"),
    labels = c("Total",
               "Major (\u2265 2 vs \u2265 2)",
               "Minor (one outlier)"),
    name = "Disagreement") +
  labs(x = expression(tau~"(ha)"), y = "Fraction of cells") +
  theme_AP() +
  theme(legend.position = c(0.65, 0.77))

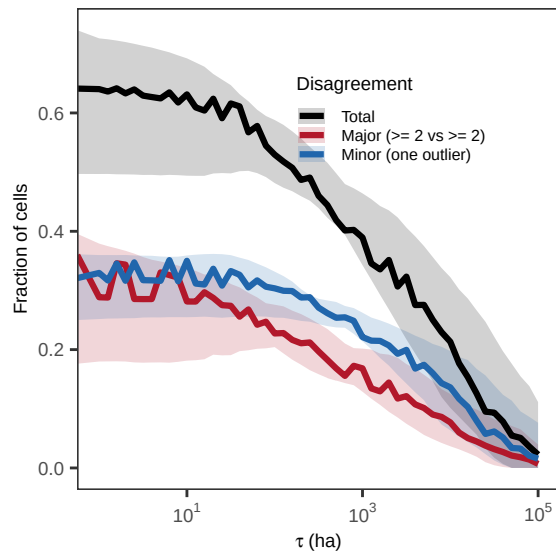
plot_uncertainty

```

```

## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation introduced :
## log-10 transformation introduced infinite values.

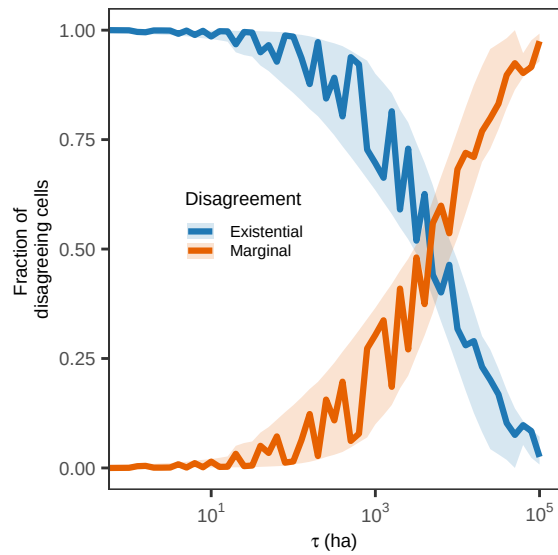
```



```
plot_agreement <- mat_sum[agreement %in% c("share_existence", "share_marginal")] %>%
  ggplot(., aes(x = tau, y = frac_median, colour = agreement, fill = agreement)) +
  geom_ribbon(aes(ymin = frac_lwr, ymax = frac_upr), alpha = 0.18, colour = NA) +
  geom_line(linewidth = 1.1) +
  scale_colour_manual(values = exist_cols,
                      labels = c(share_existence = "Existential",
                                share_marginal = "Marginal"),
                      name = "Disagreement") +
  scale_fill_manual(values = exist_cols,
                    labels = c(share_existence = "Existential",
                              share_marginal = "Marginal"),
                    name = "Disagreement") +
  scale_x_log10(labels = label_log(digits = 2)) +
  labs(x = expression(tau~"(ha)"), y = "Fraction of \ndisagreeing cells") +
  theme_AP() +
  theme(legend.position = c(0.3, 0.55))

plot_agreement
```

```
## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation introduced :
## log-10 transformation introduced infinite values.
```



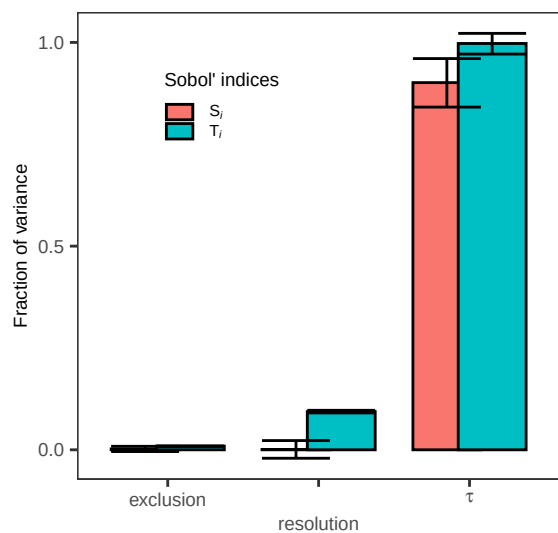
SENSITIVITY ANALYSIS

Sobol' indices -----

```
ind <- sobol_indices(matrices = matrices, N = N, params = params, Y
                    = mat[, frac_disagree], R = R, boot = TRUE)
```

```
plot_indices2 <- plot(ind) +
  theme_AP() +
  labs(x = "", y = "Fraction of variance") +
  scale_x_discrete(labels = c(exclusion = "exclusion",
                              resolution = "resolution",
                              tau = expression(tau)),
                  guide = guide_axis(n.dodge = 2)) +
  theme(legend.position = c(0.3, 0.8))
```

plot_indices2

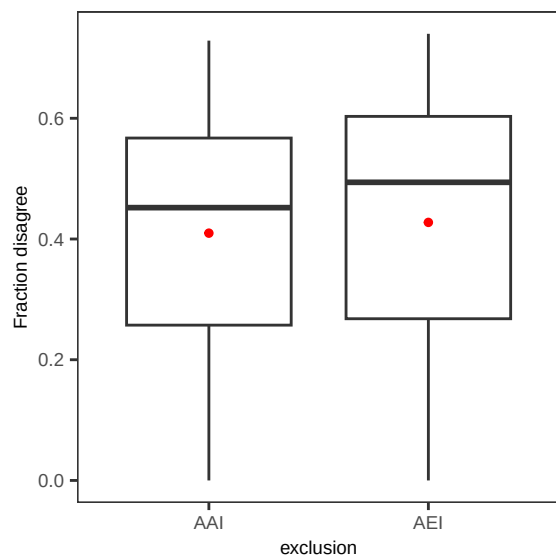


```
# SCATTERPLOTS OF OUTPUT AGAINST INPUT #####
```

```
label_map <- setNames(new_dataset_names, dataset_names)

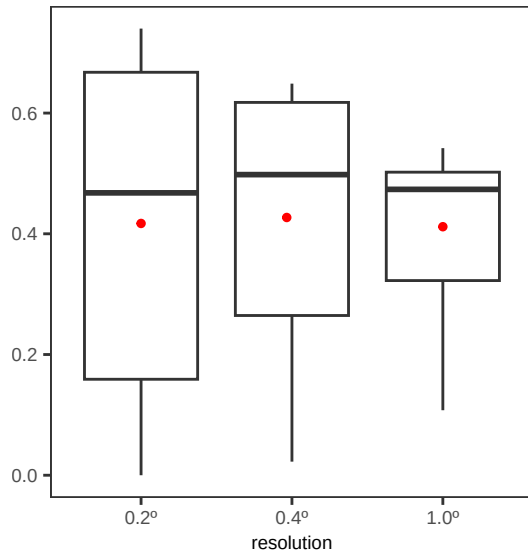
a2<- ggplot(mat[!exclusion == "all", c("exclusion", "frac_disagree")],
  aes(exclusion, frac_disagree)) +
  geom_boxplot() +
  stat_summary_bin(fun = "mean", geom = "point",
    colour = "red", size = 1) +
  scale_x_discrete(labels = label_map) +
  scale_y_continuous(breaks = breaks_pretty(n = 3)) +
  theme_AP() +
  labs(x = "exclusion", y = "Fraction disagree")
```

a2



```
b2 <- ggplot(mat[1:N, c("resolution", "frac_disagree")],
  aes(resolution, frac_disagree)) +
  geom_boxplot() +
  scale_x_discrete(labels = c(`0.2deg` = "0.2°",
    `0.4deg` = "0.4°",
    `1deg` = "1.0°")) +
  scale_y_continuous(breaks = breaks_pretty(n = 3)) +
  stat_summary_bin(fun = "mean", geom = "point",
    colour = "red", size = 1) +
  theme_AP() +
  labs(x = "resolution", y = "")
```

b2

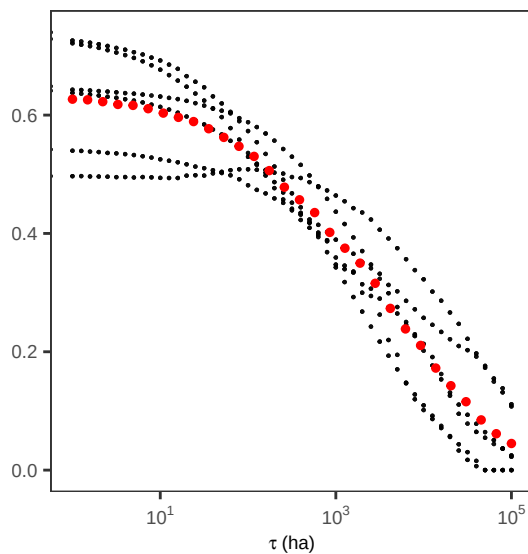


```
c2 <- ggplot(mat[1:N, c("tau", "frac_disagree")],
             aes(tau, frac_disagree)) +
  geom_point(size = 0.1, alpha = 0.1) +
  stat_summary_bin(fun = "mean", geom = "point",
                  colour = "red", size = 1) +
  scale_y_continuous(breaks = breaks_pretty(n = 3)) +
  theme_AP() +
  scale_x_log10(labels = label_log(digits = 2)) +
  labs(x = expression(tau~"("ha)"), y = "")
```

c2

```
## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation introduced :
## log-10 transformation introduced infinite values.
```

```
## Warning: Removed 157 rows containing non-finite outside the scale range
## (`stat_summary_bin()`).
```



```
# MERGE UA/SA PLOTS #####
```

```
top <- plot_grid(plot_uncertainty, plot_agreement, plot_indices2,
  ncol = 3, labels = "auto", rel_widths = c(0.4, 0.3, 0.3))
```

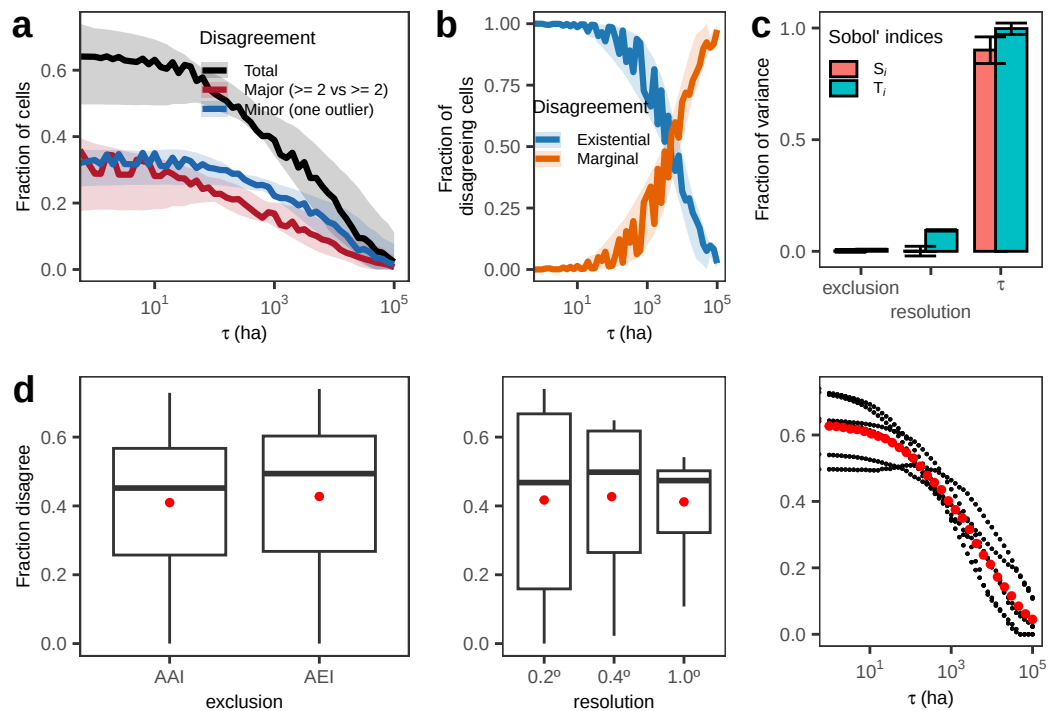
```
## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation introduced .
## log-10 transformation introduced infinite values.
## log-10 transformation introduced infinite values.
## log-10 transformation introduced infinite values.
```

```
bottom <- plot_grid(a2, b2, c2, ncol = 3, rel_widths = c(0.4, 0.3, 0.3), labels = "d")
```

```
## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation introduced .
## log-10 transformation introduced infinite values.

## Warning: Removed 157 rows containing non-finite outside the scale range
## (`stat_summary_bin()`).
```

```
plot_grid(top, bottom, ncol = 1)
```



3 The maximum threshold for existence disagreement: $\tau_{\max}$

```
# COMPUTE TAU MAX #####
#####

# Tau max is the largest irrigation threshold at which at least one dataset
# classifies the cell as essentially unirrigated while another classifies
# it as irrigated. Tau max thus quantifies how large the ontological disagreement
# on irrigation is. We also set NA to two cases:

# - Universally non-irrigated cells: all datasets report areas <= threshold.
# - Universally irrigated cells: all datasets report areas >threshold.

# NA cells are cells for which there is agreement irrigation exists or it is absent.

## Compute tau_max_results -----

tau_max_results <- rbindlist(
  parallel::mclapply(resolutions, function(res) {

    dt_res_long <- dt[resolution == res]

    # datasets present at this resolution
    ds_res <- sort(unique(dt_res_long$dataset))

    # resolution-specific detectability thresholds (in ha)-----

    zero_tol_ha <- switch(res,
      "0.2deg" = 500, # 1% of cell at 0.2°
      "0.4deg" = 2000, # 1% of cell at 0.4°
      "1deg" = 12000, # 1% of cell at 1°
      stop("Unknown resolution: ", res))

    rbindlist(
      lapply(dataset_vec, function(scn) {

        # which datasets to use -----

        if (scn == "all") {
          use_ds <- ds_res
        } else {
          use_ds <- setdiff(ds_res, scn)
        }

        # if only one dataset left, skip -----

        if (length(use_ds) < 2L) return(NULL)
```



```

# reshape to wide -----

dt_res_wide <- dcast(
  dt_res_long[dataset %in% use_ds],
  lon + lat + country + code + continent ~ dataset,
  value.var = "mha", fill = 0
)

tau_max_dt <- tau_grid[,
  compute_tau_max_existential_fun(dt_res_wide, use_ds, tau_mha, tau_ha,
    zero_tol_ha = zero_tol_ha)
]

tau_max_dt[, `:=`(resolution = res, scenario = scn)]

tau_max_dt
}),
use.names = TRUE, fill = TRUE
)
}, mc.cores = ncores),
use.names = TRUE, fill = TRUE
)

# Add the countries -----

chunk_size <- 500000L
idx <- split(seq_len(nrow(tau_max_results)),
  ceiling(seq_len(nrow(tau_max_results)) / chunk_size))

tau_max_results[, country := NA_character_]
for (i in idx) {
  tau_max_results[i, country := lonlat_to_country(lon, lat, countries_sf)]
}

# Add the continents -----

country_to_continent(tau_max_results)

# Retrieve the round with all datasets and exclude the rest -----

tau_max_all <- tau_max_results[scenario == "all"]

# DEFINE PLOTS TAU MAX #####

# Bin the tau max results -----

```

```

tau_max_results[, det_thresh:= fcase(
  resolution == "0.2deg", 500,
  resolution == "0.4deg", 2000,
  resolution == "1deg", 12000)]

tau_max_results[, tau_bin:= fcase(
  is.na(tau_max_ha), "Agreement",
  tau_max_ha <= 2 * det_thresh, "Low disagreement",
  tau_max_ha <= 5 * det_thresh, "Moderate disagreement",
  tau_max_ha <= 10 * det_thresh, "High disagreement",
  default = "Extreme disagreement"
)]

tau_max_results[, tau_bin:= factor(tau_bin, levels = c("Agreement",
  "Low disagreement",
  "Moderate disagreement",
  "High disagreement",
  "Extreme disagreement"),
  ordered = TRUE)]

# Proportion of cells disagreeing at 1% threshold-----

tau_max_results[, .N, .(tau_bin, resolution)] %>%
  .[, fraction := N / sum(N), by = resolution] %>%
  .[!tau_bin == "Agreement", sum(fraction), resolution]

##      resolution      V1
##      <char>      <num>
## 1:      0.2deg 0.5508904
## 2:      0.4deg 0.4590500
## 3:      1deg 0.3799635

thresh_df <- unique(tau_max_results[!is.na(det_thresh), .(resolution, det_thresh)])

# ANALYSIS OF DATA -----

# Fraction of high and extreme ontological disagreement
# across resolutions in each continent -----

dt_all <- tau_max_results[scenario == "all"]
dt_all[, N_total := .N, by = .(resolution, continent)]
frac_by_rcb <- dt_all[, .(frac = .N / unique(N_total)),
  .(resolution, continent, tau_bin)]

frac_by_rcb[, .(min = min(frac), max = max(frac)), .(continent, tau_bin)] %>%
  .[tau_bin %in% c("Extreme disagreement", "High disagreement")] %>%
  .[order(-min)] %>%

```

```

[, .(min = min(min), max = max(max)), .(continent)]

##      continent      min      max
##      <char>      <num>      <num>
## 1:      Asia 0.10417315 0.3561106
## 2:  Oceania 0.10869565 0.2256921
## 3:    Europe 0.08386916 0.1890844
## 4:  Americas 0.05267170 0.1500000
## 5:    Africa 0.03516442 0.1117143

# distribution of detectability limits

tau_max_results[, .N, .(tau_max_ha, resolution, scenario)]

##      tau_max_ha resolution scenario      N
##      <num>      <char>  <char> <int>
## 1:      NA      0.2deg      all 55250
## 2: 1584.893      0.2deg      all 5919
## 3: 1000.000      0.2deg      all 2969
## 4: 5011.872      0.2deg      all 7187
## 5: 3981.072      0.2deg      all 5088
## ---
## 588: 31622.777      1deg      spam 291
## 589: 25118.864      1deg      spam 281
## 590: 100000.000      1deg      spam 533
## 591: 63095.734      1deg      spam 259
## 592: 79432.823      1deg      spam 257

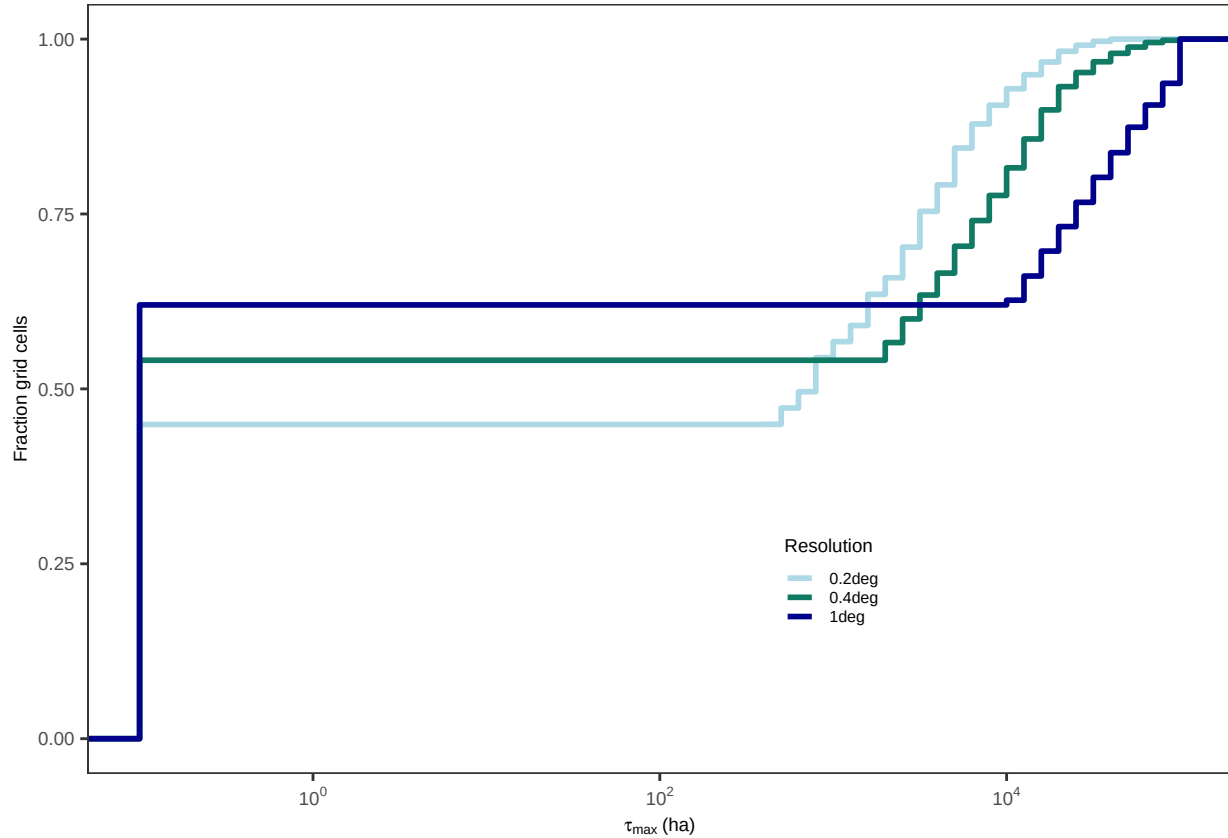
tmp <- copy(tau_max_results)
tmp[is.na(tau_max_ha), tau_max_ha:= 0.1]

plot_ecdf <- ggplot(tmp, aes(x = tau_max_ha, color = resolution)) +
  stat_ecdf(size = 1) +
  scale_x_log10(labels = label_log(digits = 2)) +
  scale_color_manual(values = res_palette) +
  labs(x = expression(tau[max]~"(ha)"), y = "Fraction grid cells",
       color = "Resolution") +
  theme_AP() +
  theme(legend.position = c(0.65, 0.25))

## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

```

```
plot_ecdf
```



How much of the world is perfectly detectable vs fundamentally ambiguous?

```
tau_max_results[, .N, by = .(resolution, tau_bin)] %>%
  .[, frac:= N / sum(N), by = resolution] %>%
  print()
```

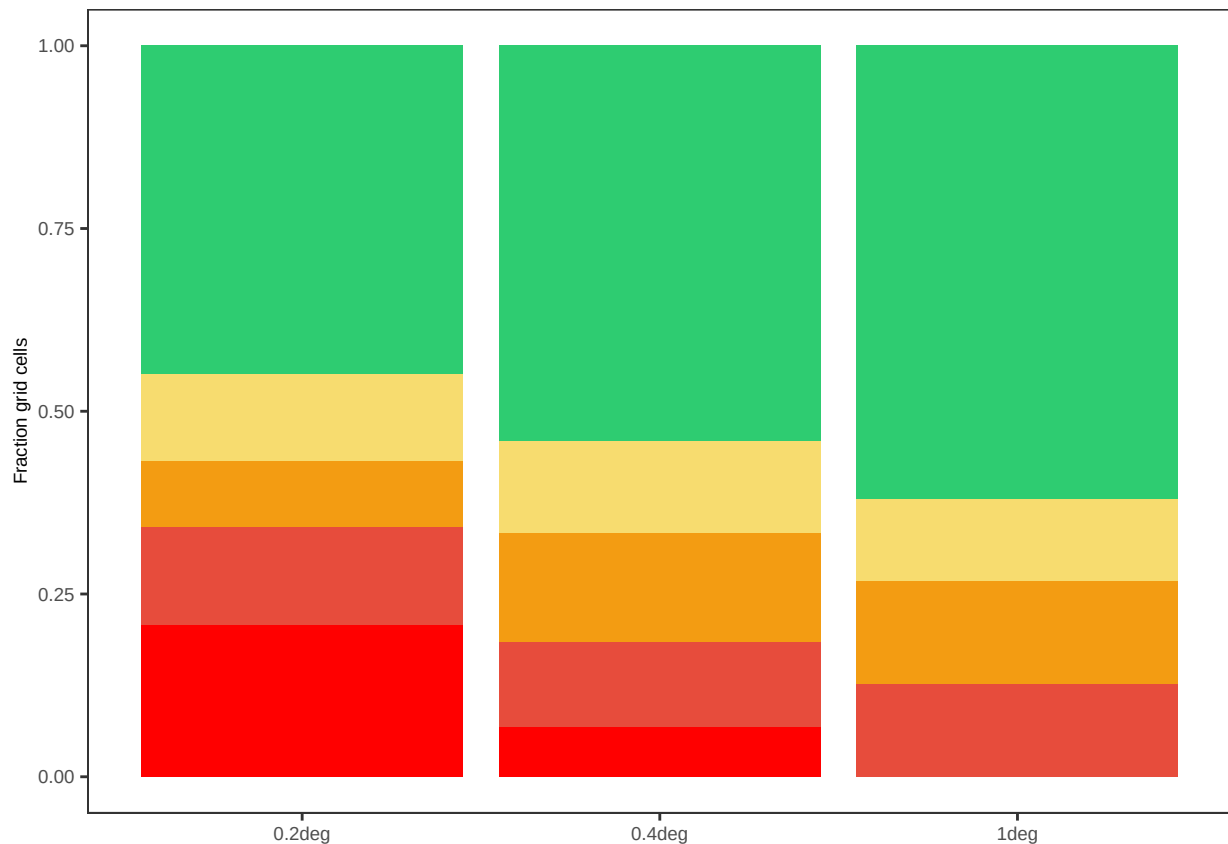
##	resolution	tau_bin	N	frac
##	<char>	<ord>	<int>	<num>
## 1:	0.2deg	Agreement	640389	0.44910962
## 2:	0.2deg	Moderate disagreement	130164	0.09128499
## 3:	0.2deg	Low disagreement	168896	0.11844803
## 4:	0.2deg	Extreme disagreement	297053	0.20832550
## 5:	0.2deg	High disagreement	189406	0.13283185
## 6:	0.4deg	Agreement	236578	0.54095002
## 7:	0.4deg	Moderate disagreement	65807	0.15047172
## 8:	0.4deg	Low disagreement	54457	0.12451925
## 9:	0.4deg	High disagreement	50769	0.11608641
## 10:	0.4deg	Extreme disagreement	29727	0.06797260
## 11:	1deg	Agreement	55027	0.62003651
## 12:	1deg	Low disagreement	9935	0.11194618
## 13:	1deg	Moderate disagreement	12605	0.14203137
## 14:	1deg	High disagreement	11181	0.12598594

```

plot_stacked <- tau_max_results[, .N, by = .(resolution, tau_bin)] %>%
  .[, frac := N / sum(N), by = resolution] |>
  ggplot(aes(resolution, frac, fill = tau_bin)) +
  geom_col() +
  scale_y_continuous() +
  scale_fill_manual(values = tau_palette, name = expression(tau[max])) +
  labs(y = "Fraction grid cells", x = NULL, fill = expression(tau[max])) +
  theme_AP() +
  theme(legend.position = "none")

```

plot_stacked



```

# Boxplots of tau max per resolution -----
plot_box <- ggplot(tau_max_results[!is.na(tau_max_ha) & tau_max_ha > 0],
  aes(resolution, tau_max_ha, fill = resolution)) +
  geom_boxplot(outlier.shape = NA, alpha = 0.85) +
  geom_hline(data = thresh_df, aes(yintercept = det_thresh, color = resolution),
    linetype = "dashed", linewidth = 0.7, show.legend = FALSE) +
  scale_y_log10(labels = scales::label_log(digits = 2),
    name = expression(tau[max] ~ "(ha)")) +
  scale_fill_manual(values = res_palette, guide = "none") +
  scale_color_manual(values = res_palette, guide = "none") +
  labs(

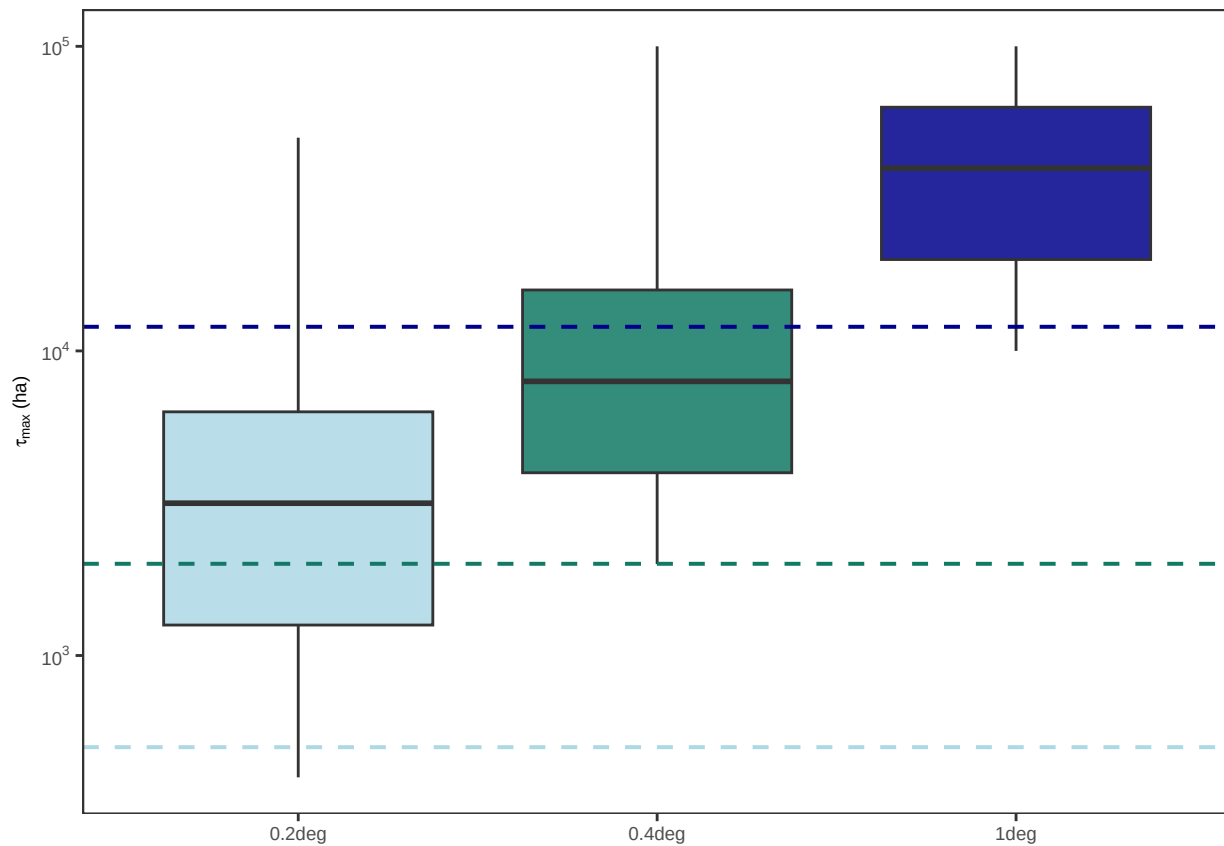
```

```

x = NULL,
) +
theme_AP() +
theme()

```

plot_box



```

# MERGE TAU MAX PLOTS #####

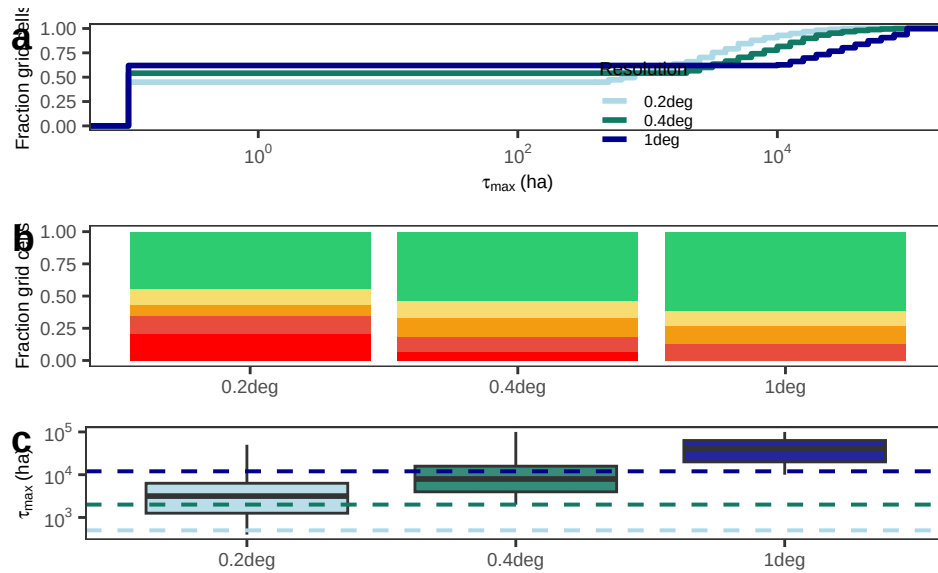
```

```

top <- plot_grid(plot_ecdf, plot_stacked, plot_box, ncol = 1,
  rel_heights = c(0.35, 0.35, 0.3),
  labels = "auto")

```

top

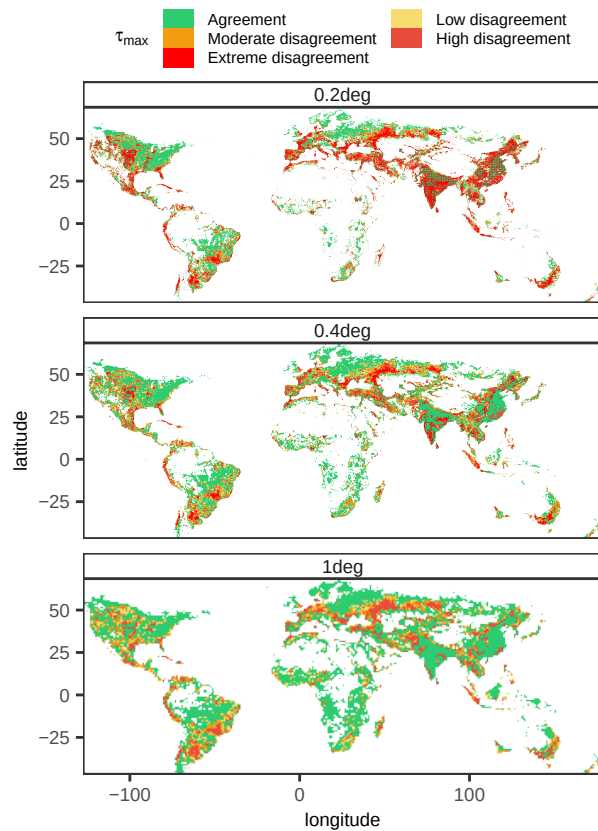


```
# The map #####

plot_raster <- ggplot(tau_max_results, aes(lon, lat, fill = tau_bin)) +
  geom_raster() +
  coord_equal(expand = FALSE) +
  facet_wrap(~ resolution, ncol = 1) +
  scale_fill_manual(values = tau_palette, name = expression(tau[max])) +
  scale_x_continuous(expand = c(0, 0)) +
  scale_y_continuous(expand = c(0, 0)) +
  theme_AP() +
  theme(panel.grid = element_blank(),
        strip.text = element_text(size = 11),
        legend.position = "top") +
  labs(x = "longitude", y = "latitude") +
  guides(fill = guide_legend(nrow = 3, byrow = TRUE))

plot_raster
```

```
## Warning: Raster pixels are placed at uneven horizontal intervals and will be shifted
## i Consider using `geom_tile()` instead.
## Raster pixels are placed at uneven horizontal intervals and will be shifted
## i Consider using `geom_tile()` instead.
```

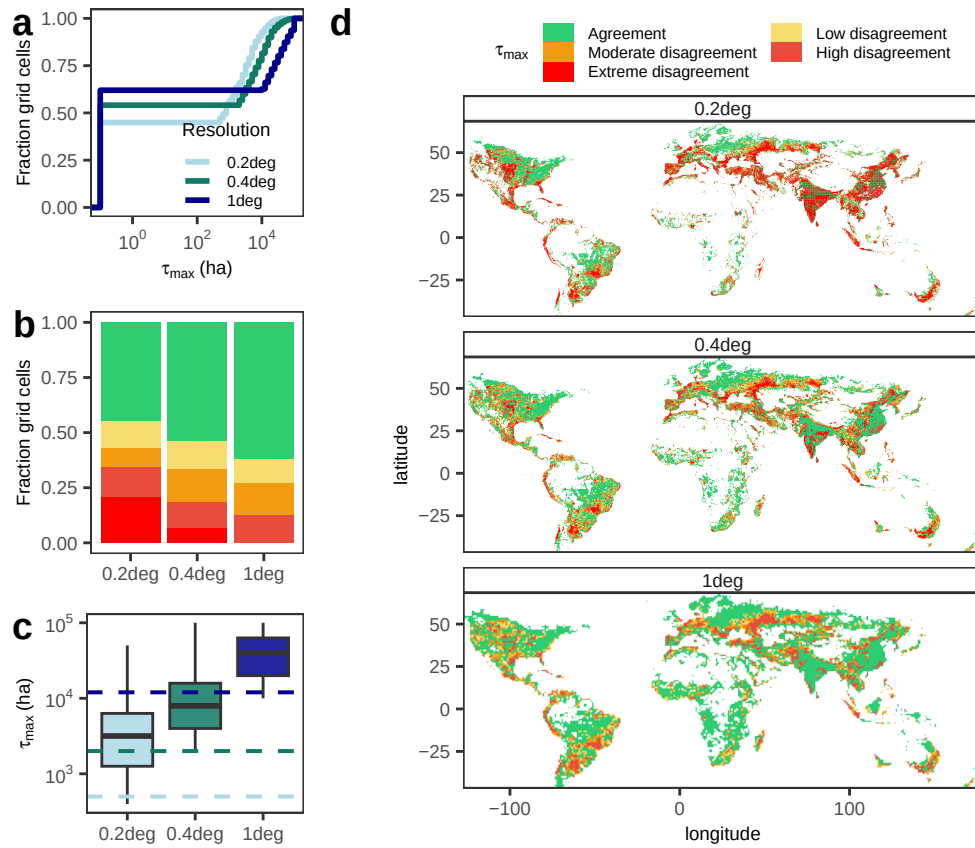


3.1 Composite figure

```
# MERGE PLOTS #####

plot_grid(top, plot_raster, rel_widths = c(0.3, 0.7), ncol = 2, labels = c("", "d"))

## Warning: Raster pixels are placed at uneven horizontal intervals and will be shifted
## i Consider using `geom_tile()` instead.
## Raster pixels are placed at uneven horizontal intervals and will be shifted
## i Consider using `geom_tile()` instead.
```

3.1.1 Which datasets drive deep disagreement?

```
# WHICH DATASETS DRIVE DEEP DISAGREEMENT? -----

# Check leave one-out scenarios -----

loo_stats_full <- tau_max_results[, .(
  median_tau = median(tau_max_ha, na.rm = TRUE),
  p75_tau = quantile(tau_max_ha, 0.75, na.rm = TRUE),
  p90_tau = quantile(tau_max_ha, 0.90, na.rm = TRUE),
  share_gt_1000 = mean(tau_max_ha >= 1000, na.rm = TRUE)),
  scenario] %>%
  .[order(-p90_tau)]

loo_stats_full
```

##	scenario	median_tau	p75_tau	p90_tau	share_gt_1000
##	<char>	<num>	<num>	<num>	<num>
## 1:	all	3981.072	10000.000	19952.62	0.8705251
## 2:	gaez_v4	3981.072	10000.000	19952.62	0.8700602
## 3:	giam	3981.072	7943.282	19952.62	0.8558969
## 4:	gmia	3981.072	10000.000	19952.62	0.8704727
## 5:	gripc	3981.072	7943.282	19952.62	0.8650565

```
## 6:      meier    3981.072 10000.000 19952.62      0.8693391
## 7: mirca2000    3981.072 10000.000 19952.62      0.8721156
## 8:  mirca_os    3981.072 10000.000 19952.62      0.8712516
## 9:   nagaraj    3981.072 10000.000 19952.62      0.8587307
## 10:      spam    3981.072 10000.000 19952.62      0.8702693
## 11:      luh2    3162.278  7943.282 15848.93      0.8537190

# In half of all grid cells worldwide, at least one dataset reports >=1000 ha
# of irrigation while another reports none. Removing most datasets does not
# change anything. Removing GIAM, Nagaraj, LUH2 and GRIPC drops the median disagreement
# and the extreme values, as they particularly increase existential contradictions
# in the upper half of the distribution.

# In which regions is each dataset structurally responsible for
# deep existential contradictions? -----

# Compute spatial delta -----

tau_all <- tau_max_results[scenario == "all",.(lon, lat, resolution,
                                             tau_all = tau_max_ha)]

# Keep only datasets of interest--

tau_loo <- tau_max_results[scenario %in% c("giam", "luh2", "nagaraj", "gripc"),
                           .(lon, lat, country, continent, resolution, scenario,
                              tau_loo = tau_max_ha)]

# Merge -----

tau_delta <- merge(tau_loo, tau_all, by = c("lon", "lat", "resolution"),
                  all.x = TRUE)

# Compute difference -----

tau_delta[, delta:= tau_all - tau_loo]

# Let us define "strong amplification" of differences -----
# Differences larger than 1000 ha.

tau_delta[, strong:= delta >= 1000]

# Compute overlap -----

influence_matrix <- dcast(tau_delta[resolution == "0.2deg"],
                        lon + lat ~ scenario, value.var = "strong",
                        fill = FALSE)

influence_matrix[, n_datasets:= rowSums(.SD), .SDcols = c("giam", "luh2", "nagaraj", "gripc")]

```

```
table(influence_matrix$n_datasets)
```

```
##
```

```
##      0      1
```

```
## 27483 18737
```

```
# 0: deep disagreement in the cell not driven by these three.
```

```
# 1: dataset-specific amplification.
```

```
# 2: partial overlap.
```

```
# 3: same hotspot amplified by all three.
```

```
# Since there are zero cells where two datasets simultaneously amplify deep  
# disagreement (> 1000 ha) or all three do, it means that their influence is  
# spatially distinct. 16,511 cells show strong amplification but always due  
# to exactly one dataset.
```

```
# Where does amplification concentrate? -----
```

```
regional_amp <- tau_delta[resolution == "0.2deg",  
                          .(median_delta = median(delta, na.rm = TRUE),  
                            share_strong = mean(delta >= 1000, na.rm = TRUE)),  
                          .(continent, scenario)] %>%  
  .[order(-median_delta)]
```

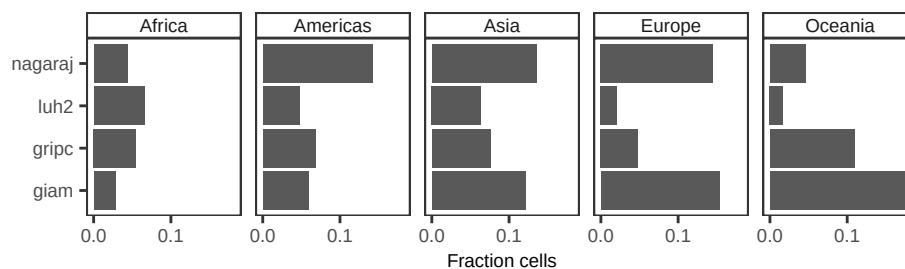
```
regional_amp
```

```
##      continent scenario median_delta share_strong  
##      <char>   <char>         <num>         <num>  
## 1:  Americas   giam          0      0.05904022  
## 2:  Americas   gripc         0      0.06811812  
## 3:  Americas   luh2          0      0.04731494  
## 4:  Americas   nagaraj        0      0.14193931  
## 5:   Europe    giam          0      0.15351445  
## 6:   Europe    gripc         0      0.04808959  
## 7:   Europe    luh2          0      0.02085566  
## 8:   Europe    nagaraj        0      0.14524721  
## 9:   Africa    giam          0      0.02829181  
## 10:  Africa    gripc         0      0.05506249  
## 11:  Africa    luh2          0      0.06565308  
## 12:  Africa    nagaraj        0      0.04397835  
## 13:   Asia     giam          0      0.12130263  
## 14:   Asia     gripc         0      0.07585118  
## 15:   Asia     luh2          0      0.06372931  
## 16:   Asia     nagaraj        0      0.13518991  
## 17: Oceania    giam          0      0.18177718  
## 18: Oceania    gripc         0      0.10928320  
## 19: Oceania    luh2          0      0.01693667
```

```
## 20: Oceania nagaraj 0 0.04547121
## continent scenario median_delta share_strong
## <char> <char> <num> <num>
```

Table shows which continents each dataset most strongly affects:
- Europe: GIAM dominates existential disagreement in Europe, LUH2 barely affects it.
- Asia: all three datasets contribute, LUH2 contributes less existential disagreement
- Ocenia: GIAM contributes the largest existential disagreement.
- Americas: Nagaraj.
- Africa: Overall amplification much lower, all contribute equally.
So overall there is no global problem dataset! And since they all show
median delta of zero everywhere, these datasets do not globally modify the
disagreement but create region-specific deep contradictions.

```
ggplot(regional_amp, aes(scenario, share_strong)) +
  geom_bar(stat = "identity") +
  coord_flip() +
  facet_wrap(~continent, ncol = 5) +
  scale_y_continuous(breaks = breaks_pretty(n = 2)) +
  theme_AP() +
  labs(x = "", y = "Fraction cells")
```



Which dataset drives the 22,615 cells? -----

```
influence_matrix[, dominant := fifelse(giam, "GIAM",
                                       fifelse(luh2, "LUH2",
                                                fifelse(nagaraj, "Nagaraj",
                                                         fifelse(gripc, "GRIPC",
                                                                    NA_character_)))]

table(influence_matrix$dominant)
```

```
##
## GIAM GRIPC LUH2 Nagaraj
## 6527 3361 2754 8259
```

3.2 How many grid cells disagree?

DETECTABILITY OF IRRIGATED AREA ACROSS MAPS
#####

```

# UNWEIGHTED: ALL CELLS COUNT EQUALLY #####
# QUESTION ANSWERED: HOW MANY GRID CELLS DISAGREE? -----

# Make dataset wide and retrieve only the 0.2deg resolution one -----

dt_wide <- dt[resolution == "0.2deg"] %>%
  dcast(., lon + lat + country + code + continent ~ dataset, value.var = "mha")

# Retrieve 0.2deg resolution from tau max dataset and remove unwanted cols -----

tau_max_results_clean <- tau_max_results[resolution == "0.2deg" & scenario == "all"]
tau_max_results_clean[, resolution:= NULL]
tau_max_results_clean[, scenario:= NULL]

# Merge -----

dt2 <- merge(dt_wide, tau_max_results_clean[, .(lon, lat, tau_max_mha)],
  by = c("lon", "lat"), all.x = TRUE)

setnames(dt2, dataset_names, new_dataset_names)

# Convert to ha for easier interpretation -----

dt2[, tau_max_ha:= tau_max_mha * 1e6]

# Which percentage of the world is in disagreement on whether there is
# or there is not irrigation? -----

frac_cells_nonid <- dt2[, mean(!is.na(tau_max_mha))]
frac_cells_nonid

## [1] 0.5737804

summary(dt2$tau_max_ha)

##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.     NA's
##   398.1  1584.9  3162.3  5575.2  6309.6 50118.7   55250

# Minimum = 398 ha, the detectability threshold for 0.2. No disagreement below it.
#
# Median disagreement (50%) is eight times the detectability threshold (400 ha)
# 75% of disagreeing cells exceed 2.5x detectability.
# Half exceed 8x detectability

# Even after imposing a conservative 1% rule some cells show contradictions at
# tens of thousands of hectares; these are large irrigation landscapes, not
# fragmented patches.

# Existential contradictions occur at scales that are agronomically and

```

```
# hydrologically meaningful and they are not confined to marginal lands. Include
# cells where irrigation is either extensive or completely absent depending on
# the dataset.Ddisagreement is not about "how much" irrigation exists but about
# whether large irrigation systems exist at all in certain locations.
```

3.3 How much irrigated land lies in disagreement?

```
# WEIGHTED: CELLS WITH LARGER IRRIGATION COUNT MORE #####
# QUESTION: HOW MUCH IRRIGATED LAND LIES IN DISAGREEMENT? -----
```

```
# Keep only cells with existential disagreement -----
# (tau_max_ha non-NA)
```

```
dt2_disagree <- dt2[!is.na(tau_max_ha)]
```

```
# Sanity check: no NA tau_max_ha left
summary(dt2_disagree$tau_max_ha)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  398.1  1584.9  3162.3  5575.2  6309.6 50118.7
```

```
# should show NA's = 0
```

```
# Compute total and non-identifiable area per dataset -----
```

```
area_nonid <- rbindlist(lapply(new_dataset_names, function(d) {
```

```
  # cells where this dataset has irrigation AND there is disagreement
```

```
  dt_d <- dt2[get(d) > 0, .(tau_max_ha, area = get(d))]
```

```
  # total irrigated area (for this dataset) in cells with disagreement
```

```
  total_area <- dt_d[, sum(area, na.rm = TRUE)]
```

```
  # irrigated area in cells where tau_max_ha is not NA; that is, in cells
  # that disagree
```

```
  nonid_area <- dt_d[!is.na(tau_max_ha), sum(area, na.rm = TRUE)]
```

```
  data.table(dataset = d, total_area_mha = total_area,
              nonid_area_mha = nonid_area, share_nonid = nonid_area / total_area)
```

```
}))
```

```
area_nonid
```

```
##      dataset total_area_mha nonid_area_mha share_nonid
##      <char>          <num>          <num>          <num>
```

```
## 1: GAEZ+2015      280.2879      154.70438      0.5519481
## 2:      GIAM      335.1898      204.36911      0.6097116
## 3:      GMIA      245.8296      131.80237      0.5361534
## 4:      GRIPC     305.9983      171.29686      0.5597967
## 5:      LUH2      262.3541       66.70482      0.2542549
## 6:      Meier     354.6320      209.34845      0.5903259
## 7: MIRCA 2000     210.5253      108.83424      0.5169652
## 8:  MIRCA OS      265.5109      140.13459      0.5277921
## 9:   Nagaraj      285.7831      186.01552      0.6508975
## 10: SPAM2010      219.5544      112.55164      0.5126368
```

```
# Depending on dataset, between 50-70% of all global irrigated areas
# lie in areas where we do not even agree on whether there is irrigation at all.
```

```
# How much tolerance do we need to make datasets agree?-----
```

```
tau_diag <- rbindlist(lapply(new_dataset_names, function(d) {

  dt_d <- dt2_disagree[get(d) > 0, .(tau_max_ha, area = get(d))]

  if (nrow(dt_d) == 0) {
    return(data.table(dataset = d,
                      tau_50 = NA_real_,
                      tau_80 = NA_real_,
                      tau_90 = NA_real_,
                      area_share_1000 = NA_real_ ))
  }

  # area-weighted empirical CDF of tau_max_ha (for this dataset in disagreeing cells)

  tmp <- dt_d[, .(area = sum(area)), by = tau_max_ha][order(tau_max_ha)]
  tmp[, cum_area := cumsum(area)]
  tmp[, cum_share := cum_area / sum(area)]

  # tau_50: minimum tau required to make 50% of irrigated area identifiable
  # tau_80: tau required for 80%
  # tau_90: tau required for 90%

  tau_50 <- tmp[cum_share >= 0.5][1, tau_max_ha]
  tau_80 <- tmp[cum_share >= 0.8][1, tau_max_ha]
  tau_90 <- tmp[cum_share >= 0.9][1, tau_max_ha]

  # share of area that is identifiable under tau_threshold_ha (e.g. 1000 ha)

  area_share_1000 <- tmp[tau_max_ha <= 1000, fifelse(.N == 0, 0,
                                                    max(cum_share, na.rm = TRUE))]]

  data.table(dataset = d,
```

```

        tau_50 = tau_50,
        tau_80 = tau_80,
        tau_90 = tau_90,
        area_share_1000 = area_share_1000)
    })

```

tau_diag

##	dataset	tau_50	tau_80	tau_90	area_share_1000
##	<char>	<num>	<num>	<num>	<num>
## 1:	GAEZ+2015	12589.254	25118.86	31622.78	0.024079566
## 2:	GIAM	15848.932	25118.86	31622.78	0.015560853
## 3:	GMIA	12589.254	25118.86	31622.78	0.020758906
## 4:	GRIPC	15848.932	25118.86	31622.78	0.017073267
## 5:	LUH2	10000.000	19952.62	25118.86	0.007732642
## 6:	Meier	12589.254	25118.86	31622.78	0.021551156
## 7:	MIRCA 2000	12589.254	25118.86	31622.78	0.025574408
## 8:	MIRCA OS	12589.254	25118.86	31622.78	0.022653569
## 9:	Nagaraj	7943.282	19952.62	25118.86	0.018838929
## 10:	SPAM2010	15848.932	25118.86	31622.78	0.019254211

tau_50 ranges from 8,000 to 16,000 ha.
tau_80 is 25,000-40,000 ha.
tau90 is often 31,000-40,000 ha.
#
At 0.2° resolution (40,000 ha per cell):
tau_50 equals 12,000-16,000 ha; that is 30-40% of the grid cell.
tau_90 equals 30,000-40,000 ha; most of the grid cell area.
#
That means that To reconcile 90% of irrigated area in disagreeing cells one
must tolerate differences approaching the size of an entire grid cell.

area_share_1000: Under a tolerance of 1000 ha, only 0.6%-2.7% of irrigated area
in disagreeing cells becomes identifiable.

PARAGRAPH?
Area-weighted quantiles of tau_max reveal that resolving even half of irrigated
area located in disagreeing cells would require tolerating discrepancies of
8,000-16,000 ha. resolving 90% would require accepting differences
exceeding 30,000 ha in most datasets. Under a tolerance of 1,000 ha,
less than 3% of irrigated area in disagreeing cells becomes identifiable.
These values indicate that disagreement persists at scales corresponding to
substantial irrigation systems rather than marginal patches.

4 Consequences for countries

4.1 Identification of irrigation hotspots

```
# DOES THE DIFFERENT DATASETS IDENTIFY THE SAME TOP IRRIGATION HOTSPOTS? #####

# Identify top 10% hotspots per dataset -----

hotspots <- lapply(new_dataset_names, function(d) {

  tmp <- copy(dt2)
  tmp[, value:= get(d)]

  # rank cells-----

  tmp <- tmp[order(-value)]
  n_hot <- ceiling(0.10 * nrow(tmp))

  tmp[, hotspot:= FALSE]
  tmp[1:n_hot, hotspot:= TRUE]

  tmp[, .(lon, lat, hotspot, dataset = d)]
})

# name the list elements by dataset -----

names(hotspots) <- sapply(hotspots, function(x) unique(x$dataset))

# extract hotspot cell IDs per dataset -----

hotspot_sets <- lapply(hotspots, function(dt) {
  dt[hotspot == TRUE, .(lon, lat)]})

# Country rankings per dataset -----

country_totals <- rbindlist(lapply(new_dataset_names, function(d) {
  dt2[, .(irrigated_area = sum(get(d), na.rm = TRUE)), country] %>%
    .[, dataset:= d]})

jaccard_cells <- function(dt1, dt2) {
  set1 <- paste(dt1$lon, dt1$lat)
  set2 <- paste(dt2$lon, dt2$lat)

  length(intersect(set1, set2)) / length(union(set1, set2))
}

# Pairwise jacqard similarity -----
```

```

datasets <- names(hotspot_sets)

jaccard_results <- rbindlist(combn(datasets, 2, simplify = FALSE,
                                  FUN = function(pair) {
    d1 <- pair[1]
    d2 <- pair[2]

    data.table(dataset_1 = d1, dataset_2 = d2,
               jaccard = jaccard_cells(hotspot_sets[[d1]], hotspot_sets[[d2]]))
  })
)

jaccard_results[order(jaccard)]

```

```

##      dataset_1 dataset_2  jaccard
##      <char>    <char>    <num>
##  1:      GIAM      LUH2 0.3018328
##  2:     GRIPC      LUH2 0.3175789
##  3:      LUH2   Nagaraj 0.3527785
##  4:      LUH2     Meier 0.3611592
##  5: GAEZ+2015      LUH2 0.3622320
##  6:      LUH2 SPAM2010 0.3861947
##  7:      GMIA      LUH2 0.3943957
##  8:      LUH2  MIRCA OS 0.3944707
##  9:      GIAM   Nagaraj 0.4022392
## 10:      LUH2  MIRCA 2000 0.4062703
## 11: GAEZ+2015      GIAM 0.4131691
## 12:     GRIPC   Nagaraj 0.4160249
## 13:      GIAM SPAM2010 0.4306368
## 14:      GIAM      GMIA 0.4361844
## 15:      GIAM     GRIPC 0.4400133
## 16:     GRIPC SPAM2010 0.4416148
## 17: GAEZ+2015     GRIPC 0.4525183
## 18:      GIAM  MIRCA 2000 0.4547189
## 19:      GIAM     Meier 0.4570899
## 20:     GRIPC     Meier 0.4571718
## 21:      GMIA     GRIPC 0.4574175
## 22:      GIAM  MIRCA OS 0.4621024
## 23:     GRIPC  MIRCA OS 0.4734868
## 24:     GRIPC  MIRCA 2000 0.4804705
## 25:   Nagaraj SPAM2010 0.4954144
## 26: GAEZ+2015   Nagaraj 0.5058373
## 27:  MIRCA OS   Nagaraj 0.5133084
## 28:  MIRCA 2000   Nagaraj 0.5171163
## 29:     Meier   Nagaraj 0.5343552
## 30:      GMIA   Nagaraj 0.5484680
## 31: GAEZ+2015 SPAM2010 0.5702259

```

```
## 32: GAEZ+2015 MIRCA 2000 0.5815287
## 33:      Meier  SPAM2010 0.5853972
## 34:      Meier MIRCA 2000 0.5887003
## 35:      Meier  MIRCA OS 0.5982000
## 36: GAEZ+2015      Meier 0.6025467
## 37: GAEZ+2015  MIRCA OS 0.6276996
## 38: GAEZ+2015      GMIA 0.6362259
## 39: MIRCA 2000  SPAM2010 0.6371558
## 40:  MIRCA OS  SPAM2010 0.6599014
## 41: MIRCA 2000  MIRCA OS 0.6619231
## 42:      GMIA      Meier 0.6752391
## 43:      GMIA MIRCA 2000 0.6842721
## 44:      GMIA  SPAM2010 0.6969499
## 45:      GMIA  MIRCA OS 0.7250649
##      dataset_1 dataset_2 jaccard
##      <char>      <char>      <num>
```

```
jaccard_results[, .(min_jaccard = min(jaccard),
                        median_jaccard = median(jaccard),
                        mean_jaccard = mean(jaccard),
                        max_jaccard = max(jaccard))]
```

```
##      min_jaccard median_jaccard mean_jaccard max_jaccard
##      <num>          <num>          <num>          <num>
## 1:    0.3018328      0.4734868      0.5021623      0.7250649
```

```
# results:
```

```
# 1: both datasets identify the same hotspots
```

```
# 0: datasets disagree completely
```

```
# 0.5: half overlap, half disagreement
```

```
# No pair of global irrigation datasets agrees strongly on where
# irrigation hotspots are.
```

```
# Weak agreement (0.35 - 0.45):
```

```
# -----
```

```
# one dataset says "this is a core irrigated region",
```

```
# the other dataset says "no, it is not".
```

```
# Moderate agreement (0.5-0.65)
```

```
# -----
```

```
# one third to one half of hotspots do not overlap
```

```
# High agreement (0.7 - 0.8)
```

```
# -----
```

```
# High agreement only between specific subsets because of shared lineage
```

```
# (GMIA and Meier; MIRCA2000 and Meier, etc)
```

4.2 National rankings

```
# NATIONAL RANKINGS USING ALL CELLS #####

# Country totals & ranks .-----

country_totals_all <- compute_country_totals(dt2, new_dataset_names)

# Pairwise rank correlations -----

rank_corr_all <- compute_rank_correlations(country_totals_all)
rank_corr_all[order(-spearman)]
```

##	dataset_1	dataset_2	kendall	spearman
##	<char>	<char>	<num>	<num>
## 1:	MIRCA 2000	MIRCA OS	0.8869269	0.9792934
## 2:	GMIA	MIRCA 2000	0.8899580	0.9764568
## 3:	GMIA	MIRCA OS	0.8782725	0.9726811
## 4:	Meier	MIRCA 2000	0.8613228	0.9701687
## 5:	GMIA	Meier	0.8605534	0.9678578
## 6:	GMIA	LUH2	0.8796674	0.9675411
## 7:	Meier	MIRCA OS	0.8402259	0.9608881
## 8:	GMIA	SPAM2010	0.8229857	0.9543192
## 9:	LUH2	MIRCA 2000	0.8326323	0.9521371
## 10:	LUH2	MIRCA OS	0.8306333	0.9516782
## 11:	MIRCA OS	SPAM2010	0.8140963	0.9469802
## 12:	LUH2	SPAM2010	0.8126740	0.9454728
## 13:	MIRCA 2000	SPAM2010	0.8033644	0.9443709
## 14:	GAEZ+2015	Meier	0.8019412	0.9432197
## 15:	GAEZ+2015	MIRCA 2000	0.8032595	0.9423526
## 16:	Meier	SPAM2010	0.7935570	0.9393322
## 17:	LUH2	Meier	0.8078651	0.9361524
## 18:	GAEZ+2015	MIRCA OS	0.7958000	0.9351266
## 19:	GAEZ+2015	GMIA	0.7845397	0.9275645
## 20:	Meier	Nagaraj	0.7802484	0.9145597
## 21:	MIRCA 2000	Nagaraj	0.7657801	0.9093070
## 22:	GMIA	Nagaraj	0.7570546	0.9069830
## 23:	GIAM	Meier	0.7434964	0.9066388
## 24:	GAEZ+2015	LUH2	0.7415172	0.9043953
## 25:	MIRCA OS	Nagaraj	0.7454631	0.9032900
## 26:	GAEZ+2015	SPAM2010	0.7309457	0.9002531
## 27:	GIAM	MIRCA 2000	0.7256531	0.8932012
## 28:	Nagaraj	SPAM2010	0.7171736	0.8807849
## 29:	GIAM	Nagaraj	0.7057269	0.8777524
## 30:	GAEZ+2015	Nagaraj	0.7107326	0.8765867
## 31:	GIAM	GMIA	0.7023198	0.8763834
## 32:	GMIA	GRIPC	0.6971987	0.8749677
## 33:	GRIPC	MIRCA OS	0.6966780	0.8748337

```
## 34:      GRIPC  MIRCA 2000 0.6958898 0.8692777
## 35:      GIAM   MIRCA OS 0.6917915 0.8674594
## 36:      GRIPC      Meier 0.6927134 0.8673210
## 37:      GIAM   SPAM2010 0.6834225 0.8653044
## 38:      GRIPC   SPAM2010 0.6796605 0.8649194
## 39:      LUH2    Nagaraj 0.7021109 0.8635624
## 40:  GAEZ+2015      GIAM 0.6798635 0.8568604
## 41:      GRIPC      LUH2 0.6783125 0.8541425
## 42:      GIAM      LUH2 0.6694764 0.8450648
## 43:      GRIPC   Nagaraj 0.6514905 0.8300450
## 44:  GAEZ+2015      GRIPC 0.6442363 0.8237471
## 45:      GIAM      GRIPC 0.6289032 0.8063916
##      dataset_1 dataset_2 kendall spearman
##      <char>      <char>      <num>      <num>
```

```
# Pairwise top-N overlap (e.g. top 20 irrigators) -----
```

```
topN_overlap_all <- compute_topN_overlap(country_totals_all,
                                         dataset_names = new_dataset_names,
                                         N = 20)
topN_overlap_all[order(-jaccard_topN)]
```

```
##      dataset_1 dataset_2      N intersection_N jaccard_topN
##      <char>      <char> <num>      <int>      <num>
##  1:      GMIA      LUH2    20          19    0.9047619
##  2:      GMIA  MIRCA 2000    20          19    0.9047619
##  3:      LUH2      Meier    20          19    0.9047619
##  4:      GIAM  MIRCA 2000    20          18    0.8181818
##  5:      GMIA      Meier    20          18    0.8181818
##  6:      LUH2  MIRCA 2000    20          18    0.8181818
##  7:      Meier  MIRCA 2000    20          18    0.8181818
##  8:      Meier   MIRCA OS    20          18    0.8181818
##  9:      GIAM      GMIA    20          17    0.7391304
## 10:      GIAM      LUH2    20          17    0.7391304
## 11:      GIAM      Meier    20          17    0.7391304
## 12:      GMIA   MIRCA OS    20          17    0.7391304
## 13:      LUH2   MIRCA OS    20          17    0.7391304
## 14:      Meier   Nagaraj    20          17    0.7391304
## 15:      Meier   SPAM2010    20          17    0.7391304
## 16:  MIRCA 2000   MIRCA OS    20          17    0.7391304
## 17:  MIRCA 2000   Nagaraj    20          17    0.7391304
## 18:   MIRCA OS   SPAM2010    20          17    0.7391304
## 19:  GAEZ+2015      GRIPC    20          16    0.6666667
## 20:      GIAM      GRIPC    20          16    0.6666667
## 21:      GIAM   MIRCA OS    20          16    0.6666667
## 22:      GIAM   Nagaraj    20          16    0.6666667
## 23:      GIAM   SPAM2010    20          16    0.6666667
## 24:      GMIA   Nagaraj    20          16    0.6666667
```

```
## 25:      GRIPC MIRCA 2000      20      16      0.6666667
## 26:      LUH2   Nagaraj  20      16      0.6666667
## 27:      LUH2   SPAM2010  20      16      0.6666667
## 28: MIRCA 2000   SPAM2010  20      16      0.6666667
## 29:      MIRCA OS   Nagaraj  20      16      0.6666667
## 30:      Nagaraj SPAM2010  20      16      0.6666667
## 31: GAEZ+2015 MIRCA 2000  20      15      0.6000000
## 32:      GMIA      GRIPC  20      15      0.6000000
## 33:      GMIA      SPAM2010  20      15      0.6000000
## 34:      GRIPC      LUH2  20      15      0.6000000
## 35:      GRIPC      Meier  20      15      0.6000000
## 36:      GRIPC      MIRCA OS  20      15      0.6000000
## 37: GAEZ+2015      GIAM  20      14      0.5384615
## 38: GAEZ+2015      GMIA  20      14      0.5384615
## 39: GAEZ+2015      MIRCA OS  20      14      0.5384615
## 40: GAEZ+2015      Nagaraj  20      14      0.5384615
## 41:      GRIPC      SPAM2010  20      14      0.5384615
## 42: GAEZ+2015      LUH2  20      13      0.4814815
## 43: GAEZ+2015      Meier  20      13      0.4814815
## 44:      GRIPC      Nagaraj  20      13      0.4814815
## 45: GAEZ+2015      SPAM2010  20      12      0.4285714
##      dataset_1 dataset_2      N intersection_N jaccard_topN
##      <char>      <char> <num>      <int>      <num>
```

```
# NATIONAL RANKINGS USING ONLY "IDENTIFIABLE" CELLS #####
```

```
# Only areas that agree -----
```

```
dt_ident <- dt2[is.na(tau_max_ha)]
```

```
# Country totals & ranks on identifiable cells only -----
```

```
country_totals_ident <- compute_country_totals(dt_ident, new_dataset_names)
```

```
# Pairwise rank correlations (identifiable-only) -----
```

```
rank_corr_ident <- compute_rank_correlations(country_totals_ident)
rank_corr_ident[order(-spearman)]
```

```
##      dataset_1 dataset_2 kendall spearman
##      <char>      <char>      <num>      <num>
## 1:      GMIA      MIRCA OS 0.8835329 0.9783609
## 2:      GMIA MIRCA 2000 0.8814086 0.9765933
## 3:      GMIA      SPAM2010 0.8545396 0.9657292
## 4: MIRCA 2000      MIRCA OS 0.8595946 0.9652307
## 5:      Meier MIRCA 2000 0.8459998 0.9635799
## 6:      GMIA      Meier 0.8436327 0.9596853
## 7:      MIRCA OS      SPAM2010 0.8333674 0.9594505
```

```

## 8:      LUH2      Nagaraj 0.8977533 0.9561767
## 9: MIRCA 2000    SPAM2010 0.8250296 0.9536111
## 10: GAEZ+2015 MIRCA 2000 0.8187728 0.9535817
## 11: GAEZ+2015    MIRCA OS 0.8204898 0.9521117
## 12: GAEZ+2015      GMIA 0.8146970 0.9500532
## 13:      Meier    SPAM2010 0.8096173 0.9466166
## 14:      Meier    MIRCA OS 0.8119047 0.9429633
## 15: GAEZ+2015      Meier 0.7973006 0.9353605
## 16: GAEZ+2015    SPAM2010 0.7717534 0.9270248
## 17:      GIAM MIRCA 2000 0.7717169 0.9150786
## 18:      GIAM    MIRCA OS 0.7601485 0.9139648
## 19:      GIAM    SPAM2010 0.7645711 0.9139506
## 20:      GIAM      Meier 0.7714894 0.9111299
## 21:      GIAM      GMIA 0.7609975 0.9067002
## 22:      GIAM    Nagaraj 0.8031987 0.9044408
## 23:      Meier    Nagaraj 0.7968289 0.8993616
## 24: GAEZ+2015      GIAM 0.7402983 0.8991512
## 25:    Nagaraj    SPAM2010 0.7761437 0.8971564
## 26:      GMIA    Nagaraj 0.7818313 0.8961145
## 27:    MIRCA OS    Nagaraj 0.7756057 0.8950570
## 28: MIRCA 2000    Nagaraj 0.7814858 0.8935640
## 29:      GMIA      LUH2 0.7711712 0.8901905
## 30:      LUH2 MIRCA 2000 0.7701480 0.8869588
## 31:    GRIPC    Nagaraj 0.7656340 0.8864691
## 32:      LUH2      Meier 0.7698723 0.8856886
## 33:      LUH2    MIRCA OS 0.7580659 0.8823827
## 34:    GRIPC    SPAM2010 0.7197618 0.8816127
## 35:      GMIA    GRIPC 0.7205550 0.8809232
## 36:      LUH2    SPAM2010 0.7535114 0.8787780
## 37: GAEZ+2015    Nagaraj 0.7444534 0.8763956
## 38:    GRIPC      Meier 0.7150236 0.8731797
## 39:    GRIPC MIRCA 2000 0.7172403 0.8706369
## 40:      GIAM    GRIPC 0.7091910 0.8680663
## 41:    GRIPC      LUH2 0.7301817 0.8674414
## 42:      GIAM      LUH2 0.7445265 0.8669448
## 43:    GRIPC    MIRCA OS 0.7031575 0.8664374
## 44: GAEZ+2015      LUH2 0.7097899 0.8530376
## 45: GAEZ+2015    GRIPC 0.6858624 0.8455006
##      dataset_1 dataset_2 kendall spearman
##      <char>      <char>      <num>      <num>

```

```

# Pairwise top-N overlap (identifiable-only) -----

```

```

topN_overlap_ident <- compute_topN_overlap(country_totals_ident,
                                           dataset_names = new_dataset_names,
                                           N = 20)
topN_overlap_ident[order(-jaccard_topN)]

```

##	dataset_1	dataset_2	N	intersection_N	jaccard_topN
##	<char>	<char>	<num>	<int>	<num>
## 1:	GIAM	GMIA	20	19	0.9047619
## 2:	GIAM	Meier	20	19	0.9047619
## 3:	GIAM	Nagaraj	20	19	0.9047619
## 4:	GMIA	MIRCA 2000	20	19	0.9047619
## 5:	GMIA	Nagaraj	20	19	0.9047619
## 6:	Meier	MIRCA OS	20	19	0.9047619
## 7:	Nagaraj	SPAM2010	20	19	0.9047619
## 8:	GAEZ+2015	MIRCA OS	20	18	0.8181818
## 9:	GIAM	LUH2	20	18	0.8181818
## 10:	GIAM	MIRCA 2000	20	18	0.8181818
## 11:	GIAM	MIRCA OS	20	18	0.8181818
## 12:	GIAM	SPAM2010	20	18	0.8181818
## 13:	GMIA	LUH2	20	18	0.8181818
## 14:	GMIA	Meier	20	18	0.8181818
## 15:	GMIA	MIRCA OS	20	18	0.8181818
## 16:	GMIA	SPAM2010	20	18	0.8181818
## 17:	GRIPC	LUH2	20	18	0.8181818
## 18:	LUH2	Nagaraj	20	18	0.8181818
## 19:	LUH2	SPAM2010	20	18	0.8181818
## 20:	Meier	MIRCA 2000	20	18	0.8181818
## 21:	Meier	Nagaraj	20	18	0.8181818
## 22:	Meier	SPAM2010	20	18	0.8181818
## 23:	MIRCA 2000	MIRCA OS	20	18	0.8181818
## 24:	MIRCA 2000	Nagaraj	20	18	0.8181818
## 25:	MIRCA OS	Nagaraj	20	18	0.8181818
## 26:	MIRCA OS	SPAM2010	20	18	0.8181818
## 27:	GAEZ+2015	Meier	20	17	0.7391304
## 28:	GAEZ+2015	Nagaraj	20	17	0.7391304
## 29:	GAEZ+2015	SPAM2010	20	17	0.7391304
## 30:	GIAM	GRIPC	20	17	0.7391304
## 31:	GMIA	GRIPC	20	17	0.7391304
## 32:	GRIPC	Meier	20	17	0.7391304
## 33:	GRIPC	MIRCA 2000	20	17	0.7391304
## 34:	GRIPC	Nagaraj	20	17	0.7391304
## 35:	GRIPC	SPAM2010	20	17	0.7391304
## 36:	LUH2	Meier	20	17	0.7391304
## 37:	LUH2	MIRCA 2000	20	17	0.7391304
## 38:	MIRCA 2000	SPAM2010	20	17	0.7391304
## 39:	GAEZ+2015	GIAM	20	16	0.6666667
## 40:	GAEZ+2015	GMIA	20	16	0.6666667
## 41:	GAEZ+2015	GRIPC	20	16	0.6666667
## 42:	GAEZ+2015	MIRCA 2000	20	16	0.6666667
## 43:	GRIPC	MIRCA OS	20	16	0.6666667
## 44:	LUH2	MIRCA OS	20	16	0.6666667
## 45:	GAEZ+2015	LUH2	20	15	0.6000000
##	dataset_1	dataset_2	N	intersection_N	jaccard_topN


```

##           <char>      <char> <num>           <int>           <num>
# HOW DO RANKINGS CHANGE WHEN NON-IDENTIFIABLE CELLS ARE MASKED? #####

# How much does each country's rank shift for each dataset? -----

rank_change <- merge(
  country_totals_all[, .(country, dataset, rank_all = rank)],
  country_totals_ident[, .(country, dataset, rank_ident = rank)],
  by = c("country", "dataset"),
  all.x = TRUE) %>%
  .[, rank_shift := rank_ident - rank_all]

rank_change

# PLOT RESULTS #####

# Arrange data -----

rank_corr_all[, pair := paste(dataset_1, dataset_2, sep = "-")]
rank_corr_ident[, pair := paste(dataset_1, dataset_2, sep = "-")]

rank_corr_long <- rbind(rank_corr_all[, .(pair, spearman, mode = "All cells")],
  rank_corr_ident[, .(pair, spearman, mode = "Identifiable cells")])

# order pairs by Spearman (all cells) so the y-axis is readable -----

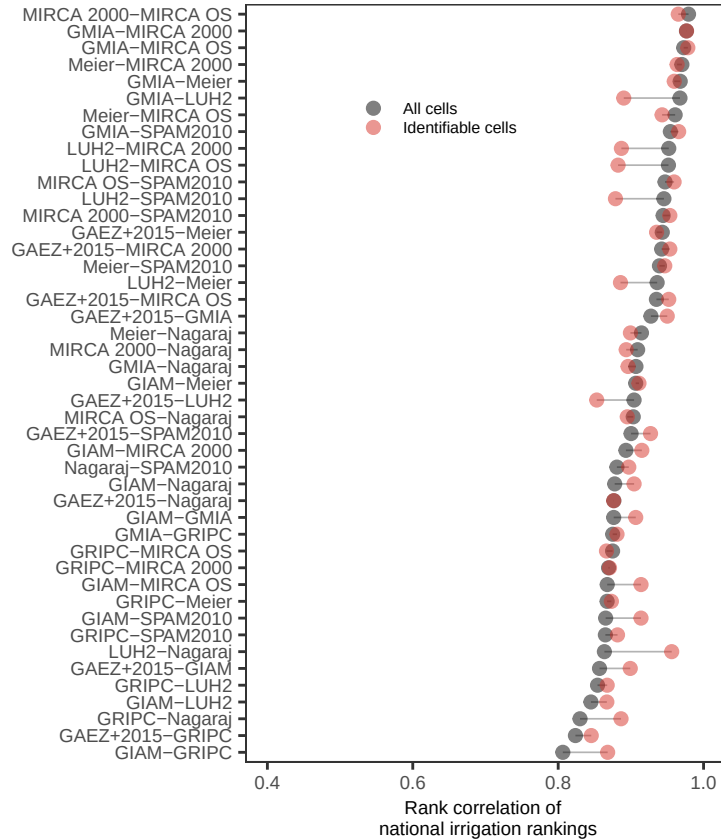
pair_order <- rank_corr_all[order(spearman), pair]
rank_corr_long[, pair := factor(pair, levels = pair_order)]

# Plot rank correlation -----

p_rankcorr <- ggplot(rank_corr_long, aes(spearman, pair)) +
  geom_segment(data = dcast(rank_corr_long, pair ~ mode, value.var = "spearman"),
    aes(x = `All cells`, xend = `Identifiable cells`,
      y = pair, yend = pair),
    inherit.aes = FALSE,
    colour = "grey70", linewidth = 0.3) +
  geom_point(aes(colour = mode), size = 2, alpha = 0.5) +
  scale_x_continuous(limits = c(0.4, 1.0)) +
  scale_color_manual(values = c("Identifiable cells" = "#d73027",
    "All cells" = "black")) +
  labs(x = "Rank correlation of \n national irrigation rankings",
    y = "",
    colour = NULL) +
  theme_AP() +
  theme(legend.position = c(0.4, 0.85))

```

p_rankcorr



```
# Only for top N countries -----

topN_overlap_all[, pair:= paste(dataset_1, dataset_2, sep = "-")]
topN_overlap_ident[, pair:= paste(dataset_1, dataset_2, sep = "-")]

topN_long <- rbind(topN_overlap_all[, .(pair, jaccard_topN, mode = "All cells")],
                  topN_overlap_ident[, .(pair, jaccard_topN, mode = "Identifiable cells")])

# order by All-cells Jaccard for consistency -----

pair_order_topN <- topN_overlap_all[order(jaccard_topN), pair]
topN_long[, pair:= factor(pair, levels = pair_order_topN)]

# Plot -----

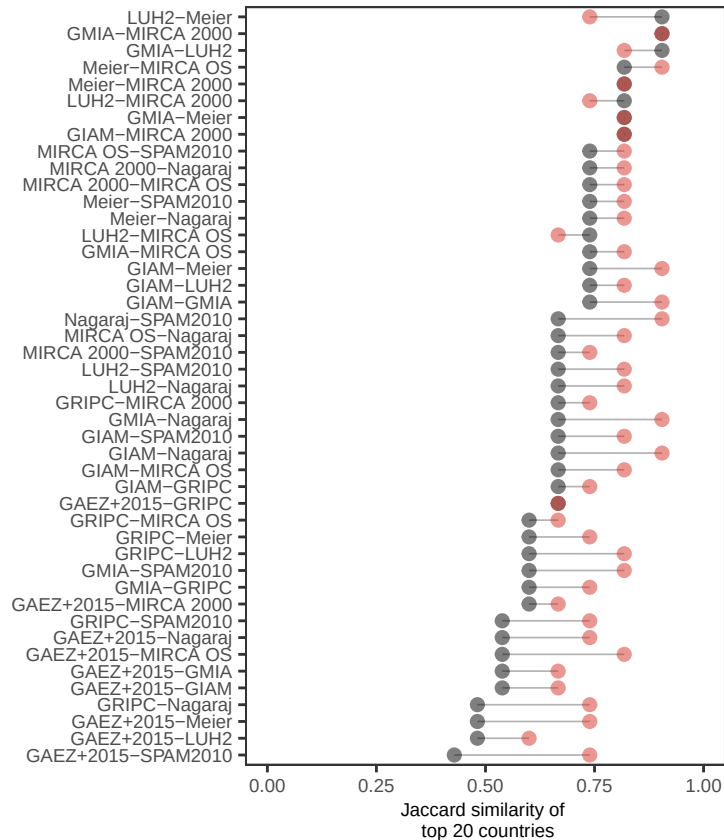
p_topN <- ggplot(topN_long, aes(x = jaccard_topN, y = pair)) +
  geom_segment(data = dcast(topN_long, pair ~ mode, value.var = "jaccard_topN"),
              aes(x = `All cells`, xend = `Identifiable cells`,
                  y = pair, yend = pair), inherit.aes = FALSE, colour = "grey70",
              linewidth = 0.3) +
  geom_point(aes(colour = mode), size = 2, alpha = 0.5) +
  scale_x_continuous(limits = c(0, 1)) +
```

```

scale_color_manual(values = c("Identifiable cells" = "#d73027",
                              "All cells" = "black")) +
labs(x = "Jaccard similarity of \n top 20 countries",
     y = "", colour = NULL) +
theme_AP() +
theme(legend.position = "none")

```

p_topN



MERGE

(a) Spearman rank correlation of national total irrigated area rankings for all
pairwise combinations of datasets. Black points show correlations computed across
all grid cells, while red points show correlations restricted to identifiable
cells (i.e., excluding cells where datasets fundamentally disagree on the
presence of irrigation). Correlations are generally high (0.7–1.0),
indicating broad agreement in country-level rankings. In some cases removing
disputed cells increases the correlation (red dot to the right of black dot),
while in others it does not (red dot to the left of black dot)

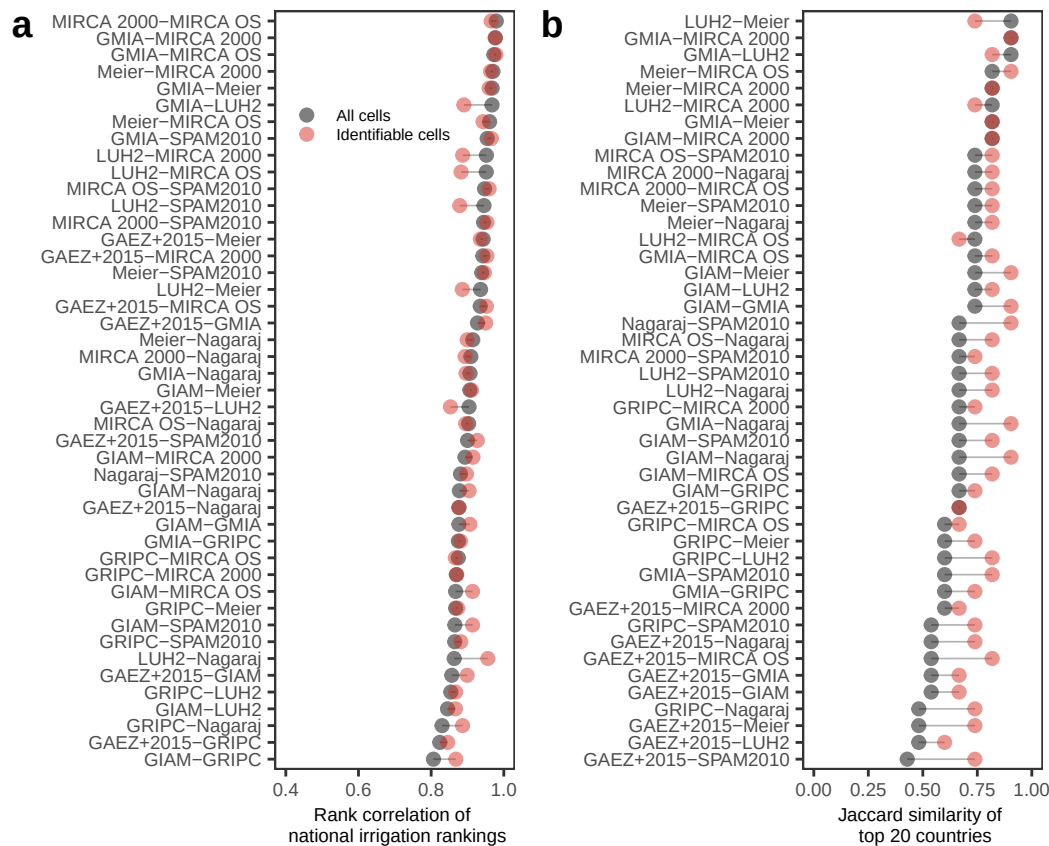
#

(b) Jaccard similarity of the top 20 irrigated countries for each dataset pair.
Black points again denote results using all cells and red points identifiable cells
only. Most of the time focusing only on identifiable cells increases the

similarity; meaning that the top 20 become more similar across datasets. In other words, once disputed cells are removed there is a more coherent "core" of globally dominant irrigation countries. Possibly this is because the strongest disagreement disproportionately affect marginal or mid-rank countries that occasionally enter or leave the top 20 depending on dataset.

SD: removing disagreeing cells stabilizes who belongs in the top twenty, but not necessarily the full ranking hierarchy across all countries.

```
plot_grid(p_rankcorr, p_topN, ncol = 2, labels = "auto")
```



4.3 Shift in country ranks

SHIFT IN COUNTRY RANKS

Total rank shift -----

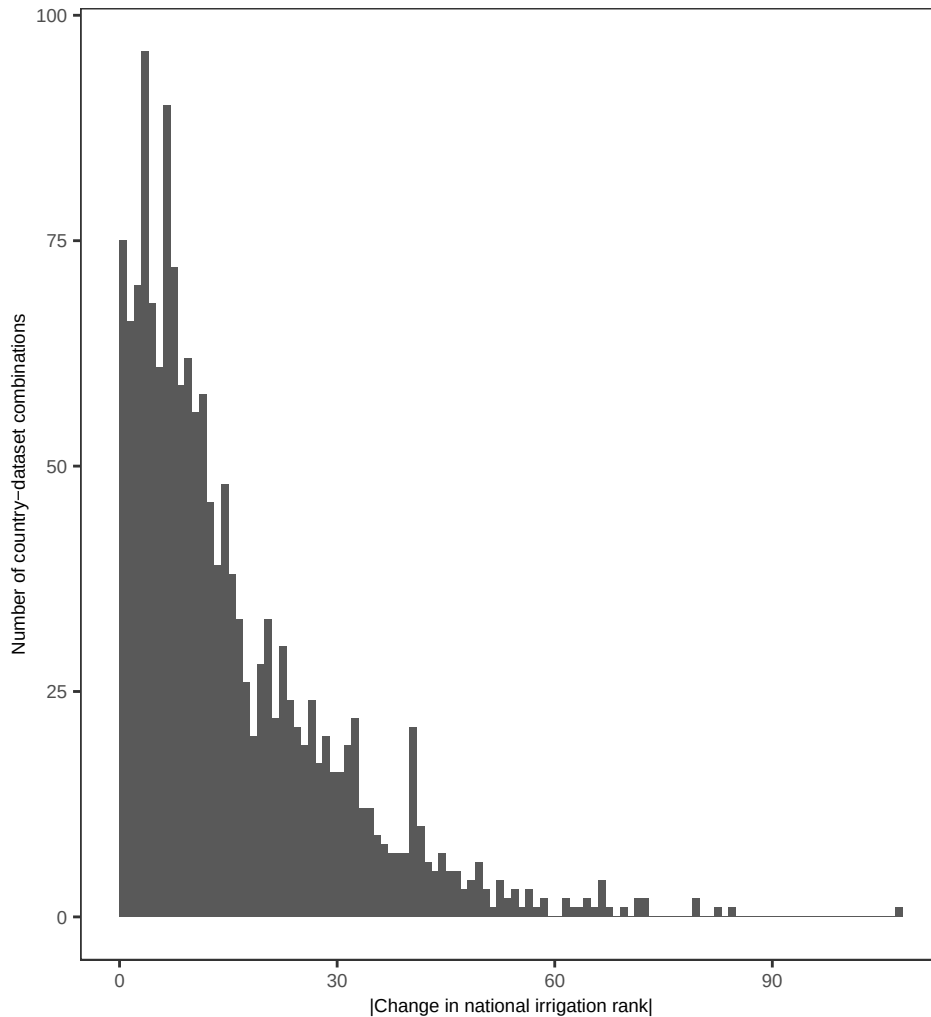
```
rank_change[, abs_shift:= abs(rank_shift)]  
rank_change_clean <- rank_change[!is.na(abs_shift)]
```

Histogram of rank shift -----

```
p_rankshift_hist <- ggplot(rank_change_clean, aes(abs_shift)) +
```

```
geom_histogram(binwidth = 1, boundary = 0, closed = "left") +
labs(x = "|Change in national irrigation rank|",
     y = "Number of country-dataset combinations") +
theme_AP()
```

p_rankshift_hist



```
## Cumulative share -----

rank_shift_stats <- rank_change_clean[, .(share_ge_1 = mean(abs_shift >= 1),
                                           share_ge_5 = mean(abs_shift >= 5),
                                           share_ge_10 = mean(abs_shift >= 10),
                                           share_ge_20 = mean(abs_shift >= 20),
                                           share_ge_50 = mean(abs_shift >= 50))]

rank_shift_stats
```

```
##   share_ge_1 share_ge_5 share_ge_10 share_ge_20 share_ge_50
##         <num>         <num>         <num>         <num>         <num>
```

```
## 1: 0.9522293 0.7611465 0.5420382 0.2923567 0.02675159
# 98.4% of country-dataset combinations change rank by at least 1 position.
# No country keeps the same rank once we enforce identifiability.
#
# 90% of countries move by at least 5 rank positions
#
# 71% of country-dataset combinations move by 10 ranks or more

# FOCUS ON THE SO-CALLED "TOP IRRIGATORS" #####

number_countries <- 20

# Countries consistently classified as top irrigators across datasets -----

top10_all <- country_totals_all[order(rank), .SD[1:number_countries], dataset] %>%
  .[, .N, by = country] %>%
  .[!country == ""] %>%
  .[N > 5]

top10_all
```

```
##          country      N
##          <char> <int>
## 1:         China     10
## 2:         India     10
## 3: United States     10
## 4:      Pakistan     10
## 5:         Iran      8
## 6:        Mexico     10
## 7:        Brazil     10
## 8:         Egypt      8
## 9:         Turkey     10
## 10:  Uzbekistan     10
## 11:         Italy      8
## 12:         Spain     10
## 13:        Russia      6
## 14:  Kazakhstan      8
## 15:        Thailand      8
## 16:  Bangladesh      8
## 17:        Vietnam      9
## 18:   Indonesia      8
## 19:         Iraq      6
```

```
# Also with the median -----

top10_median <- country_totals_all[, .(median_rank = median(rank)), country] %>%
  .[order(median_rank)] %>%
  .[1:number_countries]
```

```
top10_median
```

```
##           country median_rank
##           <char>         <num>
##  1:           China          1.0
##  2:           India          2.0
##  3:   United States          3.0
##  4:           Pakistan        4.0
##  5:           Iran           5.0
##  6:           Mexico          7.0
##  7:     Indonesia           9.0
##  8:           Thailand         9.0
##  9:     Bangladesh        11.0
## 10:           Brazil        11.5
## 11:           Vietnam        12.0
## 12:           Turkey        12.5
## 13:     Uzbekistan        12.5
## 14:           Spain         13.5
## 15:           Egypt         15.0
## 16:           Italy          15.5
## 17:           Russia         15.5
## 18:     Kazakhstan        18.5
## 19:           Iraq          20.0
## 20: Myanmar (Burma)        20.0
##           country median_rank
##           <char>         <num>
```

```
# See change of ranks in countries when only confirmed irrigation is
# accounted for -----

rank_focus <- merge(country_totals_all[, .(country, dataset, rank_all = rank)],
                    country_totals_ident[, .(country, dataset, rank_ident = rank)],
                    by = c("country", "dataset"), all = TRUE)

rank_focus[, rank_shift:= rank_ident - rank_all]
rank_focus
```

```
## Key: <country, dataset>
##           country    dataset rank_all rank_ident rank_shift
##           <char>    <char>    <num>    <num>    <num>
##  1:  Afghanistan GAEZ+2015    23.0        29         6
##  2:  Afghanistan   GIAM     37.0        30        -7
##  3:  Afghanistan   GMIA     24.0        27         3
##  4:  Afghanistan GRIPC     44.0        31        -13
##  5:  Afghanistan  LUH2     21.0        27         6
##  ---
## 1626: Åland Islands MIRCA 2000    161.5         NA         NA
## 1627: Åland Islands  MIRCA OS    161.0         NA         NA
```

```
## 1628: Åland Islands      Meier      150.0      NA      NA
## 1629: Åland Islands      Nagaraj    136.0      NA      NA
## 1630: Åland Islands      SPAM2010   150.0      NA      NA

# Summarize -----

rank_focus_summary <- rank_focus[!is.na(rank_ident), .(median_shift = median(rank_shift, na.rm = TRUE),
max_drop = max(rank_shift, na.rm = TRUE),
max_rise = min(rank_shift, na.rm = TRUE),
share_drop5 = mean(rank_shift >= 5, na.rm = TRUE),
share_drop10 = mean(rank_shift >= 10, na.rm = TRUE)),
country] %>%
.[order(-median_shift)]

# Focus on top 20 irrigation hotspots -----

focus_countries <- top10_median$country
rank_focus_summary[country %in% focus_countries]
```

##	country	median_shift	max_drop	max_rise	share_drop5	share_drop10
## 1:	Brazil	12.5	21	6	1.0	0.9
## 2:	Russia	10.0	21	1	0.8	0.5
## 3:	Indonesia	5.5	12	-2	0.8	0.1
## 4:	Mexico	3.5	7	-1	0.4	0.0
## 5:	Turkey	3.0	9	-3	0.4	0.0
## 6:	Iran	2.5	5	-8	0.1	0.0
## 7:	Kazakhstan	1.0	12	-4	0.3	0.1
## 8:	China	0.0	0	0	0.0	0.0
## 9:	India	0.0	0	0	0.0	0.0
## 10:	Myanmar (Burma)	0.0	4	-4	0.0	0.0
## 11:	Pakistan	0.0	0	-1	0.0	0.0
## 12:	United States	0.0	1	0	0.0	0.0
## 13:	Thailand	-1.0	1	-5	0.0	0.0
## 14:	Spain	-1.5	3	-6	0.0	0.0
## 15:	Italy	-3.0	-1	-8	0.0	0.0
## 16:	Vietnam	-3.0	1	-9	0.0	0.0
## 17:	Bangladesh	-5.0	-3	-14	0.0	0.0
## 18:	Iraq	-5.5	6	-9	0.1	0.0
## 19:	Uzbekistan	-5.5	-4	-11	0.0	0.0
## 20:	Egypt	-7.0	-3	-15	0.0	0.0

```
##          country median_shift max_drop max_rise share_drop5 share_drop10
##          <char>      <num>      <num>      <num>      <num>      <num>

# Positive rank shift: worse rank if only confirmed cells are accounted for.
# Negative rank shift: better rank if only confirmed cells are accounted for.

# - median_shift: median of rank_shift across datasets.
# - max_drop: maximum worsening (largest positive rank_shift).
```



```

# - max_rise: maximum improvement (most negative rank_shift).
# - share_drop5: share of dataset cases with rank_shift >= 5.
# - share_drop10: share with rank_shift >= 10.

# VIETNAM, BANGLADESH, THAILAND, INDONESIA: in 90% of cases, Vietnam's rank gets
# at least 10 places worse. Median drop is c. 40 places. Bangladesh, Indonesia,
# Thailand show similar pattern; median shifts of 26-27 ranks and 90% of cases
# with drops larger than 90 ranks.

# CHINA, INDIA, PAKISTAN, USA: Global rank is invariant to whether we use all cells
# or only agreement cells. Irrigated area dominance is robust.

# ITALY, EGYPT, UZBEKISTAN, IRAQ, SPAIN, KAZAKHSTAN: they
# improve their rank when we restrict to confirmed cells; because when we remove
# ambiguous cells in other countries, these countries become more prominent.

# Focus on the countries whose ranking drops the most -----

rank_focus_summary[order(-median_shift)][1:number_countries]

```

```

##          country median_shift max_drop max_rise share_drop5
##          <char>         <num>    <num>    <num>         <num>
##  1:      Netherlands      64.50    108.0     23.0           1.0
##  2:           Hungary      48.50     62.5     10.0           1.0
##  3:      Guatemala       41.25     72.5    -28.0           0.9
##  4: United Arab Emirates  40.50     79.0    -24.5           0.8
##  5:           Eswatini     39.00     71.0     -3.5           0.9
##  6:      Madagascar      39.00     79.5      8.0           1.0
##  7:             Mali      36.00     57.5    -13.0           0.7
##  8:           Germany      30.50     67.5     18.0           1.0
##  9:           Slovakia      30.50     52.0      1.0           0.9
## 10:           Bulgaria      27.00     52.0    -42.0           0.9
## 11:           Jamaica      27.00     40.0      3.5           0.9
## 12:      Costa Rica       26.50     49.0      7.0           1.0
## 13:      Venezuela       26.50     40.0     11.0           1.0
## 14:           Ethiopia      24.00     71.5      1.0           0.9
## 15:           Nigeria      23.50     65.5      5.0           1.0
## 16:           Ecuador      21.00     33.0     10.0           1.0
## 17:      New Zealand      21.00     31.0    -19.5           0.9
## 18:           Somalia      19.50     66.0    -33.5           0.9
## 19:           Cyprus       19.00     52.0      7.5           1.0
## 20:      Timor-Leste      17.75     48.0     -7.0           0.7
##          country median_shift max_drop max_rise share_drop5
##          <char>         <num>    <num>    <num>         <num>
##  share_drop10
##          <num>
##  1:           1.0

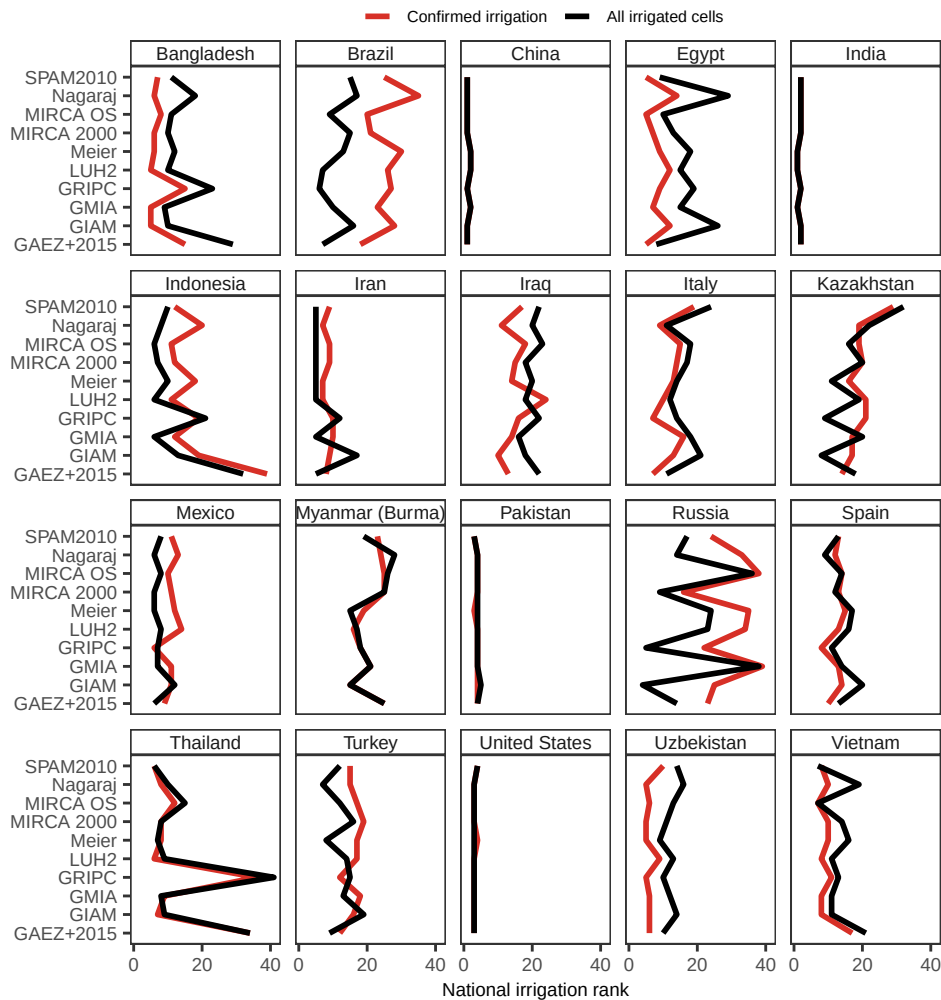
```

```
## 2:      1.0
## 3:      0.9
## 4:      0.8
## 5:      0.8
## 6:      0.9
## 7:      0.7
## 8:      1.0
## 9:      0.8
## 10:     0.9
## 11:     0.8
## 12:     0.9
## 13:     1.0
## 14:     0.9
## 15:     0.8
## 16:     1.0
## 17:     0.8
## 18:     0.9
## 19:     0.8
## 20:     0.7
##      share_drop10
##      <num>
```

```
# Plot the results -----

plot_shift_ranks <- rank_focus[country %in% focus_countries] %>%
  .[, .(country, dataset,
    `All irrigated cells` = rank_all,
    `Confirmed irrigation` = rank_ident)] %>%
  melt(., measure.vars = c("Confirmed irrigation", "All irrigated cells")) %>%
  ggplot(., aes(dataset, value, color = variable,
    group = interaction(country, variable))) +
  geom_line(linewidth = 1) +
  scale_y_reverse() +
  scale_color_manual(values = c("Confirmed irrigation" = "#d73027",
    "All irrigated cells" = "black"),
    name = "") +
  scale_y_continuous(breaks = breaks_pretty(n = 3)) +
  facet_wrap(~country) +
  labs(y = "National irrigation rank", x = NULL) +
  coord_flip() +
  theme_AP() +
  theme(legend.position = "top")
```

```
## Scale for y is already present.
## Adding another scale for y, which will replace the existing scale.
plot_shift_ranks
```



Dashed (red) line much lower than solid (black)
 # - country drops in rank when ambiguity is removed
 # - it looks large only because of non-identifiable cells
 #
 # Red equals black
 # - rank is robust to identifiability
 # - irrigation signal is spatially coherent
 #
 # Red above black
 # - country rises in rank when ambiguity is removed
 # - it was previously masked by others' ambiguous irrigation

If Red and black lines cross across datasets:
 # The ordering of countries changes
 # Rankings are dataset-dependent
 # There is no stable hierarchy
 # Crossings are visual proof of rank instability.

```

# COMPUTE AREA RETAINED AND LOST OF TOP 20 COUNTRIES #####

country_loss <- rbindlist(lapply(new_dataset_names, function(d) {

  dt_country <- dt2[country %in% top10_all$country]

  dt_country[, .(dataset = d,
    total_area = sum(get(d), na.rm = TRUE),
    retained_area = sum(get(d)[is.na(tau_max_ha)], na.rm = TRUE),
    lost_area = sum(get(d)[!is.na(tau_max_ha)], na.rm = TRUE)),
    country]
}))

country_loss[, share_lost := lost_area / total_area]

# Compute mean loss per country across datasets to reorder -----

country_order <- country_loss[, .(mean_share_lost = mean(share_lost, na.rm = TRUE)),
  by = country][order(-mean_share_lost)] # decreasing order

# Set factor levels in that order-----

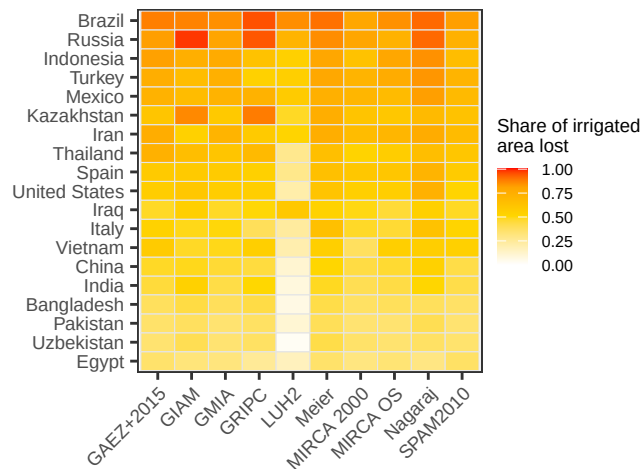
country_loss[, country_f := factor(country, levels = rev(country_order$country))]

# Plot tileplot -----

plot_tileplot <- ggplot(country_loss, aes(dataset, country_f, fill = share_lost)) +
  geom_tile(color = "grey90") +
  scale_fill_gradientn(colours = c("white", "gold", "orange", "red"),
    values = c(0, 0.5, 0.8, 1),
    limits = c(0, 1),
    name = "Share of irrigated\narea lost") +
  scale_y_discrete(name = NULL) +
  theme_AP() +
  labs(x = "", y = "") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1),
    legend.position = "right")

plot_tileplot

```



4.4 Fraction of irrigated areas lost

```
# COLLAPSE ASSESSMENT ACROSS A CONTINUUM OF AGREEMENT #####

dt_share_lost <- dcast(dt, lon + lat + country + code + continent + resolution ~ dataset,
  value.var = "mha")

# Define resolution specific thresholds -----

zero_tol_mha_fun <- function(res) {
  switch(res,
    "0.2deg" = 400 / 1e6, # 0.0004 Mha
    "0.4deg" = 1600 / 1e6, # 0.0016 Mha
    "1deg" = 10000 / 1e6, # 0.01 Mha
    stop("Unknown resolution: ", res)
  )
}

# Make copy just in case -----

dt_pres <- copy(dt_share_lost)

# Apply resolution-specific thresholds -----

dt_pres[, (dataset_names) := {
  tol_mha <- zero_tol_mha_fun(resolution[1])
  lapply(.SD, function(x) as.integer(x >= tol_mha))
},
  resolution,
  .SDcols = dataset_names]

# How many datasets say "irrigated" per cell -----
```

```

dt_pres[, n_pos:= rowSums(.SD, na.rm = TRUE), .SDcols = dataset_names]

n_datasets <- length(dataset_names) # should be 10

# Melt for k-of-n agreement -----

grid_long <- melt(dt_pres, id.vars = c("lon","lat","country","code",
    "continent","resolution","n_pos"),
    measure.vars = dataset_names,
    variable.name = "dataset",
    value.name = "present")

# Run calculations for multiple k (10 down to 5) -----

ks <- n_datasets:1
loss_by_k <- rbindlist(lapply(ks, compute_loss_for_k), use.names = TRUE)

# Summarize results -----

country_loss_by_k <- loss_by_k[, .(min = min(share_lost, na.rm = TRUE),
    max = max(share_lost, na.rm = TRUE),
    mean = mean(share_lost, na.rm = TRUE)),
    .(country, k)] %>%
    .[, allowed_disagree:= 10 - k]

# PLOT RESULTS: THE GLOBAL COLLAPSE CURVE #####

global_k <- country_loss_by_k[, .(mean_loss = mean(mean, na.rm = TRUE),
    median_loss = median(mean, na.rm = TRUE)), k] %>%
    .[order(-k)] %>%
    .[, allowed_disagree:= 10 - k]

print(global_k)

```

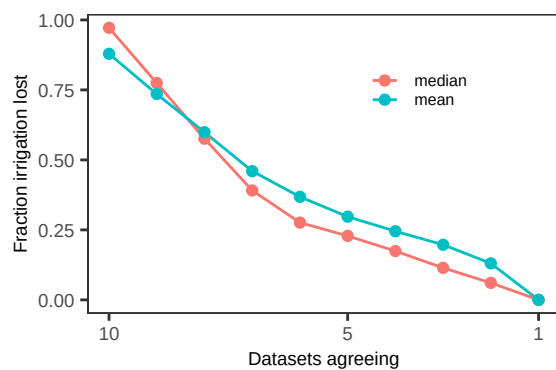
##	k	mean_loss	median_loss	allowed_disagree	
##	<int>	<num>	<num>	<num>	
##	1:	10	0.8789739	0.97198163	0
##	2:	9	0.7353161	0.77453888	1
##	3:	8	0.5989455	0.57571490	2
##	4:	7	0.4597504	0.39078182	3
##	5:	6	0.3682308	0.27614700	4
##	6:	5	0.2972035	0.22858468	5
##	7:	4	0.2451639	0.17430556	6
##	8:	3	0.1969250	0.11438299	7
##	9:	2	0.1304492	0.06068063	8
##	10:	1	0.0000000	0.00000000	9

```

plot_global_k <- global_k %>%
  melt(., measure.vars = c("median_loss", "mean_loss")) %>%
  ggplot(., aes(k, value, color = variable)) +
  geom_line() +
  geom_point() +
  scale_color_discrete(labels = c(median_loss = "median", mean_loss = "mean"),
                        name = "") +
  labs(x = "Datasets agreeing", y = "Fraction irrigation lost") +
  theme_AP() +
  scale_x_reverse(breaks = c(10, 5, 1)) +
  theme(legend.position = c(0.7, 0.8))

```

plot_global_k



COUNTRY-LEVEL COLLAPSE PROFILES

Optional: keep only countries with mean loss < 1 at k=5

```

collapse_profiles <- country_loss_by_k[country %in% top10_all$country] %>%
  .[, allowed_disagree:= 10 - k]

```

Under unanimity

```

tmp <- collapse_profiles[k == 10 & country %in% top10_all$country] %>%
  .[order(-mean)]

```

tmp

##	country	k	min	max	mean	allowed_disagree
##	<char>	<int>	<num>	<num>	<num>	<num>
## 1:	Brazil	10	0.88064269	0.9533493	0.9275510	0
## 2:	Russia	10	0.75986842	0.9877311	0.9062287	0
## 3:	Indonesia	10	0.69328494	0.9186718	0.8443967	0
## 4:	Mexico	10	0.73804971	0.8904438	0.8298258	0
## 5:	Kazakhstan	10	0.65354331	0.9279574	0.7943901	0
## 6:	Turkey	10	0.61011905	0.8732463	0.7883040	0
## 7:	United States	10	0.47315809	0.8526610	0.7551067	0
## 8:	Iran	10	0.53748783	0.8364325	0.7399565	0

```
## 9:      Thailand      10 0.42953020 0.7826087 0.6993478      0
## 10:       Spain      10 0.36942675 0.7899576 0.6815685      0
## 11:       Italy      10 0.46445498 0.7618546 0.6523922      0
## 12:     Vietnam      10 0.38235294 0.7169118 0.6246655      0
## 13:       Iraq      10 0.45000000 0.6653144 0.5841619      0
## 14:       China      10 0.30208040 0.6298919 0.5560644      0
## 15:       India      10 0.15740741 0.6092448 0.5245148      0
## 16:    Pakistan      10 0.37068063 0.5587372 0.4671962      0
## 17:       Egypt      10 0.28909953 0.5575221 0.4548616      0
## 18:   Uzbekistan      10 0.07220217 0.4711934 0.3867142      0
## 19:   Bangladesh      10 0.08823529 0.4213333 0.3637888      0
```

```
tmp[, .(n_above_90 = sum(mean > 0.9),
      n_above_80 = sum(mean > 0.8),
      n_above_50 = sum(mean > 0.5),
      pct_above_90 = 100 * mean(mean > 0.9),
      pct_above_80 = 100 * mean(mean > 0.8),
      pct_above_50 = 100 * mean(mean > 0.5))]
```

```
##      n_above_90 n_above_80 n_above_50 pct_above_90 pct_above_80 pct_above_50
##      <int>      <int>      <int>      <num>      <num>      <num>
## 1:           2           4          15      10.52632      21.05263      78.94737
```

```
# Under the most permissive criteria (5/10)
```

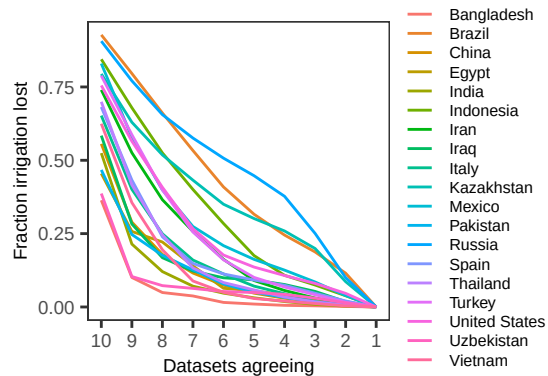
```
collapse_profiles[k == 5 & country %in% top10_all$country] %>%
  .[order(-mean)]
```

```
##      country      k      min      max      mean allowed_disagree
##      <char> <int>      <num>      <num>      <num>      <num>
## 1:      Russia      5 0.028199566 0.88873950 0.447484937      5
## 2:      Brazil      5 0.069400631 0.62105969 0.316117863      5
## 3: Kazakhstan      5 0.017716535 0.77691363 0.301456424      5
## 4:      Indonesia      5 0.040892193 0.30895091 0.175197788      5
## 5:       Mexico      5 0.031192661 0.35785686 0.162330836      5
## 6: United States      5 0.035277325 0.32532615 0.136061138      5
## 7:       Turkey      5 0.013750955 0.22883406 0.100755304      5
## 8:       Iraq      5 0.000000000 0.32962138 0.092350998      5
## 9:       Spain      5 0.005928854 0.25742574 0.091921916      5
## 10:      Iran      5 0.041157294 0.17183196 0.090127081      5
## 11:      Italy      5 0.001572327 0.20021075 0.071055620      5
## 12:     Thailand      5 0.007352941 0.12020460 0.058142125      5
## 13:      Egypt      5 0.004484305 0.16853933 0.055262476      5
## 14:    Pakistan      5 0.004280822 0.11967695 0.049999775      5
## 15:   Uzbekistan      5 0.000000000 0.11728395 0.049347680      5
## 16:       China      5 0.003233049 0.08483041 0.047069619      5
## 17:       India      5 0.003975265 0.08171255 0.031221534      5
## 18:     Vietnam      5 0.002083333 0.07843137 0.028810410      5
## 19:   Bangladesh      5 0.000000000 0.02840909 0.009946725      5
```



```
plot_collapse_profiles <- ggplot(collapse_profiles, aes(k, mean, group = country,
                                                    color = country)) +
  scale_color_discrete(name = "") +
  geom_line() +
  scale_x_reverse(breaks = c(10:1)) +
  labs(x = "Datasets agreeing", y = "Fraction irrigation lost") +
  theme_AP()
```

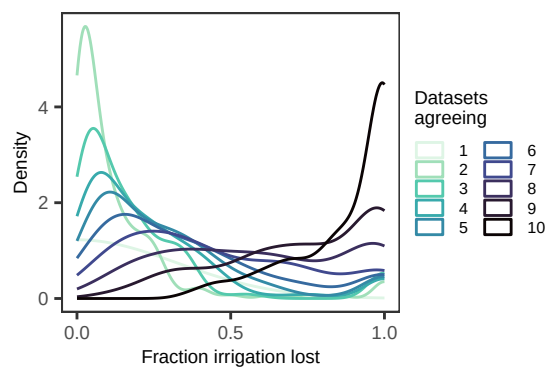
plot_collapse_profiles



Global distribution by k -----

```
plot_curves_country <- ggplot(country_loss_by_k, aes(mean, color = factor(k), group = k)) +
  geom_density() +
  labs(x = "Fraction irrigation lost", y = "Density", color = "Datasets \nagreeing") +
  theme_AP() +
  theme(legend.position = "right") +
  scale_x_continuous(breaks = breaks_pretty(n = 3)) +
  scale_color_viridis_d(option = "mako", direction = -1) +
  guides(colour = guide_legend(ncol = 2))
```

plot_curves_country



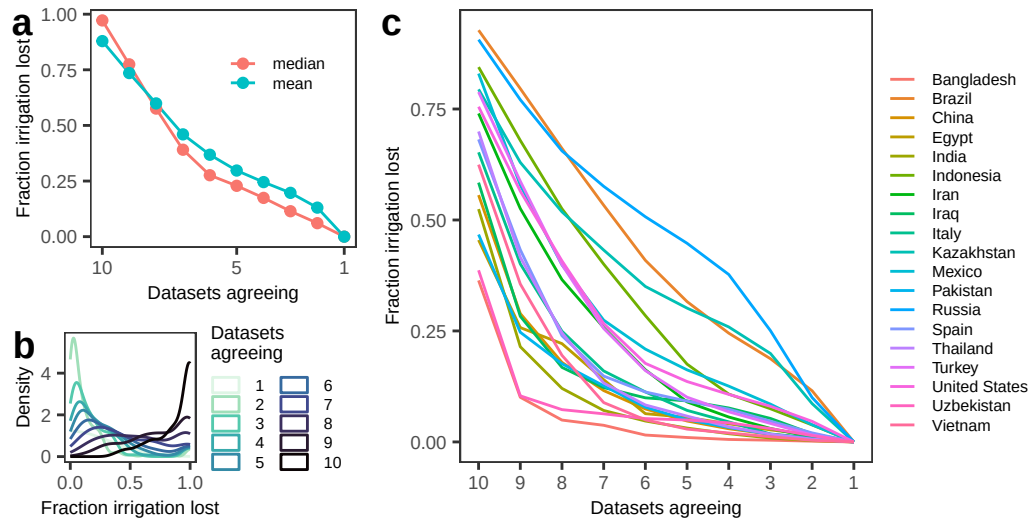
MERGE

```
left <- plot_grid(plot_global_k, plot_curves_country, ncol = 1, labels = "auto",
```

```

rel_heights = c(0.6, 0.4))
plot_grid(left, plot_collapse_profiles, ncol = 2, labels = c("", "c"),
rel_widths = c(0.35, 0.65))

```



PLOTS SHOWING K-OF-10 AGREEMENT

Prepare data -----

```

dt_plot <- dt_pres[, .N, .(resolution, n_pos)]
dt_plot[, frac:= N / sum(N), by = resolution]

```

Fraction of cells in the consensus core (>=9/10) -----

```

dt_plot[order(n_pos)] %>%
  .[n_pos >= 9] %>%
  .[, sum(frac), resolution]

```

```

## resolution      V1
##      <char>      <num>
## 1:      1deg 0.2196331
## 2:      0.4deg 0.1945520
## 3:      0.2deg 0.1620483

```

Fraction of cells irrigated by less than half of the datasets .-----

```

dt_plot[order(n_pos)] %>%
  .[n_pos < 5 & n_pos != 0] %>%
  .[, sum(frac), resolution]

```

```

## resolution      V1
##      <char>      <num>
## 1:      0.2deg 0.3057287
## 2:      0.4deg 0.2735299
## 3:      1deg 0.2205007

```

```

# Compute statistics -----

dt_summary <- dt_plot[, .(total = sum(N),
                          absence = sum(N[n_pos == 0]),
                          contested = sum(N[n_pos >= 1 & n_pos <= 8]),
                          fewer_than_half = sum(N[n_pos < 5 & n_pos > 0]),
                          presence = sum(N[n_pos >= 9])),
                        resolution]

dt_summary[, `:=`(frac_absence = absence / total,
                  frac_contested = contested / total,
                  frac_fewer_than_half = fewer_than_half / total,
                  frac_presence = presence / total)]

dt_summary

##      resolution  total absence contested fewer_than_half presence frac_absence
##      <char>    <int>   <int>    <int>         <int>    <int>         <num>
## 1:      1deg    8068     3473     2823           1779     1772         0.4304660
## 2:     0.2deg  129628   43077    65545          39631    21006         0.3323125
## 3:     0.4deg   39758   14514    17509          10875     7735         0.3650586
##      frac_contested frac_fewer_than_half frac_presence
##                  <num>                <num>         <num>
## 1:      0.3499008           0.2205007       0.2196331
## 2:      0.5056392           0.3057287       0.1620483
## 3:      0.4403894           0.2735299       0.1945520

# Probability that a dataset detects irrigation in a given cell -----

dt_k <- dt_pres[, .N, .(resolution, n_pos)]

dt_prob <- dt_k[, .(mean_k = weighted.mean(n_pos, N),
                  detection_prob = weighted.mean(n_pos, N) / 10),
                resolution]

dt_prob

##      resolution  mean_k detection_prob
##      <char>    <num>         <num>
## 1:      1deg  3.410387       0.3410387
## 2:     0.2deg  3.426945       0.3426945
## 3:     0.4deg  3.498114       0.3498114

# In how many cells do fewer than half of the datasets detect irrigation? -----

dt_weak <- dt_k[, .(total_cells = sum(N),
                    no_irrigation = sum(N[n_pos == 0]),
                    any_irrigation = sum(N[n_pos >= 1]),
                    weak_detection = sum(N[n_pos >= 1 & n_pos <= 5])), resolution]

```

```
dt_weak[, `:=`(frac_weak_all = weak_detection / total_cells,
               frac_weak_given_any = weak_detection / any_irrigation)]
```

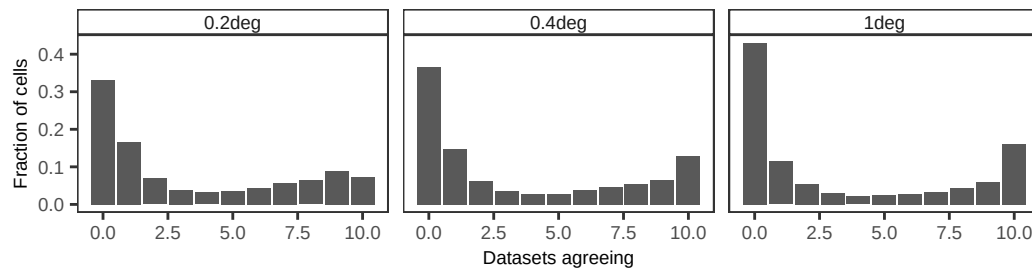
```
dt_weak
```

```
##      resolution total_cells no_irrigation any_irrigation weak_detection
##      <char>      <int>      <int>          <int>          <int>
## 1:      1deg      8068      3473          4595          1977
## 2:      0.2deg    129628    43077        86551         44255
## 3:      0.4deg    39758    14514        25244         11993
##      frac_weak_all frac_weak_given_any
##      <num>          <num>
## 1:      0.2450421      0.4302503
## 2:      0.3414000      0.5113170
## 3:      0.3016500      0.4750832
```

```
# Plot distribution -----
```

```
p1 <- plot_agreement <- ggplot(dt_plot, aes(n_pos, frac)) +
  geom_col() +
  facet_wrap(~resolution) +
  labs(x = "Datasets agreeing",
       y = "Fraction of cells") +
  theme_AP()
```

```
p1
```



```
# Prepare data for cumulative plot -----
```

```
dt_ecdf <- dt_pres[, .N, .(resolution, n_pos)]
dt_ecdf[, frac := N / sum(N), resolution]
setorder(dt_ecdf, resolution, n_pos)
dt_ecdf[, cdf := cumsum(frac), resolution]

dt_ecdf[, resolution := factor(resolution, levels = c("0.2deg", "0.4deg", "1deg"))]
```

```
# Plot -----
```

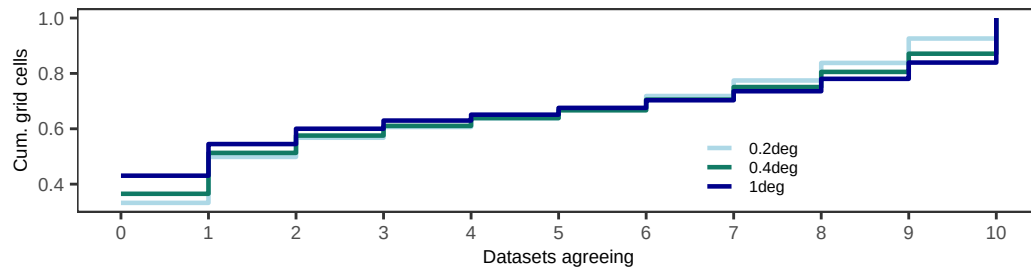
```
p2 <- ggplot(dt_ecdf, aes(n_pos, cdf, colour = resolution)) +
  geom_step(linewidth = 0.8) +
```

```

scale_x_continuous(breaks = 0:10) +
scale_color_manual(values = res_palette, name = "") +
labs(x = "Datasets agreeing", y = "Cum. grid cells",
     colour = "Resolution") +
theme_AP() +
theme(legend.position = c(0.7, 0.3))

```

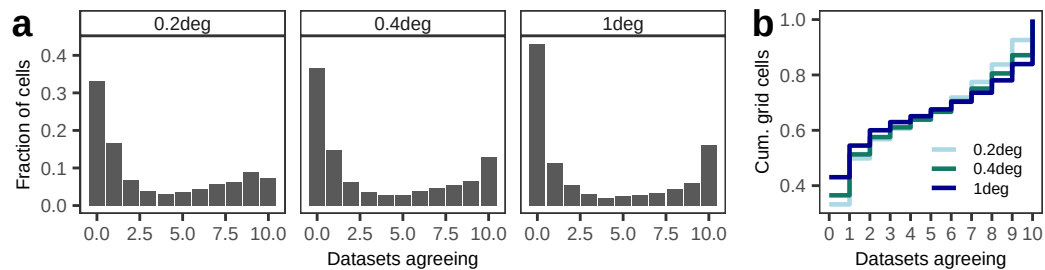
p2



```

plot_grid(p1, p2, ncol = 2, rel_widths = c(0.7, 0.3), labels = "auto")

```



```

# Plot: consensus regime plot -----

dt_regime <- dt_k[, .(absence = sum(N[n_pos == 0]),
                             contested = sum(N[n_pos >= 1 & n_pos <= 8]),
                             presence = sum(N[n_pos >= 9])), resolution]

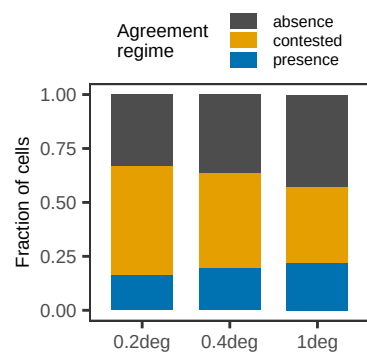
dt_regime <- melt(dt_regime, id.vars = "resolution", variable.name = "regime",
                  value.name = "N")

dt_regime[, frac := N / sum(N), by = resolution]

plot_regime <- ggplot(dt_regime, aes(resolution, frac, fill = regime)) +
  geom_col(width = 0.7) +
  scale_fill_manual(values = c(absence = "#4D4D4D",
                               contested = "#E69F00",
                               presence = "#0072B2"),
                    guide = guide_legend(nrow = 3),
                    name = "Agreement \nregime") +
  labs(x = NULL, y = "Fraction of cells") +
  theme_AP() +
  theme(legend.position = "top")

```

plot_regime



5 Consequences for crop production

5.1 Extract data from EarthStat

```
# CODE TO RETRIEVE DATA FROM EARTHSTAT AND PRODUCE CROP DATASETS #####

# PATHS-----

earthstat_root <- "HarvestedAreaYield175Crops_Geotiff"
earthstat_geotiff <- file.path(earthstat_root, "GeoTiff")

agreement_file <- "./datasets/irrigated_areas_regridded/irrigated_areas_regridded_all.csv"

out_dir <- "./datasets/earthstat_regridded"
dir.create(out_dir, recursive = TRUE, showWarnings = FALSE)

# TARGETS-----

targets <- data.table(
  res_tag = c("02", "04", "10"),
  resolution_label = c("0.2deg", "0.4deg", "1deg"),
  target_file = c(
    "mha_stack_ll_02_aligned_NEAR.tif",
    "mha_stack_ll_04_aligned_NEAR.tif",
    "mha_stack_ll_10_aligned_NEAR.tif"
  )
)

# FIND EARTHSTAT PRODUCTION FILES-----

prod_files <- list.files(
  path = earthstat_geotiff,
  pattern = "_Production\\.tif$",
  recursive = TRUE,
  full.names = TRUE
)

if (length(prod_files) == 0) stop("No *_Production.tif files found.")

earthstat_index <- data.table(
  crop = sub("_Production\\.tif$", "", basename(prod_files)),
  file = prod_files
)[order(crop)]

fwrite(earthstat_index, file.path(out_dir, "earthstat_production_file_index.csv"))

# LOAD AGREEMENT TABLE-----
```

```

agree_all <- dt_pres

# MAIN LOOP OVER TARGET RESOLUTIONS -----

ks <- 1:10

for (j in seq_len(nrow(targets))) {

  res_tag      <- targets$res_tag[j]
  res_label    <- targets$resolution_label[j]
  target_file  <- targets$target_file[j]

  message("\n=====")
  message("Processing resolution ", res_label, " (", res_tag, ")")
  message("=====\\n")

  out_dir_res <- file.path(out_dir, paste0("res_", res_tag))
  dir.create(out_dir_res, recursive = TRUE, showWarnings = FALSE)

  template_r <- rast(target_file)[[1]]
  values(template_r) <- NA_real_

  agree_dt <- copy(agree_all)[resolution == res_label, .(lon, lat, n_pos)]
  agree_dt <- unique(agree_dt)

  dup_check <- agree_dt[, .N, by = .(lon, lat)][N > 1]
  if (nrow(dup_check) > 0) {
    stop("Duplicated lon/lat rows in agreement table for resolution ", res_label)
  }

  setkey(agree_dt, lon, lat)

  #####
  # STEP 1: HARMONIZE EACH CROP TO THIS RESOLUTION
  #####

  qa_list <- lapply(seq_len(nrow(earthstat_index)), function(i) {

    crop <- earthstat_index$crop[i]
    file <- earthstat_index$file[i]

    out_file <- file.path(out_dir_res, paste0(crop, "_Production_ll_", res_tag, ".tif"))

    harmonize_one_production_to_file(
      file = file,
      crop = crop,
      template_r = template_r,

```



```

    out_file = out_file,
    method = "bilinear",
    max_valid_value = 1e9
  )
})

qa_dt <- rbindlist(qa_list)
fwrite(
  qa_dt,
  file.path(out_dir_res, paste0("earthstat_harmonized_files_", res_tag, ".csv"))
)

#####
# STEP 2: SUMMARISE PER CROP
#####

crop_results <- vector("list", length = nrow(earthstat_index))
global_results <- vector("list", length = nrow(earthstat_index))

for (i in seq_len(nrow(earthstat_index))) {

  crop <- earthstat_index$crop[i]
  file_ll <- file.path(out_dir_res, paste0(crop, "_Production_ll_", res_tag, ".tif"))

  message("Summarising ", crop, " at ", res_label, "°")

  r <- rast(file_ll)

  dt_crop <- as.data.table(as.data.frame(r, xy = TRUE, na.rm = FALSE))
  setnames(dt_crop, c("x", "y", crop), c("lon", "lat", "production_tons"))

  dt_crop <- dt_crop[!is.na(production_tons) & production_tons > 0]

  setkey(dt_crop, lon, lat)
  dt_crop <- agree_dt[dt_crop]
  dt_crop[is.na(n_pos), n_pos := 0L]

  total_all <- dt_crop[, sum(production_tons, na.rm = TRUE)]

  crop_out <- data.table(
    crop = crop,
    k = c(0L, ks),
    production_tons = NA_real_
  )

  crop_out[k == 0, production_tons := total_all]

```

```

for (kval in ks) {
  crop_out[
    k == kval,
    production_tons := dt_crop[n_pos >= kval, sum(production_tons, na.rm = TRUE)]
  ]
}

tons_k1 <- crop_out[k == 1, production_tons]
tons_k5 <- crop_out[k == 5, production_tons]

crop_out[, `:=`(
  rule = ifelse(k == 0, "all_cells", paste0("k_ge_", k)),
  resolution = res_label,
  res_tag = res_tag,
  share_of_all_cells = production_tons / total_all,
  share_of_k1 = if (!is.na(tons_k1) && tons_k1 > 0) production_tons / tons_k1 else NA_real,
  share_of_k5 = if (!is.na(tons_k5) && tons_k5 > 0) production_tons / tons_k5 else NA_real,
)]

crop_results[[i]] <- crop_out
global_results[[i]] <- crop_out[, .(k, production_tons)]

rm(r, dt_crop, crop_out)
gc()
}

#####
# STEP 3: SAVE PER-CROP RESULTS
#####

prod_k_crop <- rbindlist(crop_results)
setorder(prod_k_crop, crop, k)

fwrite(
  prod_k_crop,
  file.path(out_dir_res, paste0("earthstat_production_by_crop_k_agreement_", res_tag, ".csv")),
)

#####
# STEP 4: GLOBAL SUMMARY
#####

global_k <- rbindlist(global_results)[
  , .(production_tons = sum(production_tons, na.rm = TRUE)),
  by = k
][order(k)]

```

```

global_k[, `:=`(
  rule = ifelse(k == 0, "all_cells", paste0("k_ge_", k)),
  resolution = res_label,
  res_tag = as.integer(res_tag)
)]

global_all <- global_k[k == 0, production_tons]
global_k1 <- global_k[k == 1, production_tons]
global_k5 <- global_k[k == 5, production_tons]

global_k[, `:=`(
  share_of_all_cells = production_tons / global_all,
  share_of_k1 = if (!is.na(global_k1) && global_k1 > 0) production_tons / global_k1 else NA,
  share_of_k5 = if (!is.na(global_k5) && global_k5 > 0) production_tons / global_k5 else NA
)]

print(global_k)

fwrite(
  global_k,
  file.path(out_dir_res, paste0("earthstat_global_production_k_agreement_", res_tag, ".csv"))
)

rm(template_r, agree_dt, qa_dt, prod_k_crop, global_k, crop_results, global_results)
gc()
}

```

5.2 Plot crop heatmap

```

# PLOT HEATMAP OF CROPS #####

# Define files to retrieve -----

files <- c(
  "./datasets/earthstat_regridded/res_02/earthstat_production_by_crop_k_agreement_02.csv",
  "./datasets/earthstat_regridded/res_04/earthstat_production_by_crop_k_agreement_04.csv",
  "./datasets/earthstat_regridded/res_10/earthstat_production_by_crop_k_agreement_10.csv"
)

prod_dt <- rbindlist(lapply(files, fread), use.names = TRUE, fill = TRUE)

# TOP 25 CROPS BASED ON TOTAL PRODUCTION AT 0.2 (k = 0) -----

top25 <- prod_dt[
  resolution == "0.2deg" & k == 0,
  .(baseline_tons = production_tons),
  by = crop
]

```

```

][order(-baseline_tons)][1:25]

# PREPARE HEATMAP TABLE (LOSS VS k >= 1) -----

plot_dt <- prod_dt[
  crop %in% top25$crop & k %in% 1:10,
  .(crop, k, resolution, share_of_k1)
]

plot_dt[, loss_vs_1:= 1 - share_of_k1]

# Order crops -----

crop_levels <- rev(top25$crop)
plot_dt[, crop:= factor(crop, levels = crop_levels)]

# Order resolution -----

plot_dt[, resolution:= factor(
  resolution,
  levels = c("0.2deg", "0.4deg", "1deg")
)]

# Reverse k axis (10 → 1) -----

plot_dt[, k := factor(k, levels = rev(1:10))]

#####
# PLOT
#####

crop_labels <- c(
  "apple" = "Apple",
  "cabbage" = "Cabbage",
  "grape" = "Grape",
  "orange" = "Orange",
  "banana" = "Banana",
  "watermelon" = "Watermelon",
  "clover" = "Clover",
  "tomato" = "Tomato",
  "oilpalm" = "Oil palm",
  "grassnes" = "Grasses (NES)",
  "sweetpotato" = "Sweet potato",
  "barley" = "Barley",
  "cassava" = "Cassava",
  "soybean" = "Soybean",
  "vegetablenes" = "Vegetables (NES)",

```

```

"sugarbeet" = "Sugar beet",
"fornes" = "Fodder (NES)",
"potato" = "Potato",
"maizefor" = "Maize (forage)",
"alfalfa" = "Alfalfa",
"wheat" = "Wheat",
"rice" = "Rice",
"maize" = "Maize",
"mixedgrass" = "Mixed grass",
"sugarcane" = "Sugarcane"
)

# The fraction of a crop's total production that is excluded when requiring at
# least k irrigation maps to agree that the cell is irrigated, relative to the
# baseline where at least one map detects irrigation (k equal or greater than 1).

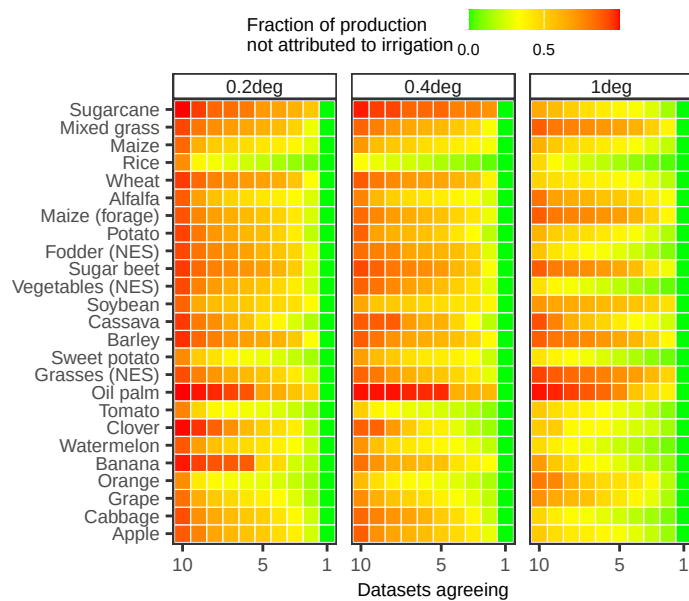
# Question that answers: "If I require k datasets to agree, how much of the
# production I previously considered 'irrigated' disappears?"

# The heatmap shows the fraction of crop production that is lost as increasingly
# stringent agreement is required among irrigation datasets, relative to the
# baseline where at least one dataset detects irrigation. This quantifies
# how sensitive the attribution of production to irrigation is to disagreement
# in irrigated-area maps.

plot_heatmap <- ggplot(plot_dt, aes(k, crop, fill = loss_vs_1)) +
  geom_tile(color = "white") +
  scale_fill_gradientn(colours = c("green", "yellow", "orange", "red"),
    values = c(0, 0.5, 1), breaks = c(0, 0.5, 1),
    name = "Fraction of production \nnot attributed to irrigation") +
  scale_y_discrete(labels = crop_labels) +
  scale_x_discrete(breaks = c("10", "5", "1")) +
  facet_wrap(~resolution, ncol = 3) +
  labs(x = "Datasets agreeing", y = NULL) +
  theme_AP() +
  theme(legend.position = "top")

plot_heatmap

```



5.3 Plot breadbasket regions

```
# DEFINE BREADBASKET REGIONS AS NUMERIC BOUNDS #####

regions_dt <- data.table(
  region = c("Indo-Gangetic Plain",
             "North China Plain",
             "US High Plains",
             "Nile Delta/Basin",
             "Mekong Delta",
             "Mediterranean Basin"),
  xmin = c(68, 113, -104, 29.5, 104.5, -6),
  xmax = c(91, 121, -96, 33.0, 106.5, 36),
  ymin = c(21.5, 34, 32, 24.0, 8.5, 30),
  ymax = c(33.0, 41, 44, 31.5, 11.5, 46)
)

# LOCATION OF HARMONIZED RASTERS #####

earthstat_dirs <- list(
  "0.2deg" = "./datasets/earthstat_regridded/res_02",
  "0.4deg" = "./datasets/earthstat_regridded/res_04",
  "1deg" = "./datasets/earthstat_regridded/res_10")

res_tags <- c("0.2deg" = "02",
              "0.4deg" = "04",
              "1deg" = "10")

# CEREAL CROPS TO KEEP #####
```

```

grain_targets <- c("wheat", "rice", "maize", "barley", "millet", "sorghum")
grain_aliases <- c("othercereals", "cerealnes")

# RUN FOR ALL THREE RESOLUTIONS #####

template_files <- list(
  "0.2deg" = "mha_stack_ll_02_aligned_NEAR.tif",
  "0.4deg" = "mha_stack_ll_04_aligned_NEAR.tif",
  "1deg" = "mha_stack_ll_10_aligned_NEAR.tif")

results_regions_all_earthstat <- rbindlist(
  lapply(names(earthstat_dirs), function(res_label) {

    folder <- earthstat_dirs[[res_label]]
    res_tag <- res_tags[[res_label]]

    grain_dt <- build_cereal_sum_table(
      folder = folder,
      res_tag = res_tag,
      grain_targets = grain_targets,
      grain_aliases = grain_aliases
    )

    template_r <- rast(template_files[[res_label]])[[1]]
    values(template_r) <- NA_real_

    out <- compute_breadbaskets_earthstat(
      res_tag = res_label,
      grain_prod_dt = grain_dt,
      dt_pres = dt_pres,
      regions_dt = regions_dt,
      template_r = template_r
    )

    rm(grain_dt, template_r)
    gc()

    out
  }),
  use.names = TRUE,
  fill = TRUE
)

## Using crops for 02: wheat, rice, maize, barley, millet, sorghum, cerealnes
## Using crops for 04: wheat, rice, maize, barley, millet, sorghum, cerealnes
## Using crops for 10: wheat, rice, maize, barley, millet, sorghum, cerealnes

```

```

# SAVE RESULTS #####

dir.create("../datasets/earthstat_regridded", recursive = TRUE, showWarnings = FALSE)

fwrite(results_regions_all_earthstat,
       "../datasets/earthstat_regridded/earthstat_breadbasket_grain_losses.csv")

# SUMMARY STATISTICS #####

print(results_regions_all_earthstat)

```

```

##           region resolution      k total_prod frac_vs_1  loss_vs_1
##           <char>    <char> <int>      <num>      <num>      <num>
##    1: Indo-Gangetic Plain    0.2deg    10  112799073 0.5417889 0.45821109
##    2: Indo-Gangetic Plain    0.2deg     9  188874014 0.9071869 0.09281306
##    3: Indo-Gangetic Plain    0.2deg     8  195009509 0.9366565 0.06334346
##    4: Indo-Gangetic Plain    0.2deg     7  198375705 0.9528248 0.04717518
##    5: Indo-Gangetic Plain    0.2deg     6  200251950 0.9618367 0.03816332
##    ---
## 176: Mediterranean Basin      1deg     5   90599945 0.8922145 0.10778553
## 177: Mediterranean Basin      1deg     4   93932840 0.9250363 0.07496369
## 178: Mediterranean Basin      1deg     3   98418519 0.9692106 0.03078940
## 179: Mediterranean Basin      1deg     2   99155152 0.9764649 0.02353515
## 180: Mediterranean Basin      1deg     1  101545030 1.0000000 0.00000000
##      frac_vs_5  loss_vs_5
##      <num>      <num>
##    1: 0.5580268 0.441973153
##    2: 0.9343762 0.065623786
##    3: 0.9647290 0.035270958
##    4: 0.9813819 0.018618094
##    5: 0.9906639 0.009336147
##    ---
## 176: 1.0000000 0.000000000
## 177:      NA      NA
## 178:      NA      NA
## 179:      NA      NA
## 180:      NA      NA

```

```

tmp_summary <- results_regions_all_earthstat[
  , .(
    min_frac_5 = min(frac_vs_5, na.rm = TRUE),
    max_frac_5 = max(frac_vs_5, na.rm = TRUE),
    min_loss_5 = min(loss_vs_5, na.rm = TRUE),
    max_loss_5 = max(loss_vs_5, na.rm = TRUE),
    min_frac_1 = min(frac_vs_1, na.rm = TRUE),
    max_frac_1 = max(frac_vs_1, na.rm = TRUE),
    min_loss_1 = min(loss_vs_1, na.rm = TRUE),
    max_loss_1 = max(loss_vs_1, na.rm = TRUE)
  )
]

```



```

),
  by = .(region, k)
] %>%
  .[k %in% 5:10]

tmp_summary[k %in% 9:10 &
  region %in% c("Nile Delta/Basin", "Indo-Gangetic Plain",
    "Mekong Delta"),
  .(region, min_loss_1, max_loss_1)]

##           region min_loss_1 max_loss_1
##           <char>      <num>      <num>
## 1: Indo-Gangetic Plain 0.05489196 0.45821109
## 2: Indo-Gangetic Plain 0.02820065 0.09281306
## 3:   Nile Delta/Basin 0.00265116 0.34278082
## 4:   Nile Delta/Basin 0.00265116 0.03917718
## 5:   Mekong Delta 0.05610146 0.47265794
## 6:   Mekong Delta 0.00000000 0.12154759
# PLOT: LOSS RELATIVE TO k >= 1 #####

plot_dt <- copy(results_regions_all_earthstat)

# Keep only the main metric
plot_dt <- plot_dt[!is.na(loss_vs_1)]

# Order regions
region_levels <- c(
  "Indo-Gangetic Plain",
  "North China Plain",
  "US High Plains",
  "Nile Delta/Basin",
  "Mekong Delta",
  "Mediterranean Basin"
)

plot_dt[, region:= factor(region, levels = region_levels)]

# Order resolutions
plot_dt[, resolution := factor(resolution, levels = c("0.2deg", "0.4deg", "1deg"))]

# Keep k numeric so reverse scale works cleanly
plot_dt[, k:= as.integer(k)]

plot_breadbaskets <- ggplot(plot_dt, aes(k, loss_vs_1, colour = resolution,
  group = resolution)) +
  geom_line() +
  geom_point() +

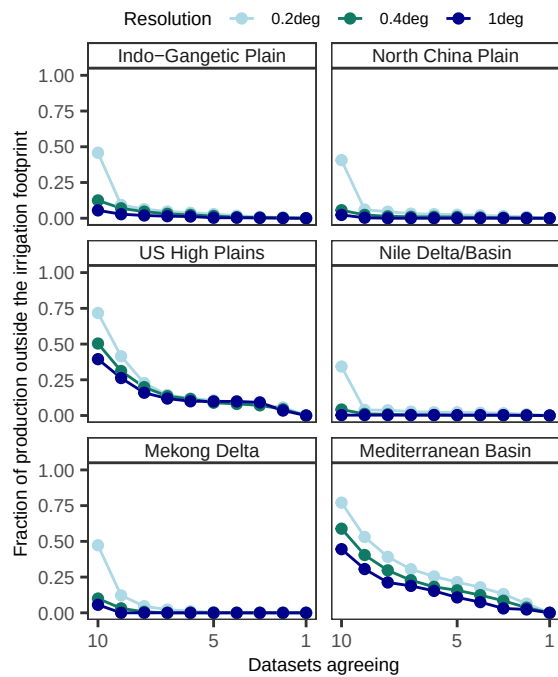
```

```

facet_wrap(~region, ncol = 2) +
scale_x_reverse(breaks = c(10, 5, 1),
               limits = c(10, 1)) +
scale_y_continuous(limits = c(0, 1),
                  breaks = c(0, 0.25, 0.5, 0.75, 1)) +
labs(x = "Datasets agreeing", y = "Fraction of production outside the irrigation footprint",
     colour = "Resolution") +
scale_color_manual(values = res_palette) +
theme_AP() +
theme(legend.position = "top")

```

plot_breadbaskets

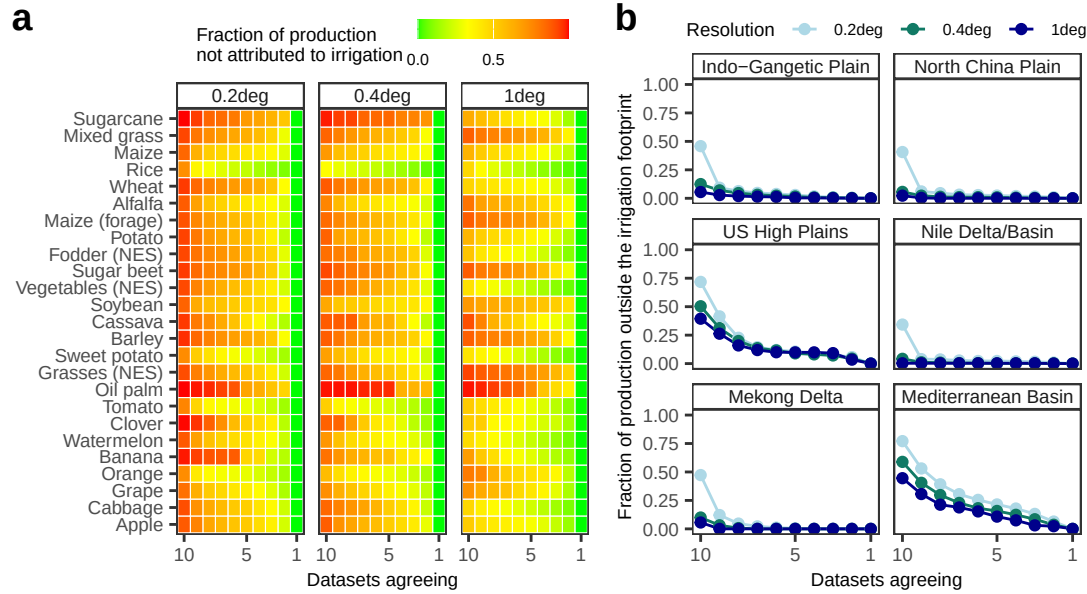


```

top2 <- plot_grid(plot_heatmap, plot_breadbaskets, ncol = 2, rel_widths = c(0.55, 0.45),
                  labels = "auto")

```

top2



5.4 Plot global

```
# LOAD GLOBAL RESULTS #####

files <- c(
  "./datasets/earthstat_regridded/res_02/earthstat_global_production_k_agreement_02.csv",
  "./datasets/earthstat_regridded/res_04/earthstat_global_production_k_agreement_04.csv",
  "./datasets/earthstat_regridded/res_10/earthstat_global_production_k_agreement_10.csv"
)

global_dt <- rbindlist(lapply(files, fread), use.names = TRUE, fill = TRUE)

# KEEP k = 10:1 #####

global_dt <- global_dt[k %in% 1:10]

# compute loss relative to k >= 1
global_dt[, loss_vs_1 := 1 - share_of_k1]

# order resolution
global_dt[, resolution := factor(resolution, levels = c("0.2deg", "0.4deg", "1deg"))]

# Print
global_dt[k %in% c(1, 5, 10), .(k, resolution, loss_vs_1)]

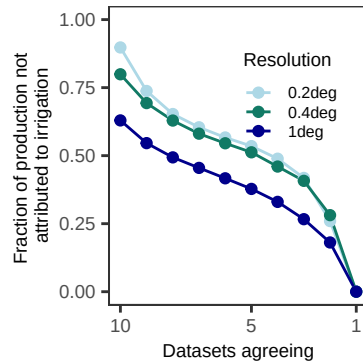
##      k resolution loss_vs_1
##    <int>    <fctr>    <num>
## 1:     1    0.2deg 0.0000000
## 2:     5    0.2deg 0.5349922
## 3:    10    0.2deg 0.8976727
```

```
## 4:      1      0.4deg 0.0000000
## 5:      5      0.4deg 0.5122408
## 6:     10      0.4deg 0.7990398
## 7:      1      1deg 0.0000000
## 8:      5      1deg 0.3777311
## 9:     10      1deg 0.6295084
```

```
# PLOT #####
```

```
plot_global <- ggplot(global_dt, aes(k, loss_vs_1, colour = resolution, group = resolution)) +
  geom_line() +
  geom_point() +
  scale_x_reverse(breaks = c(10, 5, 1),
                 limits = c(10, 1)) +
  scale_y_continuous(limits = c(0, 1),
                    breaks = c(0, 0.25, 0.5, 0.75, 1)) +
  scale_color_manual(values = res_palette) +
  labs(x = "Datasets agreeing", y = "Fraction of production not \nattributed to irrigation",
       colour = "Resolution") +
  theme_AP() +
  theme(
    legend.position = c(0.7, 0.7))
```

```
plot_global
```



6 Consequences for irrigation-induced evapotranspiration

6.1 Extract CESM data

```
# LOAD THE HARMONIZED GRIDS #####
```

```
tmpl_02 <- rast("./mha_stack_ll_02_aligned_NEAR.tif")[[1]]
tmpl_04 <- rast("./mha_stack_ll_04_aligned_NEAR.tif")[[1]]
tmpl_1 <- rast("./mha_stack_ll_10_aligned_NEAR.tif")[[1]]

templates <- list("0.2deg" = tmpl_02, "0.4deg" = tmpl_04, "1deg" = tmpl_1)
```

```

# Enforce lon/lat WGS84 on templates if needed-----

templates <- lapply(templates, function(t) {
  if (is.na(crs(t)) || crs(t) == "") crs(t) <- "EPSG:4326"
  t
})

# LOAD THE FILES #####
# Files come from the Yao et al 2025 paper (Nature Water)

# Year window -----

win <- "1985_2014_timmean"

# Base directory -----

base_dir <- "./yao_et_al_2025_datasets/water_fluxes/CESM2"

# Paste -----

irr_files <- paste0(base_dir, "/CESM2_IRR0", 1:3, "_1901_2014_QFLX_EVAP_TOT_yearmean_", win)
noi_files <- paste0(base_dir, "/CESM2_NOI0", 1:3, "_1901_2014_QFLX_EVAP_TOT_yearmean_", win)

# BUILD ET STACKS + ENSEMBLE MEAN PER RESOLUTION #####

d_list <- Map(function(irr_f, noi_f) delta_pair_resampled(irr_f, noi_f, templates),
              irr_files, noi_files)

# d_list length 3; each element is list("0.2deg"=rast, "0.4deg"=rast, "1deg"=rast)
dET_mean_rasters <- lapply(names(templates), function(res_name) {

  d_rasters <- lapply(d_list, function(one_pair) one_pair[[res_name]])
  s <- terra::rast(d_rasters) # 3-layer stack
  names(s) <- paste0("dET_IRR0", 1:3, "_mmday")

  # ensemble mean: cell-by-cell mean across 3 members
  m <- terra::app(s, fun = mean, na.rm = TRUE)
  names(m) <- paste0("CESM2_dET_mean_", win, "_", res_name, "_mmday")

  list(stack = s, mean = m)
})
names(dET_mean_rasters) <- names(templates)

# EXPORT TO DATA TABLE #####

dET_dt <- lapply(names(templates), function(res_name) {
  tmp1 <- templates[[res_name]]

```

```

out <- dET_mean_rasters[[res_name]]

dt_mean <- rast_to_dt_with_cell(out$mean, tpl, value_name = "dET_mean_mmday")
dt_mean[, resolution:= res_name]

dt_members <- rbindlist(lapply(1:terra::nlyr(out$stack), function(i) {
  rr <- out$stack[[i]]
  dt <- rast_to_dt_with_cell(rr, tpl, value_name = "dET_mmday")
  dt[, member:= names(rr)] # dET_IRR01_mmday / dET_IRR02_mmday / dET_IRR03_mmday
  dt[, resolution:= res_name]
  dt[]
}))

list(mean = dt_mean, members = dt_members)
})
names(dET_dt) <- names(templates)

# COMPUTE K-OF-10 AGREEMENT WITH CELL JOINS #####

resolutions <- c("0.2deg", "0.4deg", "1deg")
ks <- 1:10

# Compute -----

out_k_all <- rbindlist(lapply(resolutions, function(res) {

  tpl <- templates[[res]]

  # irrigation presence-----

  dt_irrig_raw <- copy(dt_pres)[resolution == res]
  dt_irrig <- pres_to_cell(dt_irrig_raw, tpl)

  # add cell area-----

  dt_area <- cell_area_dt(tpl)
  setkey(dt_irrig, cell)
  setkey(dt_area, cell)
  dt_irrig <- dt_irrig[dt_area] # add area_m2

  # increase ET mean table -----

  dt_et <- copy(dET_dt[[res]]$mean)
  valcol <- setdiff(names(dt_et), c("cell", "lon", "lat", "resolution"))[1]

  # Join: we keep all ET cells and bring irrigation info when available:
  # to summarise ONLY cells that exist in the irrigation table:

```

```

# dt_join <- dt_et[dt_irrig, on="cell"]
setkey(dt_et, cell)
dt_join <- dt_irrig[dt_et]

rbindlist(lapply(ks, function(k) {

  dt_sub <- dt_join[n_pos >= k]

  # restrict weights to cells with non-missing ET and non-missing area
  ok <- !is.na(dt_sub[[valcol]]) & !is.na(dt_sub$area_m2)
  w <- dt_sub$area_m2[ok]
  x <- dt_sub[[valcol]][ok]

  # area-weighted mean increase ET (mm/day)
  mean_aw <- if (length(x) == 0L) NA_real_ else sum(x * w) / sum(w)

  # area-weighted aggregate: mm/day * m^2 -> m^3/day
  # 1 mm = 0.001 m
  sum_m3day <- if (length(x) == 0L) NA_real_ else sum(x * w) * 1e-3

  data.table(resolution = res,
             k = k,
             n_cells_irrig_k = sum(dt_irrig$n_pos >= k, na.rm = TRUE),
             n_cells_join_k = nrow(dt_sub),
             n_cells_with_dET_k = sum(!is.na(dt_sub[[valcol]])),
             mean_dET_mmday = mean(dt_sub[[valcol]], na.rm = TRUE),
             sum_dET_mmday_unweighted = sum(dt_sub[[valcol]], na.rm = TRUE),
             mean_dET_mmday_areaweighted = mean_aw,
             sum_dET_m3day_areaweighted = sum_m3day,
             area_m2_with_dET_k = if (length(w) == 0L) 0 else sum(w)
          )
}))
}))

out_k_all

```

##	resolution	k	n_cells_irrig_k	n_cells_join_k	n_cells_with_dET_k
##	<char>	<int>	<int>	<int>	<int>
## 1:	0.2deg	1	86551	86551	86146
## 2:	0.2deg	2	65026	65026	64695
## 3:	0.2deg	3	56056	56056	55792
## 4:	0.2deg	4	51038	51038	50806
## 5:	0.2deg	5	46920	46920	46723
## 6:	0.2deg	6	42296	42296	42144
## 7:	0.2deg	7	36510	36510	36396
## 8:	0.2deg	8	29235	29235	29158
## 9:	0.2deg	9	21006	21006	20964

## 10:	0.2deg	10	9590	9590	9575
## 11:	0.4deg	1	25244	25244	25135
## 12:	0.4deg	2	19367	19367	19283
## 13:	0.4deg	3	16881	16881	16807
## 14:	0.4deg	4	15488	15488	15422
## 15:	0.4deg	5	14369	14369	14312
## 16:	0.4deg	6	13251	13251	13200
## 17:	0.4deg	7	11715	11715	11675
## 18:	0.4deg	8	9910	9910	9878
## 19:	0.4deg	9	7735	7735	7718
## 20:	0.4deg	10	5118	5118	5112
## 21:	1deg	1	4595	4595	4582
## 22:	1deg	2	3673	3673	3663
## 23:	1deg	3	3226	3226	3220
## 24:	1deg	4	2988	2988	2983
## 25:	1deg	5	2816	2816	2811
## 26:	1deg	6	2618	2618	2613
## 27:	1deg	7	2396	2396	2391
## 28:	1deg	8	2133	2133	2129
## 29:	1deg	9	1772	1772	1769
## 30:	1deg	10	1298	1298	1297
##	resolution	k	n_cells_irrig_k	n_cells_join_k	n_cells_with_dET_k
##	<char>	<int>	<int>	<int>	<int>
##	mean_dET_mmday	sum_dET_mmday_unweighted	mean_dET_mmday_areaweighted		
##	<num>		<num>		<num>
## 1:	0.02529852		2179.36620		0.02558831
## 2:	0.03176582		2055.09005		0.03191648
## 3:	0.03538052		1973.95005		0.03538168
## 4:	0.03776344		1918.60923		0.03770233
## 5:	0.04012408		1874.71758		0.04011012
## 6:	0.04332301		1825.80479		0.04337120
## 7:	0.04821179		1754.71632		0.04832421
## 8:	0.05598796		1632.49706		0.05619227
## 9:	0.06818188		1429.36490		0.06853504
## 10:	0.08271255		791.97271		0.08345616
## 11:	0.02324782		584.33387		0.02345943
## 12:	0.02916753		562.43750		0.02922414
## 13:	0.03254787		547.03199		0.03244195
## 14:	0.03477842		536.35276		0.03462010
## 15:	0.03678079		526.40665		0.03666735
## 16:	0.03922764		517.80487		0.03914187
## 17:	0.04295393		501.48715		0.04292040
## 18:	0.04846748		478.76176		0.04850110
## 19:	0.05689614		439.12441		0.05710005
## 20:	0.07209324		368.54062		0.07260780
## 21:	0.02146453		98.35049		0.02162021
## 22:	0.02635598		96.54197		0.02637286
## 23:	0.02973174		95.73620		0.02956890


```

## 24:      0.03164255                94.38972                0.03144091
## 25:      0.03344324                94.00895                0.03322915
## 26:      0.03568753                93.25152                0.03550908
## 27:      0.03845227                91.93937                0.03830816
## 28:      0.04232406                90.10792                0.04219702
## 29:      0.04798235                84.88078                0.04799651
## 30:      0.05863885                76.05459                0.05891090
##      mean_dET_mmday sum_dET_mmday_unweighted mean_dET_mmday_areaweighted
##              <num>                <num>                <num>
##      sum_dET_m3day_areaweighted area_m2_with_dET_k
##              <num>                <num>
## 1:      897374704                3.506972e+13
## 2:      849120561                2.660446e+13
## 3:      818354103                2.312931e+13
## 4:      797228362                2.114534e+13
## 5:      780279411                1.945343e+13
## 6:      761085387                1.754817e+13
## 7:      733047410                1.516936e+13
## 8:      683881025                1.217038e+13
## 9:      601363946                8.774547e+12
## 10:     333696792                3.998468e+12
## 11:     961188517                4.097238e+13
## 12:     926238757                3.169431e+13
## 13:     902973514                2.783352e+13
## 14:     887304997                2.562976e+13
## 15:     872410097                2.379256e+13
## 16:     859248031                2.195214e+13
## 17:     833991565                1.943112e+13
## 18:     798030563                1.645387e+13
## 19:     734575092                1.286470e+13
## 20:     618966902                8.524799e+12
## 21:     1008992705                4.666896e+13
## 22:     990272480                3.754893e+13
## 23:     982727869                3.323518e+13
## 24:     970732988                3.087483e+13
## 25:     968022615                2.913173e+13
## 26:     961187451                2.706878e+13
## 27:     949610682                2.478873e+13
## 28:     932414251                2.209669e+13
## 29:     880927203                1.835398e+13
## 30:     792833208                1.345818e+13
##      sum_dET_m3day_areaweighted area_m2_with_dET_k
##              <num>                <num>

```

```

# PLOT #####

```

```

# Condition ifesle -----

```

```

total_col <- if ("sum_dET_m3day_areaweighted" %in% names(out_k_all)) {
  "sum_dET_m3day_areaweighted"
} else if ("sum_dET_mmday_unweighted" %in% names(out_k_all)) {
  "sum_dET_mmday_unweighted"
} else {
  stop("out_k_all has neither sum_dET_m3day_areaweighted nor sum_dET_mmday_unweighted")
}

# Convert m3/day to km3/day -----

out_plot <- copy(out_k_all)

if (total_col == "sum_dET_m3day_areaweighted") {
  out_plot[, total_km3day := get(total_col) / 1e9]
  total_ylab <- "Total irrigation-\ninduced ET (km³ d⁻¹)"
} else {
  out_plot[, total_km3day := get(total_col)]
  total_ylab <- "Total irrigation-induced \nET signal (sum of mm/day)"
}

# Keep only k = 5, ..., 10-----

out_plot <- out_plot[k %in% 1:10]

# Plot mean increase in irrigation-induced ET -----
# (average irrigation-induced evapotranspiration change per grid cell)

plot_mean_ET <- ggplot(out_plot, aes(k, mean_dET_mmday, colour = resolution,
                                     group = resolution)) +
  geom_line() +
  geom_point() +
  labs(x = "Datasets agreeing",
       y = expression(atop("Mean " * Delta * ET[irr], "(mm d"^{-1} * ")")) +
  scale_x_reverse(breaks = c(10, 5, 1)) +
  scale_color_manual(values = res_palette) +
  theme_AP() +
  theme(legend.position = "none")

# Plot total irrigation induced ET -----
# (global irrigation-induced evapotranspiration signal, obtained by integrating
# over all retained cells; e.g., total amount of water transferred to the
# atmosphere per day because of irrigation)

plot_total_ET <- ggplot(out_plot, aes(k, total_km3day, colour = resolution, group = resolution,
                                     geom_line() +

```

```

geom_point() +
labs(x = "Datasets agreeing", y = total_ylab) +
scale_x_reverse(breaks = c(10, 5, 1)) +
scale_color_manual(values = res_palette) +
theme_AP() +
theme(legend.position = "none")

# Plot area with irrigation-induced ET retained under k-----

out_plot <- copy(out_k_all)[k %in% 1:10]

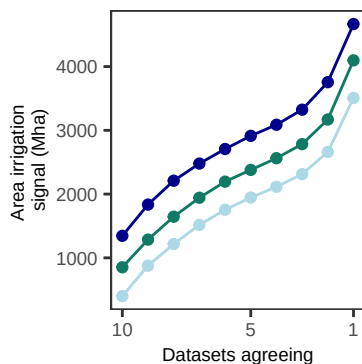
# convert m2 to Mha -----
out_plot[, area_Mha:= area_m2_with_dET_k / 1e10]

# also show contraction relative to k=5 within each resolution
out_plot[, area_frac_vs_k5:= area_m2_with_dET_k / area_m2_with_dET_k[k == 5], by = resolution]

# Total land area of grid cells with reported irrigated areas
plot_area <- ggplot(out_plot, aes(k, area_Mha, colour = resolution, group = resolution)) +
  geom_line() +
  geom_point() +
  labs(x = "Datasets agreeing", y = "Area irrigation \n signal (Mha)") +
  scale_x_reverse(breaks = c(10, 5, 1)) +
  scale_color_manual(values = res_palette) +
  theme_AP() +
  theme(legend.position = "none")

plot_area

```



```

# Also in case: fraction of area retained vs k=5
plot_area_fraction <- ggplot(out_plot, aes(k, area_frac_vs_k5, colour = resolution, group = resolution)) +
  geom_hline(yintercept = 1, linewidth = 0.4) +
  geom_line() +
  geom_point() +
  labs(x = "Datasets agreeing",
       y = expression("Area retaining "~Delta*ET~" (fraction of k=5)")) +
  scale_x_reverse(breaks = 10:1) +

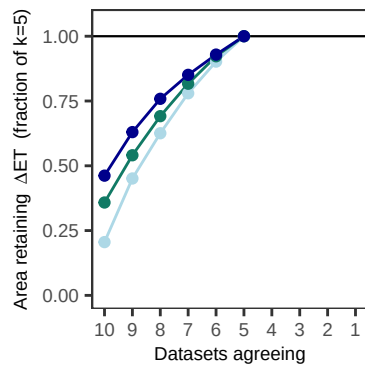
```

```
scale_y_continuous(limits = c(0, 1.05)) +
scale_color_manual(values = res_palette) +
theme_AP() +
theme(legend.position = "none")
```

```
plot_area_fraction
```

```
## Warning: Removed 12 rows containing missing values or values outside the scale range
## (`geom_line()`).
```

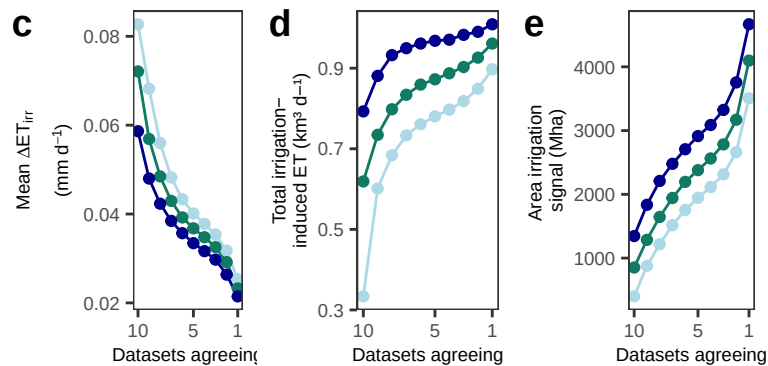
```
## Warning: Removed 12 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



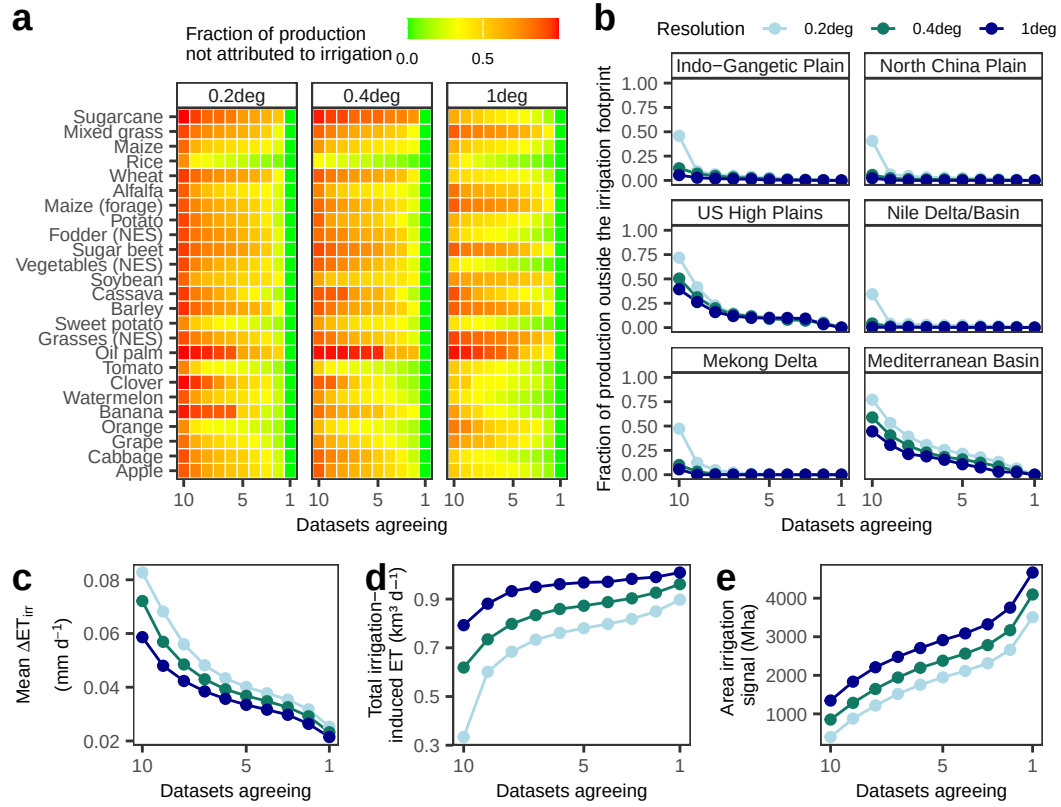
6.2 Merge figures

```
bottom <- plot_grid(plot_mean_ET, plot_total_ET, plot_area, labels = c("c", "d", "e"),
                    ncol = 3)
```

```
bottom
```



```
plot_grid(top2, bottom, ncol = 1, rel_heights = c(0.685, 0.325))
```



7 Final figures

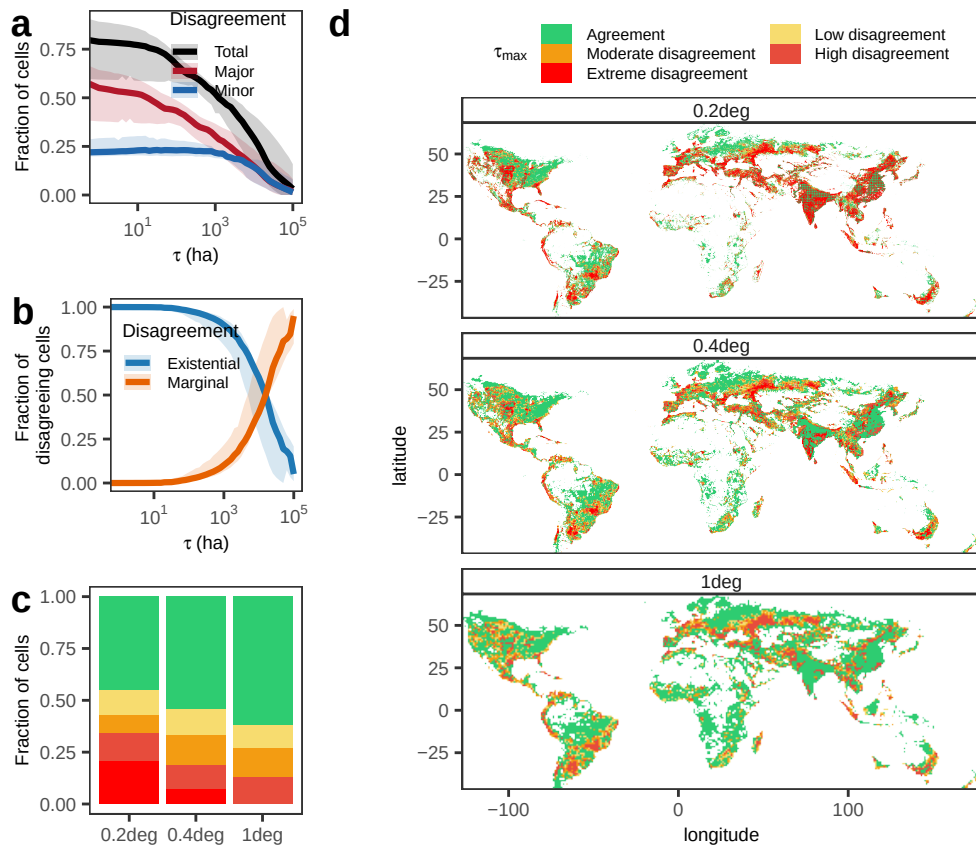
```
left <- plot_grid(plot_uncertainty1, plot_agreement1,
                  plot_stacked +
                    labs(y = "Fraction of cells"), ncol = 1, labels = "auto")
```

```
## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation introduced :
## log-10 transformation introduced infinite values.
## log-10 transformation introduced infinite values.
## log-10 transformation introduced infinite values.
```

```
plot_grid(left, plot_raster, ncol = 2, rel_widths = c(0.3, 0.7), labels = c("", "d"))
```

```
## Warning: Raster pixels are placed at uneven horizontal intervals and will be shifted
## i Consider using `geom_tile()` instead.
```

```
## Warning: Raster pixels are placed at uneven horizontal intervals and will be shifted
## i Consider using `geom_tile()` instead.
```



```
left <- plot_grid(plot_uncertainty1, plot_agreement1,
                  plot_regime, ncol = 1, labels = c("a", "b", "d"))
```

```
## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation introduced :
## log-10 transformation introduced infinite values.
## log-10 transformation introduced infinite values.
## log-10 transformation introduced infinite values.
```

```

middle <- plot_grid(p1 + facet_wrap(~resolution, ncol = 1),
  plot_stacked +
    labs(y = "Fraction of cells"),
    ncol = 1, rel_heights = c(0.7, 0.3), labels = c("c", "e"))
plot_grid(left, middle, plot_raster, ncol = 3, rel_widths = c(0.26, 0.26, 0.48), labels = c("a", "b", "f"))

```

```

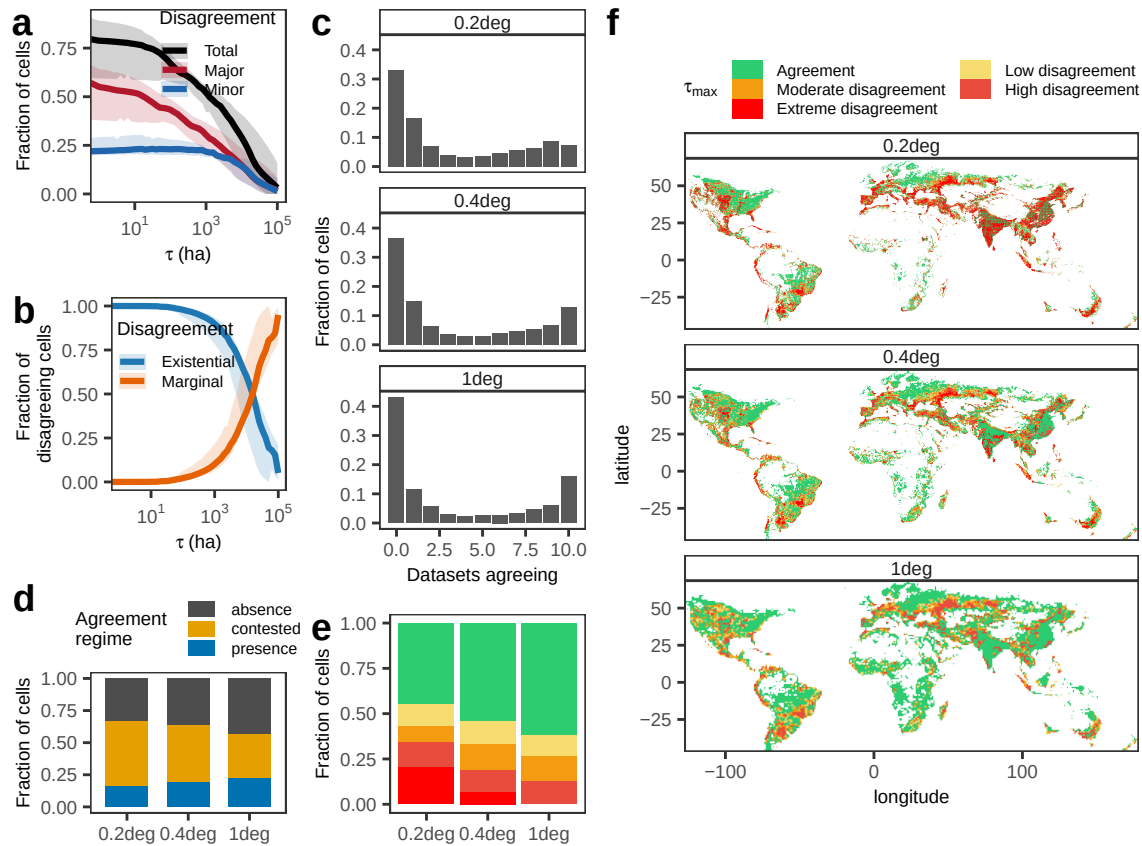
## Warning: Raster pixels are placed at uneven horizontal intervals and will be shifted
## i Consider using `geom_tile()` instead.

```

```

## Warning: Raster pixels are placed at uneven horizontal intervals and will be shifted
## i Consider using `geom_tile()` instead.

```



```

top <- plot_grid(plot_indices1 +
  labs(x = "", y = "Fraction \n variance") +
  scale_fill_discrete(name = "Sobol' \n indices") +
  theme(legend.position = c(0.4, 0.7)),
  a1 + labs(x = "exclusion", y = "Fraction \n disagree"),
  b1, c1, ncol = 4, labels = c("a", "", "", ""),
  rel_widths = c(0.23, 0.37, 0.2, 0.2))

```

```

## Scale for fill is already present.

```

```

## Adding another scale for fill, which will replace the existing scale.

```

```

## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation introduced

```

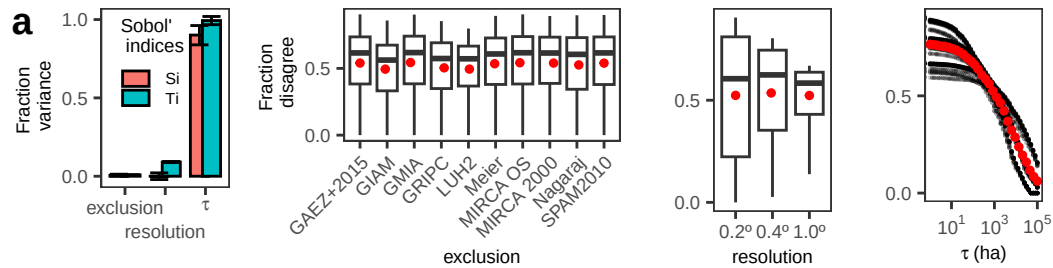
```

## log-10 transformation introduced infinite values.

```

```
## Warning: Removed 157 rows containing non-finite outside the scale range
## (`stat_summary_bin()`).
```

top



```
middle <- plot_grid(plot_indices2 +
  labs(x = "", y = "Fraction \n variance") +
  scale_fill_discrete(name = "") +
  theme(legend.position = "none"),
  a2 + labs(x = "exclusion", y = "Fraction \n disagree"),
  b2, c2, ncol = 4, labels = c("b", "", "", ""),
  rel_widths = c(0.23, 0.37, 0.2, 0.2))
```

```
## Scale for fill is already present.
```

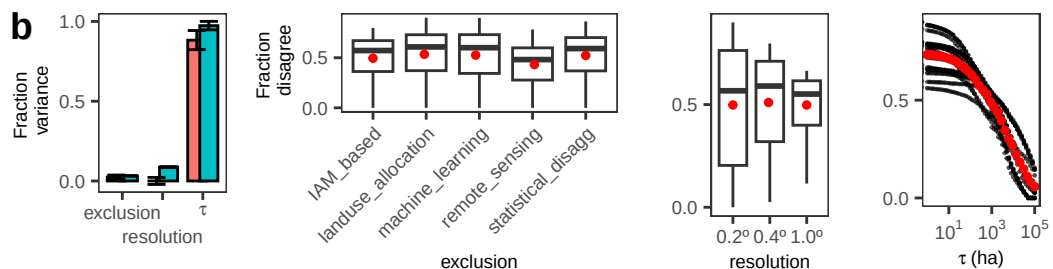
```
## Adding another scale for fill, which will replace the existing scale.
```

```
## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation
## introduced infinite values.
```

```
## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation
## introduced infinite values.
```

```
## Warning: Removed 157 rows containing non-finite outside the scale range
## (`stat_summary_bin()`).
```

middle



```
bottom <- plot_grid(plot_indices3 +
  labs(x = "", y = "Fraction \n variance") +
  scale_fill_discrete(name = "") +
  theme(legend.position = "none"),
  a3 + labs(x = "exclusion", y = "Fraction \n disagree"),
  b3, c3, d3, ncol = 5, labels = c("c", "", "", "", ""),
  rel_widths = c(0.22, 0.34, 0.16, 0.15, 0.15))
```

```
## Scale for fill is already present.
```



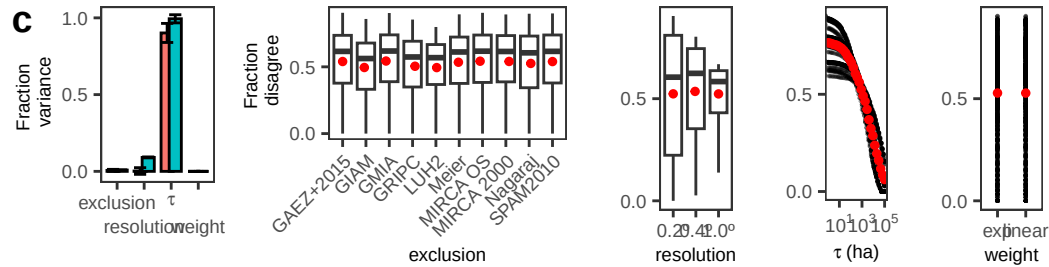
```
## Adding another scale for fill, which will replace the existing scale.

## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation
## introduced infinite values.

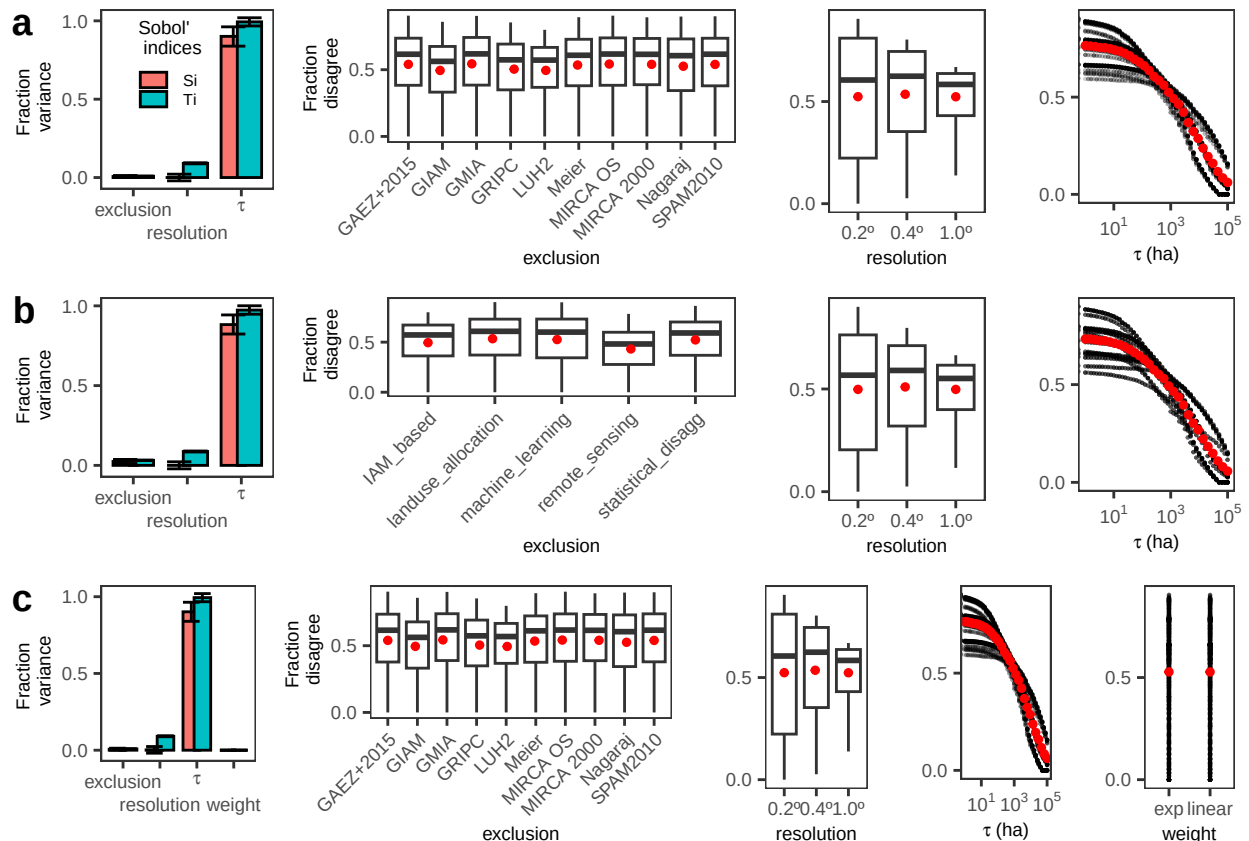
## Warning in scale_x_log10(labels = label_log(digits = 2)): log-10 transformation
## introduced infinite values.

## Warning: Removed 157 rows containing non-finite outside the scale range
## (`stat_summary_bin()`).
```

bottom



```
plot_grid(top, middle, bottom, ncol = 1)
```



```
dt_pres[, classification := fifelse(
  n_pos == 0, "absence",
  fifelse(n_pos <= 8, "contested", "presence")
)]
```

```

plot_classification <- ggplot(dt_pres, aes(lon, lat, fill = classification)) +
  geom_raster() +
  coord_equal(expand = FALSE) +
  facet_wrap(~ resolution, ncol = 1) +
  scale_fill_manual(values = c(absence = "#4D4D4D",
                              contested = "#E69F00",
                              presence = "#0072B2")) +
  scale_x_continuous(expand = c(0, 0)) +
  scale_y_continuous(expand = c(0, 0)) +
  theme_AP() +
  theme(panel.grid = element_blank(),
        strip.text = element_text(size = 11),
        legend.position = "top") +
  labs(x = "longitude", y = "latitude") +
  guides(fill = guide_legend(nrow = 3, byrow = TRUE))

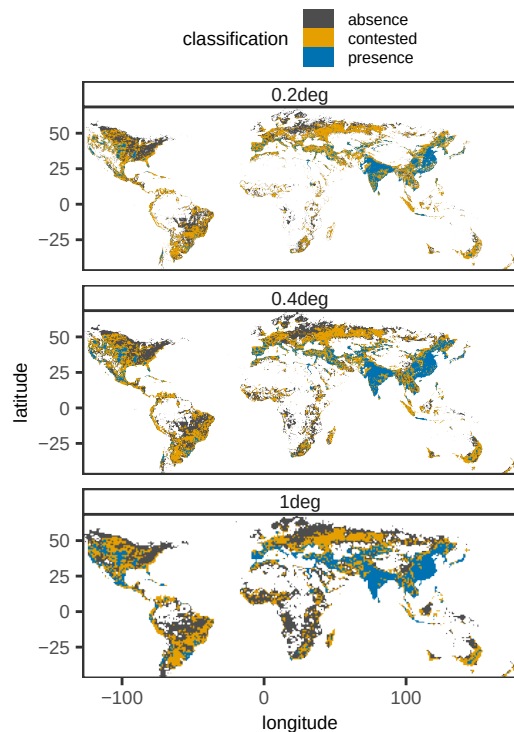
```

plot_classification

```

## Warning: Raster pixels are placed at uneven horizontal intervals and will be shifted
## i Consider using `geom_tile()` instead.
## Raster pixels are placed at uneven horizontal intervals and will be shifted
## i Consider using `geom_tile()` instead.

```



```

plot_grid(plot_raster, plot_classification, ncol = 2, labels = "auto")

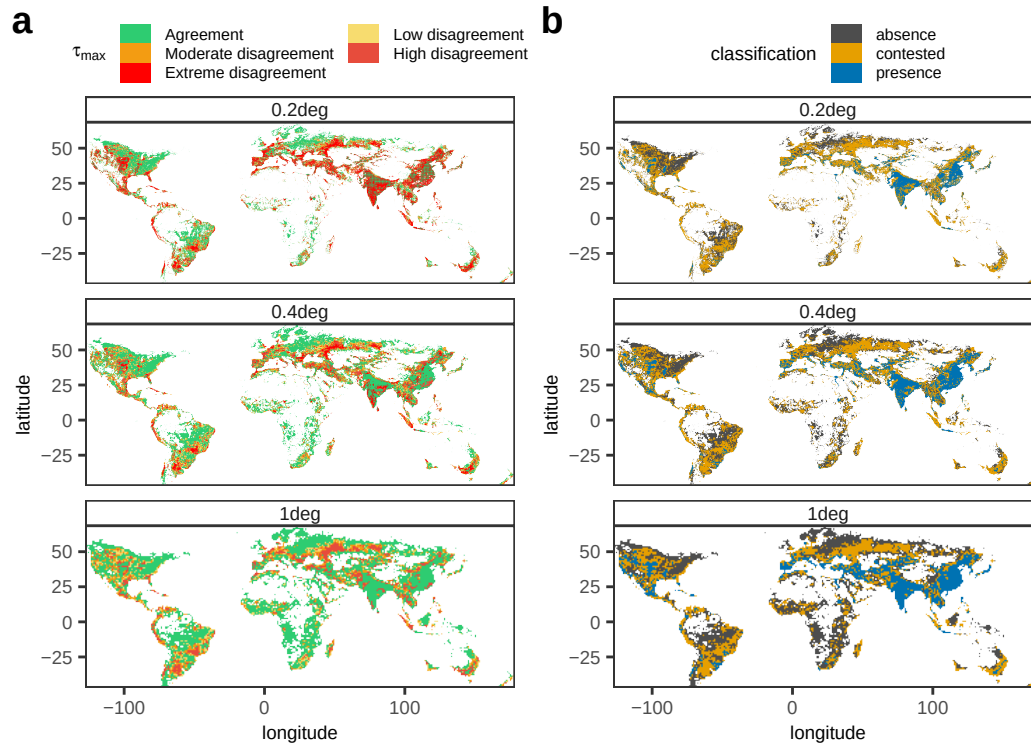
```

```

## Warning: Raster pixels are placed at uneven horizontal intervals and will be shifted
## i Consider using `geom_tile()` instead.

```

```
## Raster pixels are placed at uneven horizontal intervals and will be shifted
## i Consider using `geom_tile()` instead.
## Raster pixels are placed at uneven horizontal intervals and will be shifted
## i Consider using `geom_tile()` instead.
## Raster pixels are placed at uneven horizontal intervals and will be shifted
## i Consider using `geom_tile()` instead.
```



8 Is disagreement caused by different irrigation conceptualizations?

```
# STUDY OF DEFINITION HETEROGENEITY #####
#####

# Read the survey spreadsheet -----

dt_definitions <- as.data.table(readxl::read_excel("dataset_survey_definitions.xlsx"))

setnames(dt_definitions, c("dataset", "citation_full", "paddy_rice_raw",
                           "temporal_raw", "justification", "irrigation_target"))

# Homogenise paddy rice treatment -----

dt_definitions[, paddy_rice:= fcase(grepl("^Included", paddy_rice_raw), "included",
                                   grepl("^Separated", paddy_rice_raw), "separated",
                                   grepl("^Unclear", paddy_rice_raw), "unclear")]
```

```

# Homogenise temporal concept -----

dt_definitions[, temporal:= fcase(grepl("^Annual", temporal_raw), "annual",
                                   grepl("^5-year", temporal_raw), "multi_year",
                                   grepl("^Reference", temporal_raw), "reference_period",
                                   grepl("^Seasonal", temporal_raw), "seasonal",
                                   grepl("^Unclear", temporal_raw), "unclear")]

# Keep clean columns -----

dt_clean <- dt_definitions[, .(dataset, paddy_rice, temporal,
                               irrigation_target, justification)]

# SETUP #####

attrs <- c("irrigation_target", "paddy_rice", "temporal")
ds_names <- dt_clean$dataset
n_perm <- 10000

# Base pairwise disagreement (pooled)-----

pairs <- CJ(ds1 = ds_names, ds2 = ds_names)[ds1 < ds2]
dt_sub <- dt_pres[n_pos > 0]
pairs[, disagreement:= mean(dt_sub[[ds1]] != dt_sub[[ds2]]), by = .(ds1, ds2)]

# Resolution-stratified pairwise disagreement-----

res_pairs <- dt_pres[n_pos > 0, {
  p <- CJ(ds1 = ds_names, ds2 = ds_names)[ds1 < ds2]
  sub <- .SD
  p[, disagreement:= mean(sub[[ds1]] != sub[[ds2]]), by = .(ds1, ds2)]
  p
}, by = resolution]

# RUN ALL INDIVIDUAL TESTS #####

results <- lapply(attrs, run_test,
                  pairs_dt = pairs, res_dt = res_pairs,
                  dt_meta = dt_clean, n_perm = n_perm)

## Warning in wilcox.test.default(d, s, alternative = "greater"): cannot compute
## exact p-value with ties
## Warning in wilcox.test.default(d, s, alternative = "greater"): cannot compute
## exact p-value with ties
## Warning in wilcox.test.default(d, s, alternative = "greater"): cannot compute
## exact p-value with ties

```

```
names(results) <- attrs
```

```
# Summary table-----
```

```
summary_tbl <- rbindlist(lapply(results, function(r) {
  data.table(attribute = r$attribute,
    pooled_W = r$pooled_W,
    pooled_p = r$pooled_p,
    obs_delta = r$obs_delta,
    perm_p = r$perm_p)
}))
print(summary_tbl)
```

```
##           attribute pooled_W pooled_p  obs_delta perm_p
##           <char>    <num>    <num>      <num>  <num>
## 1: irrigation_target      233 0.6538862 -0.004393000 0.4709
## 2:      paddy_rice      263 0.4154462 -0.008059414 0.5144
## 3:      temporal      253 0.5044833 -0.012013015 0.5300
```

```
# Resolution-stratified table -----
```

```
res_summary <- rbindlist(lapply(results, function(r) {
  r$res_wilcox[, attribute := r$attribute]
}))
print(res_summary)
```

```
## resolution n_same n_diff median_same median_diff      W  p_value
##      <char>  <int>  <int>      <num>      <num> <num>    <num>
## 1: 0.2deg    20    25    0.3419198    0.3457730 234.0 0.6454975
## 2: 0.4deg    20    25    0.2913762    0.2564966 248.0 0.5225355
## 3: 1deg      20    25    0.2454842    0.1923830 243.5 0.5635208
## 4: 0.2deg    22    23    0.2879978    0.3551894 275.0 0.3143284
## 5: 0.4deg    22    23    0.2764419    0.2643797 240.0 0.6192544
## 6: 1deg      22    23    0.2581066    0.2082699 214.0 0.8151239
## 7: 0.2deg    22    23    0.3504350    0.3287426 265.0 0.3979971
## 8: 0.4deg    22    23    0.3325741    0.2564966 238.0 0.6362766
## 9: 1deg      22    23    0.2806311    0.2047878 217.0 0.7963956
##           attribute
##           <char>
## 1: irrigation_target
## 2: irrigation_target
## 3: irrigation_target
## 4:      paddy_rice
## 5:      paddy_rice
## 6:      paddy_rice
## 7:      temporal
## 8:      temporal
## 9:      temporal
```

```

# OMNIBUS: any mismatch + dose-response #####

# Tag resolution pairs with all match columns -----

omni <- copy(res_pairs)
for (a in attrs) {
  lookup <- dt_clean[, .(dataset, val = get(a))]
  omni[lookup, on = .(ds1 = dataset), v1:= i.val]
  omni[lookup, on = .(ds2 = dataset), v2:= i.val]
  omni[, (paste0(a, "_match")):= fifelse(v1 == v2, "same", "different")]
  omni[, c("v1", "v2"):= NULL]
}

match_cols <- paste0(attrs, "_match")

# Number of mismatches (0-3)-----

omni[, n_mismatches:= Reduce(`+`, lapply(.SD, function(x)
  as.integer(x == "different"))),
  .SDcols = match_cols]

# Any mismatch-----

omni[, any_mismatch:= fifelse(n_mismatches > 0, "different", "same")]

# Omnibus Wilcoxon-----

s_omni <- omni[any_mismatch == "same", disagreement]
d_omni <- omni[any_mismatch == "different", disagreement]
wt_omni <- wilcox.test(d_omni, s_omni, alternative = "greater")

# Spearman dose-response-----

cor_omni <- cor.test(omni$n_mismatches, omni$disagreement,
  method = "spearman", alternative = "greater")

## Warning in cor.test.default(omni$n_mismatches, omni$disagreement, method =
## "spearman", : Cannot compute exact p-value with ties
cat("\n-- Omnibus Wilcoxon (any mismatch) --\n")

##
## -- Omnibus Wilcoxon (any mismatch) --
cat(sprintf("  W = %.1f, p = %.4f\n", wt_omni$statistic, wt_omni$p.value))

##  W = 1069.5, p = 0.4588

```

```

cat("\n-- Spearman dose-response --\n")

##
## -- Spearman dose-response --
cat(sprintf("  rho = %.4f, p = %.4f\n", cor_omni$estimate, cor_omni$p.value))

##  rho = -0.0741, p = 0.8033
# Omnibus permutation test (resolution-blocked)-----

obs_omni <- omni[, .(
  delta = mean(.SD[any_mismatch == "different", disagreement]) -
    mean(.SD[any_mismatch == "same", disagreement])
), by = resolution][, mean(delta, na.rm = TRUE)]

null_omni <- replicate(n_perm, {
  tmp <- copy(omni)
  for (a in attrs) {
    pl <- data.table(dataset = ds_names, val = sample(dt_clean[[a]]))
    tmp[pl, on = .(ds1 = dataset), v1:= i.val]
    tmp[pl, on = .(ds2 = dataset), v2:= i.val]
    tmp[, (paste0(a, "_match")):= fifelse(v1 == v2, "same", "different")]
    tmp[, c("v1", "v2"):= NULL]
  }
  tmp[, n_m:= Reduce(`+`, lapply(.SD, function(x) as.integer(x == "different"))),
    .SDcols = match_cols]
  tmp[, am:= fifelse(n_m > 0, "different", "same")]
  tmp[, .(delta = mean(.SD[am == "different", disagreement]) -
    mean(.SD[am == "same", disagreement])
  ), by = resolution][, mean(delta, na.rm = TRUE)]
})

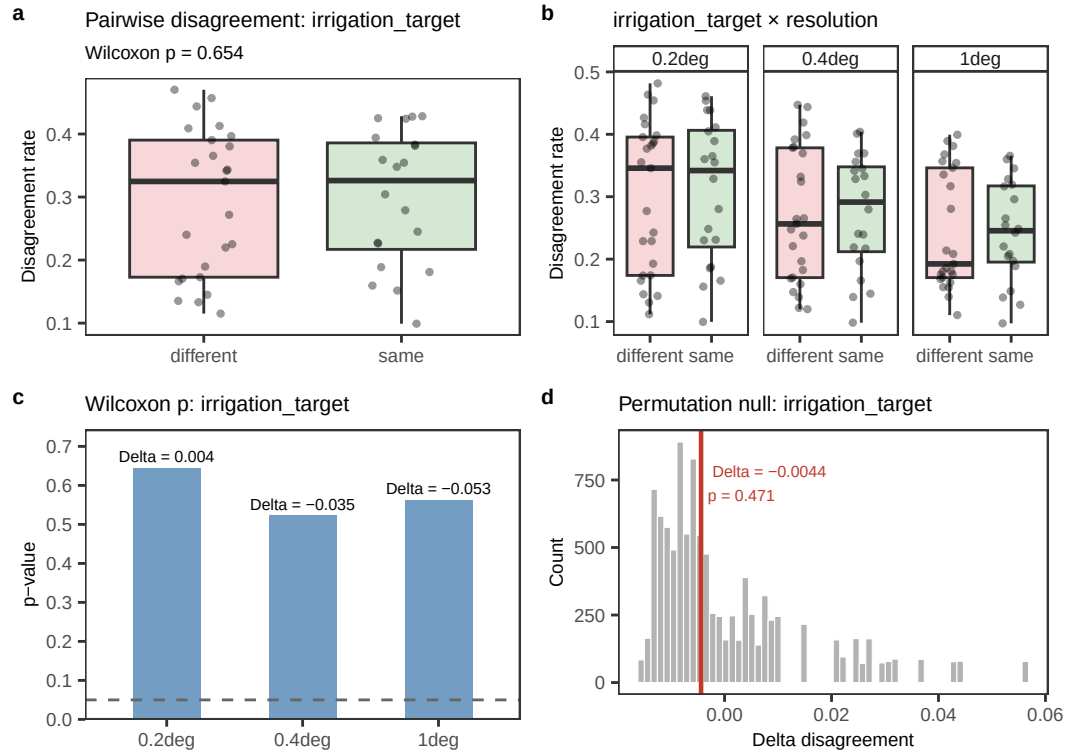
perm_p_omni <- mean(null_omni >= obs_omni)
cat(sprintf("\n-- Omnibus permutation --\n  obs Delta = %.4f, p = %.4f\n",
  obs_omni, perm_p_omni))

##
## -- Omnibus permutation --
##  obs Delta = 0.0024, p = 0.4478
# Plot results test -----

lapply(results, plot_attr)

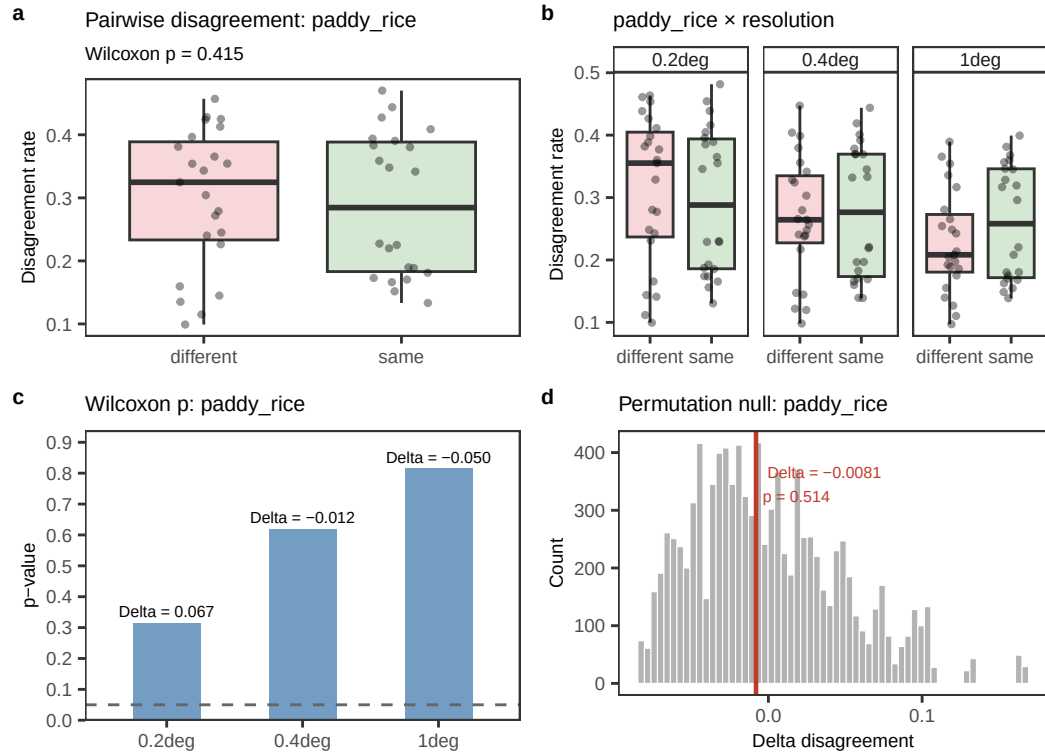
## $irrigation_target

```



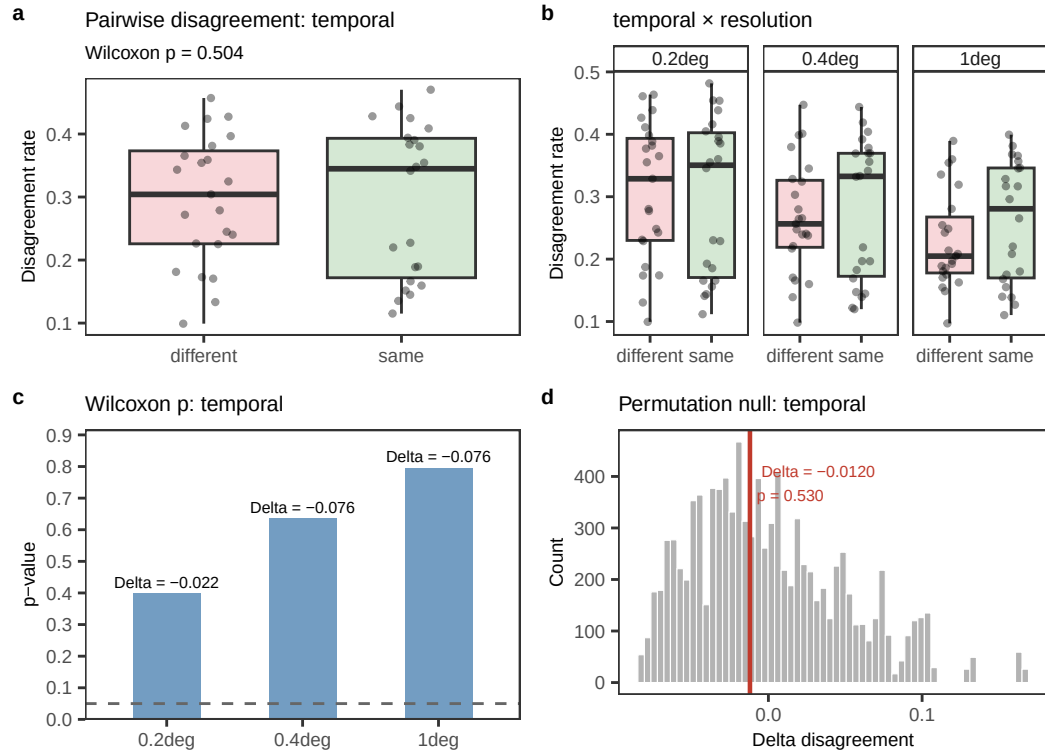
##

\$paddy_rice



##

\$temporal



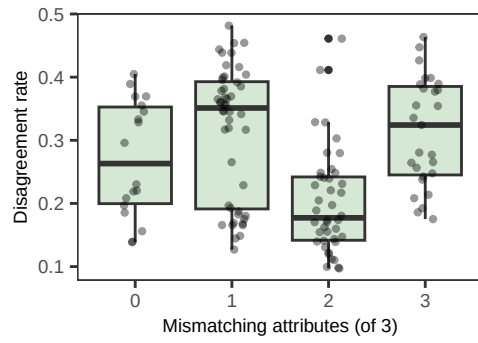
Omnibus plots -----

```
p_dose <- ggplot(omni, aes(x = factor(n_mismatches), y = disagreement)) +
  geom_boxplot(outlier.size = 0.8, fill = "#D5E8D4") +
  geom_jitter(width = 0.15, alpha = 0.4, size = 0.8) +
  labs(title = "Disagreement vs. number of \ndefinitional mismatches",
       subtitle = sprintf("Spearman rho = %.3f, p = %.3f", cor_omni$estimate, cor_omni$p.value),
       x = "Mismatching attributes (of 3)", y = "Disagreement rate") +
  theme_AP() + theme(plot.subtitle = element_text(size = 7))

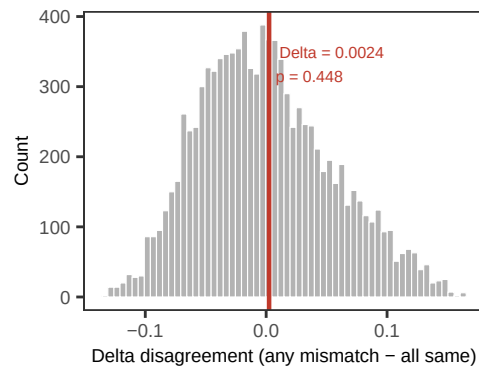
p_omni_null <- ggplot(data.table(delta = null_omni), aes(x = delta)) +
  geom_histogram(bins = 60, fill = "grey70", colour = "white", linewidth = 0.2) +
  geom_vline(xintercept = obs_omni, colour = "#C0392B", linewidth = 0.8) +
  annotate("text", x = obs_omni, y = Inf,
         label = sprintf("Delta = %.4f\np = %.3f", obs_omni, perm_p_omni),
         vjust = 2, hjust = -0.1, colour = "#C0392B", size = 2.2) +
  labs(title = "Omnibus permutation null \n(all attributes, resolution-blocked)",
       x = "Delta disagreement (any mismatch - all same)", y = "Count") +
  theme_AP()

plot_grid(p_dose, p_omni_null, ncol = 2, labels = "auto", label_size = 8)
```

a Disagreement vs. number of definitional mismatches
Spearman rho = -0.074, p = 0.803



b Omnibus permutation null
(all attributes, resolution-blocked)



9 Assessment of non-identifiability

```
# INSTABILITY UNDER SMALL PERTURBATIONS #####

# From long to wide dataset -----

dt_wide <- dcast(dt, lon + lat + country + code + continent +
               resolution ~ dataset, value.var = "mha")

# Run analysis -----
# 1% and 2% of the grid cell

inst_res <- run_instability_analysis(dt = dt_wide, dataset_names = dataset_names,
                                   tau_vals = c(1, 2), include_full = TRUE,
                                   include_loo = TRUE)

instability_dt <- inst_res$instability
classification_dt <- inst_res$classifications

instability_sum_all <- summarize_instability(instability_dt)
instability_sum_rel <- summarize_instability_relevant(instability_dt)

print(instability_sum_all)
```

```
## resolution n_cells frac_always_absent frac_always_present frac_unstable
##          <char>   <int>             <num>             <num>             <num>
## 1:      0.2deg  129628             0.3548770           0.3902243           0.2548986
## 2:      0.4deg   39758             0.3156597           0.4543237           0.2300166
## 3:       1deg   8068              0.2791274           0.5249132           0.1959593
## mean_flip_rate median_flip_rate q05_flip_rate q25_flip_rate q75_flip_rate
##          <num>             <num>             <num>             <num>             <num>
## 1:      0.05682169           0                0                0           0.04545455
## 2:      0.05049070           0                0                0           0.00000000
## 3:      0.04826024           0                0                0           0.00000000
## q95_flip_rate
##          <num>
## 1:      0.4545455
## 2:      0.4545455
## 3:      0.4545455
```

```
print(instability_sum_rel)
```

```
## resolution n_cells frac_unstable mean_flip_rate median_flip_rate
##          <char>   <int>             <num>             <num>             <num>
## 1:      0.2deg   83626             0.3951164           0.08807885           0
## 2:      0.4deg   27208             0.3361144           0.07378010           0
## 3:       1deg    5816              0.2718363           0.06694698           0
## q05_flip_rate q25_flip_rate q75_flip_rate q95_flip_rate
```

```
##           <num>           <num>           <num>           <num>
## 1:         0             0         0.09090909         0.4545455
## 2:         0             0         0.04545455         0.4545455
## 3:         0             0         0.04545455         0.5000000

# CONVERGENCE TEST #####

# Run on exact combinations -----

tau_vals <- c(1, 2)

res_exact_all <- rbindlist(lapply(tau_vals, function(tt) {
  run_by_resolution(dt = dt_wide, dataset_names = dataset_names,
    tau_percent = tt, mode = "exact")
}), fill = TRUE)

res_exact_all_sum <- summarize_agreement_vs_information(res_exact_all)

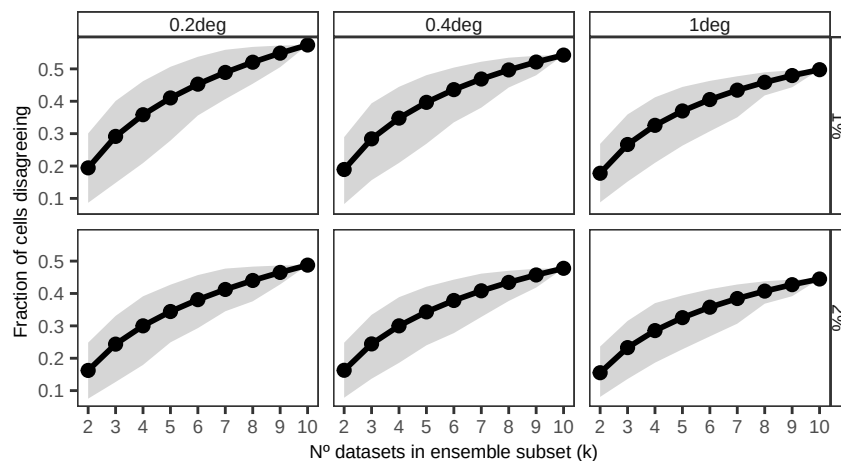
# Prettier labels-----

res_exact_all_sum[, tau_percent:= paste(tau_percent, "%", sep = "")]

# Plot results -----

p_disagree_exact <- ggplot(res_exact_all_sum, aes(k, frac_disagree_mean)) +
  geom_ribbon(aes(ymin = frac_disagree_q05, ymax = frac_disagree_q95), alpha = 0.2) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  facet_grid(tau_percent~ resolution) +
  scale_x_continuous(breaks = 2:length(dataset_names)) +
  labs(x = "N° datasets in ensemble subset (k)",
    y = "Fraction of cells disagreeing") +
  theme_AP()

p_disagree_exact
```



10 Irrigation masks

```
# PRODUCE AND EXPORT IRRIGATION MASKS #####
```

```
irrigation_mask_dt <- dt_pres[, .(lon, lat, country, code,  
                                continent, resolution, n_pos)]
```

```
setnames(irrigation_mask_dt, "n_pos", "k")
```

```
dir.create(here("datasets", "output"), recursive = TRUE, showWarnings = FALSE)
```

```
res_labels <- c("0.2deg" = "02", "0.4deg" = "04", "1deg" = "1")
```

```
irrigation_mask_dt[, {  
  fwrite(.SD, here("datasets", "output", paste0("irrigation_mask_", res_labels[resolution], ".  
}, by = resolution]
```

```
## Empty data.table (0 rows and 1 cols): resolution
```

11 Session information

```
# SESSION INFORMATION #####
```

```
sessionInfo()
```

```
## R version 4.5.2 (2025-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS Tahoe 26.3.1
##
## Matrix products: default
## BLAS: /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framework
## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: Europe/London
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods    base
##
## other attached packages:
## [1] benchmarkme_1.0.8  wesanderson_0.3.7  countrycode_1.6.1
## [4] sf_1.1-0           rnatralearnth_1.1.0 terra_1.8-93
## [7] here_1.0.2         sensobol_1.1.6     lubridate_1.9.4
## [10] forcats_1.0.1      stringr_1.6.0      dplyr_1.1.4
## [13] purrr_1.2.1        readr_2.2.0        tidyr_1.3.2
## [16] tibble_3.3.1       ggplot2_4.0.1.9000 tidyverse_2.0.0
## [19] cowplot_1.2.0.9000 scales_1.4.0        data.table_1.18.2.1
##
## loaded via a namespace (and not attached):
## [1] benchmarkmeData_1.0.4 gtable_0.3.6      xfun_0.55
## [4] lattice_0.22-7        tzdb_0.5.0        vctrs_0.7.1
## [7] tools_4.5.2           Rdpack_2.6.4      generics_0.1.4
## [10] parallel_4.5.2        proxy_0.4-29      pkgconfig_2.0.3
## [13] Matrix_1.7-4          KernSmooth_2.23-26 RColorBrewer_1.1-3
## [16] S7_0.2.1              lifecycle_1.0.5   compiler_4.5.2
## [19] farver_2.1.2          tinytex_0.58      codetools_0.2-20
## [22] htmltools_0.5.9       class_7.3-23      yaml_2.3.12
## [25] pillar_1.11.1         classInt_0.4-11   iterators_1.0.14
## [28] foreach_1.5.2         tidyselect_1.2.1  digest_0.6.39
## [31] stringi_1.8.7         labeling_0.4.3    rprojroot_2.1.1
## [34] fastmap_1.2.0         grid_4.5.2        cli_3.6.5
## [37] magrittr_2.0.4        e1071_1.7-17      withr_3.0.2
## [40] timechange_0.3.0      httr_1.4.8        rmarkdown_2.30
## [43] otel_0.2.0            hms_1.1.4         evaluate_1.0.5
```

```
## [46] knitr_1.51          rbibutils_2.4          doParallel_1.0.17
## [49] rlang_1.1.7           Rcpp_1.1.1             glue_1.8.0
## [52] DBI_1.3.0             rstudioapi_0.17.1      R6_2.6.1
## [55] units_1.0-0
```

```
## Return the machine CPU -----
```

```
cat("Machine:      "); print(get_cpu())$model_name)
```

```
## Machine:
```

```
## [1] "Apple M1 Max"
```

```
## Return number of true cores -----
```

```
cat("Num cores:    "); print(parallel::detectCores(logical = FALSE))
```

```
## Num cores:
```

```
## [1] 10
```

```
## Return number of threads -----
```

```
cat("Num threads: "); print(parallel::detectCores(logical = FALSE))
```

```
## Num threads:
```

```
## [1] 10
```